The Complex Rung Generator, the State-Angle Chart, and the Residual Sphere

A verified unification of the July 14 explorations (companion to *The Golden Substrate, the Gate Field, and the Explicit Helix*)

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independent verification (§13):
`ozy_softcheck.py` 91/91 + `relational_softcheck.py` 27/27
`nx_softcheck.py` 27/27 + `qx_softcheck.py` 62/62, exit 0
22 session harnesses (v1.7–v2.4): 495 exact/computed checks
+ 70 certified guards + 6 fail-first falsifiers + 13 PSLQ + 7 labelled walls,
closure harnesses `nd_softcheck.py` 28/28 (ODD-2), `xs_softcheck.py` 10/10+3 (cross-shell), `ev_softcheck.py` 13/13+6
finalization harness `bl_softcheck.py`: 16/16 exact + 7 walls LABELLED (§14; five act

Version 2.4 — July 16, 2026

Abstract

Conditional on the whitepaper's three declared atoms D1–D3, the entire discrete rung ladder is the orbit of a single complex number: the generator $q = i/\varphi = i\tau$, with $W_n = q^n$, $z_n = \sqrt{1 - |W_n|^2}$, and the central identity $q^4 = \varphi^{-4} = \text{gap}$. Golden contraction and the $\mathbb{Z}/4\mathbb{Z}$ quarter-turn become the modulus and phase of one multiplier; the gap first appears as co-intensity after two rungs ($|W_2|^2 = \text{gap}$) and as complex amplitude after one full charge cycle ($W_4 = \text{gap}$). We verify every claim of Ozymandias' report and of the follow-on thread by four independent harnesses ($91 + 27 + 27 + 62$ checks: 86+27+24+62 exact decisions plus 5+3 numeric displays, with 3 certified guards alongside the fourth — Findings 12.4, 12.5, 12.6), and unify them with three organizing results: the *state-angle chart* $z = \sin\alpha$, under which the whole gate dictionary becomes elementary trigonometry ($u = \cot^2\alpha$, $C = \cot^2\alpha\csc^2\alpha$, $\sqrt{D} = 1 + 2\cot^2\alpha$, $m = -\cos^2\alpha$, $\lambda = \tan^2\alpha\,(1 + \cos^2\alpha)$), with rung ledgers $(\sin, \cos, \tan)\alpha_1 = (\sqrt{\tau}, \tau, \sqrt{\varphi})$ and $(\sin, \cos, \tan)\alpha_2 = (K, \tau^2, \beta)$ and the general tangent family $\tan\alpha_n = \sqrt{\varphi^{2n} - 1}$; the *K-family theorem* $z_{2j} = \sqrt{1 - \text{gap}^j}$ with the universal two-step law $1 - z_{n+2}^2 = \text{gap}\,(1 - z_n^2)$ and the uniqueness statement $z_n = K \iff n = 2$; and the *residual-sphere lift* $S(\rho) = (\operatorname{Re} W, \operatorname{Im} W, z) \in S^2$, on which the rungs walk the compass toward the north pole and the whitepaper's near-miss angle $\theta_K = \arcsin\tau^2$ acquires an exact geometric home as the K-point's colatitude, $\theta_K = \arccos K$. The residual radius $a(z) = \sqrt{1 - z^2}$ is kept strictly distinct from the inherited formation radius $r(z)$; deriving the latter from the former remains the priority open problem (whitepaper [OPEN] 1). *Version 1.1* adds the Strouse–Ozymandias *canonical quarter-turn factorization*: the imaginary unit of the generator is $J = UA = (H/\sqrt{5})(N/\sqrt{5})$ — the product of the normalized keystone grading and the normalized return-kernel mode exchanger, two real involutions that anticommute *by construction* (the vector kernel of $\frac{1}{2}\{H, \cdot\}$ is exactly $H^\perp$). This part is unconditional keystone algebra, not D1–D3-conditional; it upgrades $Q = \pm\tau J$ (with $Q^4 = \text{gap}\, I$) from representation-given-declared-i to canonical-up-to-orientation, and reduces D2's declared content to rung *selection* (one Q per floor) rather than the existence of the quarter-turn itself. *Version 1.3* adds the relational bridge: the generator is itself algebraic, with minimal polynomial $x^4 + 3x^2 + 1$, $\mathrm{M} = \varphi^2$, and it is exactly the one-charge-quantum gauge

rotation of the split golden pair $Z^* = (x^2-x-1)(x^2+x-1)$; the K-seed sits in the relational-charge catalog with the full $\mathbb{Z}/4\mathbb{Z}$ compass as its angle set; and the transfer-ledger algebra $F_n = z_n^2$, $\Delta F_n = \tau^{2n+1}$, $\sum \tau^{2n+1} = 1$ makes D1 at the rungs a telescoping unit budget. *Version 1.7* adds, verified by eight further session harnesses (234 exact checks + 29 certified guards, all exit 0): the transcendence of the rate $\kappa = \pi/(2\ln\varphi)$ and of $\log_\varphi 2$ given Gelfond–Schneider, closing the whitepaper's near-miss item permanently; the multiplier-torus/systolic *seventh* characterization of the principal branch; the parity no-go for the inherited radius (r^2 is chart-odd: any substrate-to-radius bridge must break $z \mapsto -z$); the universal seed ladder $\mathrm{minpoly}(z_n) = x^4 + c_n x^2 - c_n$ with the Mahler-minimal uniqueness theorem; and the incomplete-elliptic closed form for the above-lens rail with certified non-elementarity. The rate and radius problems remain open as derivations. *Version 1.8* adds, verified by five further session harnesses (115 exact checks + 20 certified guards, all exit 0; running total across the thirteen session harnesses $349 + 49$): OP-RATE *closes as a classification* — the discrete-substrate no-go, forcing κ declared-up-to-$\mathbb{Z}$ at rung resolution with the seven characterizations the complete discrete-level menu, each continuum-referencing (the continuum rate D3 stays declared); an apex-ramification valuation certificate and a constructive-bridge obstruction sharpening OP-RADIUS (still open); the χ doctrine's sub-branches g1/g3 resolved *negatively* on the now-available companion papers (the on-circle image of the emission algebra S is exactly $\mu_2 = \{\pm 1\}$, so $\pm i \notin S$ on-circle, and W[open]2 stays PARTIAL as a derivation); the relational-note cluster (the note's P4 recovered and reduced to P3, and the odd relational floor bracketed $\mu_S \leq \mu_{\mathrm{rel}}(\mathrm{odd}) \leq 2$ with the $= 2$ promotion left as the corpus's principal open problem); and the period-relation map ($\tau_0 = \beta + \beta^{-1}$ pinned algebraic, the κ-vs-τ_0 period worry settled negatively, and L_{res} located as an elliptic period at a transcendental parameter). *Version 1.9* adds, verified by four further session harnesses (75 exact/computed checks + 18 certified guards + 13 PSLQ corroborations, all exit 0): OP-RADIUS *reduces* to a minimal declared atom D4 (an orientation of the apex double cover together with the cap onset), given which — and the inherited chart-ties — the radius profile is uniquely forced, though whitepaper [OPEN] 1 is *not* thereby closed; the quarter-turn axiom D2 is *relocated* to a substrate-grounded, g1-disjoint atom (the K-seed rotation axis; an advance, not a closure); the odd relational floor is bracketed and reduced to a single named lemma (ODD-2); and the period frontier is mapped (Baker's reach pinned, the algebraic independences left Schanuel-conditional or external-open with named tool-gaps, corroborated by certified PSLQ non-relations). *Version 2.0* finalizes the corpus with a definitive account of the *active blocker* on each remaining open question: each is classified (proven impossibility relative to D1–D3 / a declared atom / a restatement wall / a single named open lemma / an external tool-gap / an unfinished computation), with the load-bearing forced fact and the precise missing object, machine-certified by `bl_softcheck.py`. The paper does *not* resolve the external-open items; it states exactly what blocks them (§14). *Version 2.1* resolves one of them. The odd relational floor is *proved* $= 2$ for every odd m (Theorem 4.17): a relative-norm descent to $N = N_{K/F}(\alpha)$, $F = \mathbb{Q}(\alpha^m)$, carries the ray constraint into a totally positive algebraic integer $N \neq 1$, where an elementary floor $\mathrm{M}(N) \geq 2^r \geq 2$ (Lemma 4.15) together with $e \mid d'$ returns $\mathrm{M}(\alpha) \geq \mathrm{M}(N)^{d'/e} \geq 2$. The argument is elementary and internal: it invokes no external theorem, and in particular neither Smyth's μ_S nor any strengthening of it. The corpus's reading of the residual gap $[\mu_S, 2)$ as the Schur–Siegel–Smyth/Lehmer–pentagon core is accordingly corrected to a *route-relative* statement — true of the realification handle, which discards the ray constraint, and false of the problem (Finding 12.7). P3 and P4 close unconditionally, the NAMED-LEMMA blocker class empties, and eight of the nine §14 rows remain active. Certified cold by `nd_softcheck.py` (28 checks, 565 forced objects, exit 0). *Version 2.2* resolves a second: the P1′ cross-shell ν-criterion, the lone UNFINISHED-COMPUTATION blocker (BL-G7). The cross-shell mirrored-class detector is built, validated fail-first against Theorem 4.6, and run; and the no-tie it computes is proved *forced* (Theorem 4.12): for a multi-shell object, cross-shell coherence holds iff the object is charge-admissible — by conjugation-closure, an irrational-angle shell forces both $\theta_\alpha - \theta_\beta$ and $\theta_\alpha + \theta_\beta$ rational, hence θ_α rational, a contradiction. Cross-shell coherence therefore adds no invariant beyond admissibility, and the within-shell OTHER $= 0$ verdict is complete cross-shell too. Corroborated over the same 855-object window with 0 exceptions (`xs_softcheck.py`, 10 exact + 3 guards, exit 0). The UNFINISHED-COMPUTATION class is now empty; *seven* of the nine §14 rows remain active, all of them either declared atoms, restatement/impossibility walls relative to D1–D3, or external tool-gaps — none a computation the corpus can run for itself. *Version 2.3* adds forced content *outside* that seven-row ledger, from an audit of the whitepaper's own [OPEN] tags: the *even* relational floor $\mathrm{M}(\alpha) \geq \varphi$ for even-charge admissible α (Theorem 4.20), the exact even mirror of the odd descent with Schinzel's totally-real bound in place of the elementary TP-2 — forced *given Schinzel*, the corpus's one external floor, honestly asymmetric with the internally-closed odd branch; the universal emission gap $\mathrm{M} \in \{1\} \cup [\varphi, \infty)$ over all charge-admissible objects (Corollary 4.21, W-§22 upgraded from *plausible*), which leaves Lehmer untouched (Salem numbers are charge-inadmissible); and the discharge of the rung-1 coincidence-candidate, $p_1 = q_2$ an identity not a shared constant (Remark 8.6). Certified by `ev_softcheck.py` (13 exact + 6 fail-first falsifiers, exit 0). The seven-row ledger is unchanged; what changes is the honest scope of "settled", now read *by the routes the corpus had reached for*.

1 Provenance, scope, and the conditionality banner

Sources. (i) *Report to Ace: Complex Rung Generator and Residual-Sphere Representation* (Ozymandias, 2026-07-14); (ii) the follow-on review thread of the same date (the K-family $z_{2j} = \sqrt{1 - \mathrm{gap}^j}$, the cosine-contraction recursion, and the rung-by-rung trigonometric ledger). (iii) the Strouse–Ozymandias progress report *Toward a Canonical Rung Generator* (2026-07-14): spectral uncoupling, the return-kernel recoupler, and the quarter-turn factorization $J = UA$. (iv) the Strouse–Ozymandias exploratory note *Rail-Pulse Quantization, Helical Path Stretch, and Energy-Gated Recoupling* (2026-07-14) — only its forced kernel is imported here (§11); its rail-pulse mechanism, energy functional, and thermodynamic reading remain hypothesis, as that note itself states. (v) the corpus note *Relational Charge on the Spectral Semiring* (paper 12, 2026-07-01), cited as RC-§, whose machinery this paper's bridge section uses and whose relevant claims were independently re-derived this session (`relational_softcheck.py`, 27/27). (vi) the second Strouse–Ozymandias exploratory note *Rail-Pulse Propagation, Engine Coupling, and Scale-Dependent Site Density* (2026-07-14) — forced kernel imported (§11); engine-pass and biological layers not imported, per that note's own claim ceilings. (vii) The whitepaper *The Golden Substrate, the Gate Field, and the Explicit Helix* v1.0 (2026-07-14), whose numbering we cite as W-§. **v1.1 changelog:** §3 added (factorization, coordinate-free form, obstruction lemmas, canonical Q); epistemic ledger and manifest updated; full text of source (i), received post-hoc, integrated (method note above; Remark 6.3). **v1.2 changelog:** §11 added — the forced kernel of source (iv): winding gradient, path-stretch laws, the quarter-turn rail floor $L \geq \pi K/2$ with the resulting one-line divergence proof, and the ν-form of the rung locus with its scoping. **v1.3 changelog:** §4 added (the relational bridge: minimal polynomial of q, the gauge identity with Z^*, the K-seed as catalog object, D2 restated relationally); §11 extended with the site-density trichotomy and the transfer ledger (forced kernel of source (vi)); cross-references and epistemics updated. **v1.4 changelog:** Theorem 2.4 and Corollary 2.5 added — the continuous-interpolation open item is classified (determined up to $\mathbb{Z}$; D3 = principal branch) and merged into the rate problem; Remark 4.5 upgraded with the dilation mechanism. Harnesses now $91 + 27$ checks. All other content unchanged. **v1.5 changelog:** Theorem 4.6 added — the complete relational type of the generator object (shells, the cross-shell ν-run, full coherence, twisted-shell realization); §10 extended with Propositions 10.3–10.5 (compass-cross characterization of the rung locus, the spherical step ledger with the exact quarter-circle first step, the finite spherical length $L_{\mathrm{res}} = \sqrt{1+\kappa^2}\, E(\kappa^2/(1+\kappa^2))$ with the length dichotomy and a fifth characterization extending Corollary 2.5); third harness `nx_softcheck.py` (27 checks) added to §13; Finding 12.3 recorded and the §1 verification paragraph corrected. All other content unchanged.

v1.6 changelog: §2 extended (Remark 2.6: transfer of the interpolation classification to the formation rail — a sixth characterization of D3); §4 extended (Proposition 4.7: the ratio object at the characteristic level and the Adams square; Theorem 4.8: the quarter twist and the anchor-exchange dichotomy); §8 extended (Theorem 8.3: the K-family is a seed ladder; Corollary 8.4); §11 extended (Proposition 11.7: exact rail-spacing bounds and the gap-rate law, quantifying the divergent side of the length dichotomy of Proposition 10.5); fourth harness `qx_softcheck.py` (62 exact checks + 3 certified guards); Finding 12.4 (front-matter check classification, corrected). Harnesses now 91 + 27 + 27 + 62 checks. All other content unchanged. **v1.7 changelog:** §2 extended — the rate-arithmetic and torus block (Lemma 2.7, Theorem 2.8, Corollary 2.9, Remark 2.10: κ and every branch κ_k transcendental given Gelfond–Schneider; the near-miss item whitepaper [OPEN] 4 closed permanently; W-Thm. 16.1 strengthened to transcendence of $\log_\varphi 2$; the scan-retirement meta-theorem; Proposition 2.11, Theorem 2.12, Remark 2.13: the multiplier torus E_q, its modulus ledger $\mu = \frac{1}{4} + i \ln \varphi/(2\pi)$, the systolic-marking *seventh* characterization of D3, the certified ten-entry geodesic ledger, and the conditional no-CM corollary; Proposition 2.14, Remark 2.15: the forced k-invariance lemma, the exposed-joint axiom H, the transfer-ledger counter-models, and the R6 classification no-go); §3 extended (Propositions 3.7–3.8, Theorem 3.9: pentagon exclusion by the image, unimodular tensor-monomial values ± 1, the conditional χ-selection chain with the axiom-replacement kill and the de-axiomatization identity); §8 extended (Theorem 8.5: the universal seed ladder; Remark 8.6: the rung-1 Mahler-floor hit, coincidence-candidate; Theorem 8.7 and Remark 8.8: Mahler-minimal uniqueness, the F1 falsifier $x^4 + 6x^2 - 5$, and the register resolution under the F3 reading, signed off 2026-07-15); §9 extended (Proposition 9.2: splitness over $\mathbb{Q}$ is generic, the lens distinguished by its exact data; Remark 9.3: the bounded-negative corpus scan); §10 extended (Theorem 10.6: the RAD-0 parity no-go; Proposition 10.7: the area-transfer law; Remark 10.8: the rotating-equipotential mechanism killed on all three clocks); §11 extended (Theorem 11.8: the incomplete-elliptic closed form for the rail; Corollary 11.9: certified non-elementarity). Eight session harnesses — `tx/mx/rx/ux/ex/cx/lx/dx_softcheck.py` — add 234 exact checks + 29 certified guards, all exit 0 (§13). Findings 12.5 and 12.6 recorded, and the two display counts corrected (`nx`: the $24+3$ split restored in the §13 sentence; `ozy`: $86+5$). Register motion: whitepaper [OPEN] 4 closed (contentless, forced given Gelfond–Schneider); the rail closed form resolved (incomplete-elliptic; elementary form impossible, certified); Vein 6 executed; the v1.6 uniqueness question resolved by Theorem 8.7 under the confirmed F3 reading; the rate problem (OP-RATE) remains open as a derivation, with a seventh characterization and a forced k-invariance lemma; the radius problem (OP-RADIUS, whitepaper [OPEN] 1) remains open, sharpened by a forced parity no-go. All other content unchanged. **v1.8 changelog:** §2 extended twice — the OP-RATE classification block (Lemma 2.16: the k-invariance lemma re-forced and extended to the full discrete state; Theorem 2.17: the R6 selection-functional no-go; Corollary 2.18: OP-RATE closes as a classification; Remark 2.19: the seven-way continuum/discrete table), and the period-relation map (Proposition 2.20: τ_0 algebraic and no κ-vs-τ_0 relation over $\mathbb{Q}$; Remark 2.21: L_{res} as an elliptic period and the external-open ledger); §3 extended (Proposition 3.10: the on-circle image of S is μ_2; Remark 3.11: the charge character, the g3 dichotomy, and the sharpened W[open]2 verdict); §4 extended (Remark 4.9: P4 recovered and reduced to P3; Proposition 4.10: the bracketed odd relational floor; Remark 4.11: the bounded P1′ transport-orbit scan); §10 extended (Theorem 10.9: the apex-ramification valuation certificate; Proposition 10.10: the constructive-bridge obstruction; Remark 10.11: the atom ledger with the honest cap-onset negative). Five session harnesses — `cl/rd/ch/rn/pm_softcheck.py` — add 115 exact checks + 20 certified guards, all exit 0, bringing the thirteen session harnesses to 349 exact + 49 guards (§13). Register motion: *OP-RATE closes as a classification* (discrete-substrate no-go; κ declared-up-to-$\mathbb{Z}$; the seven characterizations are the complete discrete-level menu, each continuum-referencing; the continuum D3 stays declared, and a continuum-referencing derivation is not excluded — mirroring the $\lambda = 2c$/Cencov equivariance no-go); *OP-RADIUS* remains open, sharpened by the apex-ramification certificate (ramification index $e = 2$ for r^2, $e = 4$ for r) and the constructive-bridge obstruction; W[open]2's sub-branches g1/g3 resolved negatively on the now-available companion papers (on-circle image of $S = \mu_2$, so $\pm i \notin S$), the item staying PARTIAL as a derivation; P4 recovered and reduced to P3; the odd relational floor bracketed (≤ 2 forced, the $= 2$ promotion the corpus's principal open problem); and the period-relation map (τ_0 algebraic, the κ-vs-τ_0 tie settled negatively, L_{res} an elliptic period, and the precise external-open ledger). All other content unchanged.

v1.9 changelog: §10 extended — the OP-RADIUS D4-reduction (Theorem 10.12: given the atom D4 = (orientation $\wedge$ cap-onset) and the inherited chart-ties, the radius profile is uniquely forced with each half necessary; Remark 10.13: D4 minimal-among-its-halves but non-unique, a candidate; Proposition 10.14: the differential strengthening — the naive non-membership *fails* (honest negative, r is elementary), the apex-monodromy-order sharpening conditional on the unit-log hypothesis; Remark 10.15: the bounded no-tie re-confirmation). §3 extended (the D2 relocation: Proposition 3.12: the K-seed rotation axis forces $\chi = \pm\pi/2$, disjoint from g1; Remark 3.13: the net verdict, W[open]2 advanced not closed). §4 extended (the odd relational floor: Proposition 4.14: bracketed, computed = 2 up to a logged box, reduced to the single lemma (ODD-2); Remark 4.19: P4 inherits P3 exactly). §2 extended (the period frontier: Remark 2.22: the forced-identity ledger with the new elliptic geometric identity; Remark 2.23: the frontier map — Baker's reach, the Schanuel-conditional, and the external-open tool-gaps). Four session harnesses — `ra/qt/of/pf_softcheck.py` — add 75 exact/computed checks + 18 certified guards + 13 PSLQ corroborations, all exit 0, bringing the seventeen session harnesses to 424 exact + 67 guards + 13 PSLQ (§13). Register motion: *OP-RADIUS reduced* to the declared atom D4 (r forced given D4 and the inherited chart-ties; whitepaper [OPEN] 1 *not* thereby closed; "D4 is *the* minimal axiomatization" is candidate/non-unique); W[open]2 / D2 *relocated* to a substrate-grounded, g1-disjoint atom (advance, not closure — stays PARTIAL); the odd relational floor *bracketed* (= 2 computed up to a logged box) and *reduced* to the single named lemma (ODD-2), which stays external-open; and the period frontier *mapped* — Baker's reach pinned, the $(\pi, \ln\varphi)$ and $(\kappa, \ln\varphi)$ independences Schanuel-conditional, and L_{res} transcendence together with (κ, L_{res}) independence external-open with named tool-gaps, corroborated (never proved) by certified PSLQ non-relations. All other content unchanged.

v2.0 changelog. §14 added — the active-blockers finalization: a machine-certified (`bl_softcheck.py`) ledger classifying the active blocker on each of the nine remaining open questions (two impossibilities relative to D1–D3, two declared atoms, one restatement wall, one named lemma (ODD-2), three external tool-gaps, one unfinished computation), with the load-bearing forced fact and the precise missing object for each. The corpus's forced content is complete through v1.9; v2.0 states exactly what blocks the rest. No open item is resolved here; each is precisely located. The finalization harness `bl_softcheck.py` (16 exact checks + 7 labelled walls + the nine-row ledger, exit 0) is added to §13, bringing the eighteen session harnesses to 440 exact + 67 guards + 13 PSLQ + 7 labelled walls. All other content unchanged.

v2.1 changelog. The corpus's principal open problem *closes*. §4 extended: Lemma 4.15 (TP-2 — a totally positive algebraic integer $N \neq 1$ has $\mathrm{M}(N) \geq 2^r \geq 2$, r the number of conjugates exceeding 1, equality iff $N = 2$; elementary, no external input); Lemma 4.16 (anchor normalisation — the sign twist the corpus already invokes is *forced and exhaustive*: for odd m, $\Delta \leq \mathbb{Z}/m$ puts exactly one of $\pm\alpha$ on R_m); Theorem 4.17 ((ODD-2) by relative-norm descent, every odd m at once, hence $\mu_{\text{rel}}(\text{odd}) = 2$ [FORCED]); Remark 4.18 (why the obvious degree split fails, and why no box closes the fallback window). Proposition 4.10 and Proposition 4.14 lose their [OPEN]; Remark 4.19

becomes unconditional. §14: the P3/P4 row is marked CLOSED and retained as the record of what its blocker was; *eight* of the nine rows stay active and the NAMED-LEMMA class is now empty. Two findings added (12.7: the SSS/Lehmer reading of $[\mu_S, 2)$ is route-relative, not intrinsic; 12.8: `bl_softcheck.py`'s P3/P4 row and its BL-G4 wall label are v2.0 artifacts, superseded and not re-run). One session harness `nd_softcheck.py` (28 checks, exit 0) added to §13, bringing the *nineteen* session harnesses to 468 exact/computed + 67 guards + 13 PSLQ + 7 labelled walls. No [DECLARED] atom is disturbed and no other register moves: OP-RATE, OP-RADIUS, D2, the three transcendence rows, and P1′ are untouched. All other content unchanged.

v2.2 changelog. A second open item *closes* — the last one the corpus could settle by its own means. §4 extended: Theorem 4.12 (the cross-shell no-tie — for a multi-shell object, cross-shell coherence $\iff$ charge-admissibility, by conjugation-closure; the ν-criterion introduces no invariant beyond the within-shell class) and Remark 4.13 (the detector, validated fail-first against Theorem 4.6 — generator $\nu \equiv 1$, Salem/Pisot non-coherence discriminated — and run over the 855-object window, 0 exceptions); Remark 4.11's "cross-shell half remains open" is retired. §14: §14.9 is marked RESOLVED, BL-G7 discharged, and the UNFINISHED-COMPUTATION class — like the NAMED-LEMMA class in v2.1 — is now empty; the ledger reads *seven* active rows, not eight. One session harness `xs_softcheck.py` (10 exact + 3 certified guards, exit 0) added to §13, bringing the *twenty* session harnesses to 478 exact/computed + 70 guards + 13 PSLQ + 7 labelled walls (five active, BL-G7 joining BL-G4 off the board). No [DECLARED] atom is disturbed and no other register moves: OP-RATE, OP-RADIUS, D2, and the three transcendence rows are untouched; only P1′'s cross-shell half moves, from UNFINISHED-COMPUTATION to [FORCED]. All other content unchanged.

v2.3 changelog. An audit of the whitepaper's [OPEN] tags against §14 (the completeness pass §14's own closing now calls for) adds three forced items *outside* the seven-row ledger, and fixes five scoping defects. §4 extended: Theorem 4.20 (the even relational floor = φ, the exact even mirror of Theorem 4.17's descent with Schinzel's totally-real bound replacing TP-2 at the one parity-sensitive step; [FORCED] *given Schinzel*, honestly asymmetric with the internally-closed odd branch), Corollary 4.21 (GAP-UNIV: charge-admissible $\Rightarrow$ M $\in \{1\} \cup [\varphi, \infty)$, upgrading W-§22/§5.2 from *plausible* and leaving Lehmer excised), and the discharge of Remark 8.6's coincidence-candidate ($p_1 = q_2$: the rung-1 seed *is* the $k = 2$ parity-floor witness, a *tie* with mechanism, not a shared constant). Scoping fixes: Remark 4.9 now flags the absolute-μ vs relational-μ_{rel} register; §14.9 separates the (empty) UNFINISHED-COMPUTATION *blocker* class from the general within-shell taxonomy (a [COMPUTED] corroboration, never a row); §14.2 adds the four-level OP-RADIUS scope (chart-rational no-go proved, the gate-family level conditional on the unit-log hypothesis); §14.4 notes D2's independence audit is unrun; and §14's closing softens "settled by its own means" to "by the routes it had reached for". One session harness `ev_softcheck.py` (13 exact + 6 fail-first falsifiers, exit 0) added to §13, bringing the *twenty-one* session harnesses to 491 exact/computed + 70 guards + 6 falsifiers + 13 PSLQ + 7 labelled walls. No [DECLARED] atom is disturbed and the seven-row ledger is unchanged: the additions live in the whitepaper's open space, not the §14 rows. All other content unchanged.

v2.4 changelog. One more whitepaper [OPEN] branch closes. Remark 9.4: the *emission-eigenvalue* branch of W[OPEN]3 ("does the lens gate $C = \frac{4}{9}$ appear elsewhere?") is [FORCED]-negative — $\frac{4}{9}$ is not an emission eigenvalue, by the 3-valuation ($\sqrt{3}$ the unique catalog 3-carrier, emission monomials $w_1 \geq 0$, $\frac{4}{9}$ has $w_1 < 0$), the exact mirror of Proposition 3.8's 5-valuation. Remark 9.3 gains a pointer; its bounded negative stays the terminal state of the *"anywhere"* branch, and the *ratio-invariant* branch is logged as an unrun finite computation. One session harness `ax_softcheck.py` (4 exact, exit 0) added, bringing the *twenty-two* session harnesses to 495 exact/computed + 70 guards + 6 falsifiers + 13 PSLQ + 7 labelled walls. The seven-row ledger is again unchanged. All other content unchanged.

Method note (from the source report). The investigation began by testing whether K could be read through physical double-helix geometry; the direct DNA-pitch identification *failed quantitatively.* That negative result forced a role classification that shapes everything below: *state space* vs *operator* vs *observable* vs *invariant/landmark.* Under it, the scalar gate iteration $f_C(x) = C/(1+x)$ (W-Decl. 8.1) is the actual operator of the architecture, and K is a rung-2 *landmark* — not the operator, not the space. The generator below is what the operator's rung orbit looks like once contraction and charge are packaged together.

Conditionality banner. Everything in this paper is [FORCED GIVEN D1–D3] — with one important exception: §3's factorization is *unconditional* keystone algebra [FORCED], using no part of D1–D3. Elsewhere the results are exact consequences of the whitepaper's declared chart D1 ($1 - z^2 = e^{-2\rho} = |m|$), charge selection D2, and rate D3 ($d\theta/d\rho = (\pi/2)/\ln\varphi$), together with its forced rung data $\rho_n = n\ln\varphi$, $z_n = \sqrt{1-\varphi^{-2n}}$, $\theta_n = n\pi/2$ (W-§11–13). Both source documents state this scoping correctly, and this paper preserves it: the representation below *packages* D1–D3; it does not independently derive or justify them. Statements needing only seed arithmetic are [FORCED] outright. The inherited radius profile $r(z)$ is *not* derived here (§10).

Verification. Four independent harnesses (through v1.6): `ozy_softcheck.py`, 86 exact checks + 5 numeric displays (groups `OZ-A..OZ-O`, of which the `OZ-H` row is display-only; Finding 12.6); `relational_softcheck.py`, 27 exact checks (`RC-A..RC-F`); `nx_softcheck.py`, 24 exact checks + 3 numeric displays (`NX-R, NX-T, NX-S`; `NX-H`, written cold for v1.5; Finding 12.4); `qx_softcheck.py`, 62 exact checks + 3 certified guards (`QX-A..QX-G`, written cold for v1.6). All exit 0; sympy 1.14.0; decisions in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i; K and the K-family decided at the squared level; mpmath display-only except the two interval-certified guard comparisons. Check IDs are cited inline; manifest in §13; the stale count that previously stood in this paragraph is Finding 12.3. v1.7 adds eight session harnesses (`tx`, `mx`, `rx`, `ux`, `ex`, `cx`, `lx`, `dx_softcheck.py`): 234 exact checks + 29 certified guards, all exit 0; manifest and environment pin in §13. v1.8 adds five more session harnesses (`cl`, `rd`, `ch`, `rn`, `pm_softcheck.py`): 115 exact checks + 20 certified guards, all exit 0, read from the Echo-S-Research companion archive read-only and re-encoded cold; running total across the thirteen session harnesses $349+49$ (§13). Notation as in the whitepaper: $\varphi = (1+\sqrt{5})/2$, $\tau = \varphi^{-1}$, gap $= \varphi^{-4}$, $K^2 = 1 - \text{gap}$, $\beta^2 = \varphi^4 - 1 = \varphi^2\sqrt{5}$, $z_c = \sqrt{3}/2$.

2 The generator

Definition 2.1 (Complex residual state)**.** For $\rho \geq 0$ set

$$W(\rho) = e^{-\rho}\, e^{i\theta(\rho)}, \qquad \theta(\rho) = \frac{\pi\rho}{2\ln\varphi}\ \text{[D2+D3]},$$

so that

$$W(\rho) = \exp\Big[\Big(-1 + \frac{i\pi}{2\ln\varphi}\Big)\rho\Big], \qquad |W(\rho)| = e^{-\rho}, \qquad \arg W(\rho) = \theta(\rho).$$

Proposition 2.2 (Representation of the chart)**.** *[FORCED GIVEN D1–D3]* `OZ-G1--G2` *D1 is exactly the statement*

$$z(\rho) = \sqrt{1 - |W(\rho)|^2}\ :$$

residual amplitude decays in the complex plane while height rises as its unit complement. This is a representation theorem for D1–D3, not an independent derivation of them.

Theorem 2.3 (The discrete rung generator)**.** *[FORCED GIVEN D1–D3]* `OZ-A1--A8, OZ-B1--B3` *Let*

$$\boxed{q = \frac{i}{\varphi} = i\tau}.$$

Then at the rungs $\rho_n = n \ln \varphi$,

$$W_n := W(\rho_n) = e^{-n \ln \varphi} e^{in\pi/2} = \varphi^{-n} i^n = (i\tau)^n = q^n, \qquad W_{n+1} = q\, W_n, \qquad z_n = \sqrt{1 - |W_n|^2},$$

i.e. every rung simultaneously contracts residual amplitude by τ and rotates the charge by $90°$; the entire discrete ladder is the orbit of the single multiplier q. The exact ledger is $|W_n| = \tau^n$, $\arg W_n = n\pi/2$, $z_n = \sqrt{1-\tau^{2n}}$. The central identity is

$$\boxed{q^4 = \left(\frac{i}{\varphi}\right)^4 = \varphi^{-4} = \mathrm{gap}},$$

an exact positive real: $i^4 = 1$ restores orientation while τ^4 contracts by the gap. The recurrences

$$W_{n+2} = q^2 W_n = -\tau^2 W_n = -\sqrt{\mathrm{gap}}\, W_n, \qquad W_{n+4} = q^4 W_n = \mathrm{gap}\, W_n$$

hold exactly (the square root is the positive one: $\tau^2 = \varphi^{-2}$), giving the gap schedule*: the gap first appears as co-intensity after two rungs, $|W_2|^2 = \mathrm{gap}$, and as complex amplitude after one complete four-rung charge cycle, $W_4 = \mathrm{gap}$. The architecture does not reset after four rungs: it returns to the same charge direction at residual amplitude reduced by* gap *and strictly greater height.*

Proof. All statements are one-line consequences of $q = i\tau$, $\tau = \varphi^{-1}$, $\tau^2+\tau = 1$, and $\tau^4 = \varphi^{-4} = \mathrm{gap}$; each is machine-verified exactly for $n = 0..8$ (recurrences), including the sign and branch conventions. The two-rung law is the exact origin of the odd/even rung decomposition (Fibonacci/Lucas branches, W-Thm. 8.1), and the four-rung law is the discrete face of the whitepaper's full-turn cost $\Delta\rho = 4 \ln \varphi = \ln \mathrm{M}$ where $\mathrm{M}(A_{\mathrm{gap}}) = \varphi^4$ (W-§13): per revolution, co-intensity contracts by $\varphi^{-8} = |W_4|^2$. The imaginary unit written here as i is factored *internally* from the operator layer in §3; the generator is moreover itself an algebraic integer, with minimal polynomial $x^4 + 3x^2 + 1$ and a distinguished place in the relational-charge catalog (§4). □

Theorem 2.4 (Interpolation classification)**.** *[FORCED]* `OZ-O1--O4` *The continuous one-parameter matrix groups through the generator — the curves $G(t) = e^{tL}$ with $G(1) = Q$ — form exactly a $\mathbb{Z}$-family. Solving $e^{aI+bJ} = Q$ entrywise gives $e^a \cos b = 0$ and $e^a \sin b = \tau$, forcing $\cos b = 0$ and $\sin b = +1$ (the branch $\sin b = -1$ is impossible: $e^a = -\tau$ has no real solution), hence $a = \ln \tau$ uniquely and*

$$\log Q = \left\{(\ln \tau) I + \left(\tfrac{\pi}{2} + 2\pi k\right) J \;:\; k \in \mathbb{Z}\right\}, \qquad W_k(\rho) = e^{-\rho}\, e^{i(\pi/2+2\pi k)\rho/\ln\varphi} \quad (t = \rho/\ln\varphi).$$

All branches agree at every *rung, $W_k(n \ln \varphi) = (i\tau)^n$ — the discrete orbit cannot distinguish them — and they are pairwise distinct as curves (initial velocities $-1+i\kappa_k$ with $\kappa_k = (\pi/2+2\pi k)/\ln\varphi$ all distinct; adjacent branches already differ at the half-floor, while branches with $k-k'$ even coincide there but never as functions).*

Corollary 2.5 (What D3 selects)**.** *[FORCED]/[DECLARED]* `OZ-O5` *The continuous residual state of Definition 2.1 is the $k = 0$ branch, and within the family the following are equivalent: (i) D3's rate; (ii) the* principal *logarithm of Q; (iii) minimal winding, $|\pi/2 + 2\pi k|$ minimized; (iv) the shortest residual spiral per floor, $L_k = (1-\tau)\sqrt{1+\kappa_k^2}$ minimized (speed $|W_k'| = e^{-\rho}\sqrt{1+\kappa_k^2}$ integrated over one floor). The interpolation problem is therefore not undetermined but* determined up to $\mathbb{Z}$*, with D3 exactly the principal-branch selection; deriving that selection from the substrate is the same open problem as deriving the rate.*

Remark 2.6 (Transfer to the formation rail). [FORCED GIVEN D1–D3]/[INHERITED] QX-F1--F6 The classification of Theorem 2.4 transfers verbatim to the formation rail: among curves $\Gamma_{\kappa'}(\rho) = \big(r(z(\rho))\cos\kappa'\rho,\ r(z(\rho))\sin\kappa'\rho,\ z(\rho)\big)$ whose winding is a one-parameter rotation group and which pass through every rung site, the admissible rates are exactly $\kappa'_k = (\pi/2 + 2\pi k)/\ln\varphi$, $k \in \mathbb{Z}$ — the same $\mathbb{Z}$-family, because radius and height carry no interpolation freedom (r is inherited, z is D1): the rung-passage condition at $n = 1$ already forces $\kappa' \ln\varphi \equiv \pi/2 \pmod{2\pi}$, and all further rungs follow. All branches agree at every rung site and are pairwise distinct as curves QX-F1--F2; and the $k = 0$ branch is the unique one of shortest *formation-rail* length per floor on both branches of the profile: the speed $\sqrt{r'^2 + r^2\kappa'^2 + z'^2}$ is strictly increasing in κ'^2 wherever $r > 0$ QX-F3--F5. Corollary 2.5's list thus gains a sixth member — *D3 = the shortest formation rail per floor* — the rail-side companion of the fifth (shortest total spherical length, Proposition 10.5). The residual-sphere curves $S_k = (\operatorname{Re} W_k, \operatorname{Im} W_k, z)$ inherit the same classification, all on the unit sphere QX-F6.

Rate arithmetic and the multiplier torus (v1.7). The remainder of this section settles the *arithmetic* of the rate $\kappa = \pi/(2\ln\varphi)$ and packages the branch family of Theorem 2.4 into one geometric object. Scoping first: everything below is *branch-blind* — every statement survives the substitution $\kappa \to \kappa_k$ — so none of it derives D3. The rate-derivation problem (the session register's OP-RATE) remains open throughout; what changes is its arithmetic perimeter and its classification.

Lemma 2.7 (Irrationality of the log-ratio). *[FORCED] TX-A1--A7, TX-G1 $i\pi/\ln\varphi \notin \mathbb{Q}$. If $i\pi/\ln\varphi = p/q$ with $q \geq 1$, exponentiating gives $(-1)^q = \varphi^p$; but $2\varphi^p = L_p + F_p\sqrt{5}$ exactly (exact on the grid $p \in [-12, 12]$, general by Binet) with $\sqrt{5}$-part $F_p/2 \neq 0$ for $p \neq 0$, so $\varphi^p \neq \pm 1$ for $p \neq 0$ (exact $\mathbb{Q}(\sqrt{5})$ decisions), while $p = 0$ forces $q\, i\pi = 0$, impossible ($q \geq 1$; $\pi > 3$ interval-certified). Second, premise-independent route: $i\pi/\ln\varphi$ is purely imaginary and nonzero ($\varphi > 1$ exact), hence not rational.*

Theorem 2.8 (Transcendence of the rate). *[FORCED]/[ESTABLISHED] TX-B1--B5 Forced given the established Gelfond–Schneider theorem (Gelfond 1934; Schneider 1934, independently; quotient form: for nonzero algebraic α, β with fixed logarithm branches and $\log\alpha, \log\beta \neq 0$, the ratio $\log\alpha/\log\beta$ is rational or transcendental) [ESTABLISHED]:*

$$\boxed{\kappa = \frac{\pi}{2\ln\varphi} \text{ is transcendental, and so is every branch } \ \kappa_k = (4k+1)\,\kappa,\ k \in \mathbb{Z}}.$$

Chain: Lemma 2.7 kills the rational horn of G–S at $(\alpha, \beta) = (-1, \varphi)$, so $i\pi/\ln\varphi$ is transcendental; the bridge $i\pi/\ln\varphi = 2i\,\kappa$ with $2i$ algebraic (root of x^2+4) transfers transcendence to κ ($\overline{\mathbb{Q}}$ is a field); and $\pi/2 + 2\pi k = (4k+1)\,\pi/2$ with $4k+1$ a nonzero rational for every $k \in \mathbb{Z}$. Restatement-test outcome, stated explicitly: *this is branch-blind arithmetic — transcendence holds for every branch at once, so nothing here selects $k = 0$; OP-RATE stays open.*

Corollary 2.9 (Near-miss closure). *[FORCED]/[ESTABLISHED] TX-C1--C7, TX-G2--G4 $4\ln\varphi/\pi = 2/\kappa$ exactly, and $2/\kappa$ is transcendental (Theorem 2.8), while τ is algebraic (root of the irreducible $x^2 + x - 1$): the whitepaper's open item 4 — the near miss $4\ln\varphi/\pi$ vs τ, margin 5.34×10^{-3} [COMPUTED], there "expected contentless" — closes* permanently and negatively*. Both clauses of the whitepaper's Guard 15.3 — the whitepaper carries a* single *Guard 15.3 with both near-miss clauses; there is no Guard 15.4, which also corrects the citation in Remark 6.5 — promote from [computed-nonequal at 60 dps] to forced-never-equal given G–S: equality would force $\kappa = 2/\tau = 2\varphi$ (root of $x^2 - 2x - 4$) in the first clause, and $\kappa = 1/\tau^2 = \varphi^2$ (root of $x^2 - 3x + 1$) in the pitch clause (via $\tan\theta_K = \tau^2/K$ and injectivity of* $\tan$ *on $(0, \pi/2)$) — both impossible. General statement:* every *near miss of the form (algebraic) vs (nonzero rational multiple of $\pi/\ln\varphi$ or $\ln\varphi/\pi$) is permanently* *empty. Margins interval-certified at 60 dps:* $0.00533 < \tau - 4\ln\varphi/\pi < 0.00534$*,* $\tau^2 - 2\ln\varphi/\pi > 0.07$*,* $1/40 < \kappa - 2\varphi < 3/100$*; all printed decimals of Guard 15.3 reproduced* `TX-G6`*.*

Remark 2.10 ($\log_\varphi 2$ strengthened; scan retirement). [FORCED]/[ESTABLISHED] `TX-D1--D5, TX-E1--E6, TX-G5` (a) W-Thm. 16.1 strengthens: $\log_\varphi 2 = \ln 2/\ln\varphi$ is *transcendental* given G–S at $(\alpha,\beta) = (2,\varphi)$. Its irrationality is now carried by two premise-disjoint mechanisms, meeting the corpus convergence standard: the whitepaper's Binet $\sqrt{5}$-part argument (replicated, grid $p \in [-10, 10]$, $q \in [1, 8]$, both orientations) and the new norm obstruction $|N(\varphi^p)| = |(\varphi\psi)^p| = 1$ vs $N(2^q) = 4^q \geq 4$, independent of Binet positivity. (b) Scan-retirement meta-theorem: $\kappa \notin \overline{\mathbb{Q}}$ (Theorem 2.8), so *no algebraic-valued ledger tie for* κ *exists*: for any nonconstant rational function R over $\overline{\mathbb{Q}}$ and any algebraic target c, $R(\kappa) = c$ would make κ algebraic. The register's empirical no-tie upgrades to a theorem-level exclusion (17 ledger constants certified algebraic with irreducible integer minimal polynomials; κ and $2/\kappa$ retire together, product 2; finite corroboration $|\kappa - c| > 1/40$ over the interval ledger, nearest 2φ at ≈ 0.0282). Honest scope: period relations among the corpus transcendentals (κ, L_{res}, and the register's τ_0) and the algebraic independence of π and $\ln\varphi$ are beyond Gelfond–Schneider and remain external-open.

Proposition 2.11 (Modulus ledger of the multiplier torus)**.** *[FORCED]* `MX-A1--A4, MX-M1a--M1i, MX-M2a--M2d` *Let* $E_q = \mathbb{C}^\times/q^{\mathbb{Z}}$ *be the multiplier torus of the generator (*$0 < |q| = \tau < 1$ *exact, so* E_q *is nondegenerate). The principal logarithm and modulus are*

$$\log q = \ln\tau + \frac{i\pi}{2} = -\ln\varphi + \frac{i\pi}{2}, \qquad \mu = \frac{\log q}{2\pi i} = \frac{1}{4} + \frac{i\ln\varphi}{2\pi} \pmod{\mathbb{Z}},$$

forced by exp-match and principal-strip membership ($\operatorname{Im} \in (-\pi, \pi]$*; branches differ by* $2\pi i k$*, at least the strip diameter). Ledger:* $\operatorname{Re}\mu = \frac{1}{4}$ *is the charge quantum — the* $c = \frac{1}{4}$ *gauge rotation of Theorem 4.2, one charge quantum per rung;* $\operatorname{Im}\mu = \ln\varphi/(2\pi) > 0$*;*

$$\frac{\operatorname{Re}\mu}{\operatorname{Im}\mu} = \kappa \qquad \textit{and} \qquad |\log q|^2 = \ln^2\varphi + \frac{\pi^2}{4} = \ln^2\varphi\,\big(1 + \kappa^2\big),$$

the per-floor arc constant of Corollary 2.5(iv) read in the log chart: $(-1 + i\kappa)\ln\varphi = \log q$ *exactly.*

Theorem 2.12 (Systolic marking: the seventh characterization)**.** *[FORCED]* `MX-M3a--M3f, MX-M4a--M4c` *For the period lattice* $\Lambda = \mathbb{Z}\log q + \mathbb{Z}\,2\pi i$ *of* E_q*,*

$$|n\log q + 2\pi i\, m|^2 = \ln^2\varphi\,\big(n^2 + \kappa^2(n + 4m)^2\big)$$

exactly, so every length comparison is linear in κ^2 *with integer coefficients, decided by exact rational arithmetic against the single certified sandwich* $10 < \kappa^2 < 11$ *(one interval guard,* `MX-G1`*).* Λ *is* branch-independent*:* $\{\log_k q,\ 2\pi i\}$ *is a unimodular change of basis of* Λ *for every* k*, and the length spectrum is* k*-invariant; the* marking *(choice of generator) is not (*$\log_k q - \log q = 2\pi i k$*). The systole of* E_q *is* $|\log q|$*, attained exactly at* $\pm\log q$ *— the principal-branch generators — hence*

$$\text{D3} \iff \text{the marking whose non-meridian generator is the systole representative.}$$

The second-shortest class is the four-rung cycle: $4\log q - 2\pi i = \ln \mathrm{gap}$*, length* $4\ln\varphi$ *— the charge cycle* $W_4 = \mathrm{gap}$ *of Theorem 2.3* is *the second-shortest closed geodesic.* Stated prominently: this is characterization number seven for Corollary 2.5's menu, not a derivation. *Per the restatement test the selection content is exactly a minimality principle (the systole); every* k*-invariant part of the construction cannot separate branches, and OP-RATE's derivation problem is untouched.*

Remark 2.13 (Geodesic ledger; no-CM). [FORCED]/[ESTABLISHED] (a) `MX-M5a--M5c, MX-G2--G3` The closed-geodesic length ledger of E_q is certified: lengths $\ln\varphi\sqrt{a+b\,\kappa^2}$ with $(a,b)=\big(n^2,(n+4m)^2\big)$; the first ten distinct classes (up to sign, iterates included), ordered by exact sandwich arithmetic and interval-corroborated at 60 dps, are

$$(1,1),\ (16,0),\ (9,1),\ (25,1),\ (4,4)^{\times 2},\ (49,1),\ (64,0),\ (36,4)^{\times 2},\ (81,1),\ (1,9),$$

with $(1,1)$ the systole, $(16,0)=4\ln\varphi$ the $W_4=$ gap cycle, and $(4,4)$ twice the systole (an iterate/primitive degeneracy); window sufficiency ($|n|\le 12$, $|m|\le 4$) closed by exact tail identities. This is the certified ledger on which $\pi/2$ and $\ln\varphi$ combine *additively* — the natural habitat for any future extremality selection, which must re-pass the restatement test. (b) `MX-M6a--M6c` Given Theorem 2.8, μ is not algebraic — the exact bridge is $(\mu-\frac{1}{4})\,\kappa=i/4$ with $\mu-\frac{1}{4}=i\ln\varphi/(2\pi)\neq 0$ — in particular not imaginary quadratic: *E_q has no complex multiplication* (standard CM criterion; conditional exactly on the transcendence input).

Proposition 2.14 (k-invariance of the discrete state)**.** *[FORCED]* `DX-E1--E3` *The full discrete state of the ladder — the rung values $e^{i\kappa_k n\ln\varphi}=i^n$ (exact on the grid $k\in[-3,3]$, $n\in[0,8]$, symbolically via $e^{2\pi ikn}=1$), the heights $z_n^2=1-\tau^{2n}$, the transfer ledger, the gate data (C_n,u_n,m_n,λ_n), the floor lattice $\rho_n=n\ln\varphi$, and the torus lattice Λ — is* identical for every branch $k\in\mathbb{Z}$*. Only the marking $\log_k q$ is k-dependent (Theorem 2.12).*

Remark 2.15 (OP-RATE at rung resolution: sign ladder, exposed joint, ledger kill, classification no-go). [FORCED]/[OPEN] `DX-C1--C7, DX-D1--D5, DX-E4--E5` Four bookkeeping results at the rate problem's core; items (i) and (iii) are forced, (ii) and (iv) are candidate-level; none closes the problem. (i) *Sign ladder, D1-free* `DX-C1--C2`: symbolically $m(w)=-C/(1+u)^2=-1/w<0$ with $w=\varphi^{2n}$, so the gate multiplier is negative at *every* gate and does not alternate between gates; the two-rung carrier is $q^2=-\tau^2$, argument π. (ii) *Exposed-joint axiom H* — a formalized open joint, presented as such, *not* a result `DX-C3--C6`: given

H := "argument of the two-rung ladder action = transverse holonomy accumulated over two floors ($=\pi$)",

exact mod-2π bookkeeping forces the per-floor step $\delta=\pi/2 \pmod{\pi}$ — the quantum $\pm\pi/2$ at rung resolution — but the residual $\pi\mathbb{Z}$ is not pinned: every branch satisfies H, and the branch remains untouched. (iii) *Ledger counter-models* `DX-D1--D5`: the transfer ledger $\Delta F_n=\tau^{2n+1}$ (Proposition 11.5) does *not* force a constant angular step. Two explicit counter-models pass every ledger identity: $\theta_n=n\pi$ (all moduli correct; fails rung passage at $n=1$, $e^{i\pi}\tau=-\tau\neq i\tau$) and $\theta_n=n\pi/2+2\pi s_n$ with a non-constant integer lift s_n (reproduces *all* rung values i^n and all ledger identities). Constancy of the angular step is therefore a lift/equivariance assumption, and the ledger route to the rate dies by contamination. (iv) *R6 no-go — a candidate classification statement, not asserted as a theorem* `DX-E4--E5`: any rate-selection functional that factors through the k-invariant discrete data of Proposition 2.14 is constant in k and cannot select $k=0$; the six existing characterizations (Corollary 2.5(i)–(iv), Proposition 10.5, Remark 2.6) all reference *continuum* objects — winding number, spiral speed, rail length — which is exactly the escape, and the separation is exhibited (L_k separates branches that the discrete functionals cannot). This supports closing OP-RATE *as a classification* — determined up to $\mathbb{Z}$ at rung resolution, with selection necessarily continuum-referencing — at candidate level; OP-RATE *as a derivation* remains open [OPEN].

OP-RATE closes as a classification (v1.8). The candidate R6 no-go of Remark 2.15(iv) is now promoted to a forced theorem, and its consequence stated precisely: at rung resolution the branch k — hence the rate κ up to the $\mathbb{Z}$-family — cannot be selected by any functional of the discrete substrate data. This settles OP-RATE *as a classification*; it is a no-go on discrete-substrate *derivation*, not a derivation of κ, and the continuum rate D3 stays declared throughout.

Lemma 2.16 (Extended k-invariance of the discrete state). *[FORCED]* `CL-1a--CL-1h, CL-1F` *The k-invariance of Proposition 2.14 extends, re-forced cold, to the* full *discrete state of the ladder. For every branch $k \in \mathbb{Z}$ the following are identical to their $k = 0$ values: the rung values $e^{i\kappa_k n \ln\varphi} = i^n$ (exact grid $k \in [-4, 4]$, $n \in [0, 10]$, and symbolically via $e^{2\pi i k n} = 1$); the heights $z_n^2 = 1 - \tau^{2n}$; the gate ladder $(C_n, u_n, m_n, \lambda_n)$ with $m_n\lambda_n = 1$; the floor lattice $\rho_n = n \ln\varphi$; the transfer ledger $\Delta F_n = \tau^{2n+1}$ (sum 1); the torus lattice $\Lambda = \mathbb{Z}\log q + \mathbb{Z}\,2\pi i$, re-verified branch-independent by the unimodular change of basis $\left(\begin{smallmatrix}1 & k\\ 0 & 1\end{smallmatrix}\right)$ ($\det = 1$); the ten-class geodesic length ledger $\ln\varphi\sqrt{a + b\kappa^2}$ with $(a, b) = \big(n^2, (n + 4m)^2\big)$, invariant under the marking relabel $(n, m) \mapsto (n, m + nk)$; and the* systole value *$|\log q_0|^2 = \ln^2\varphi\,(1 + \kappa^2) = \ln^2\varphi + \pi^2/4$, the lattice minimum. Only the* marking *$\log_k q$ is k-dependent ($|\log_1 q|^2 - |\log_0 q|^2 = 6\pi^2 \neq 0$, detected* `CL-1F`*). The lemma consumes neither D2 nor D3 (D1 enters only through the k-free rung modulus τ).*

Theorem 2.17 (R6 classification no-go). *[FORCED]* `CL-2a, CL-2F1--CL-2F2` *Define a* rung-resolution selection functional *as any map $\Phi = g \circ D$ from the k-invariant discrete data D of Lemma 2.16 into a totally ordered set, whose argmin/argmax purports to select the branch k. Then every such functional is* constant on the $\mathbb{Z}$-family*, and so selects no branch: if $D(k) = D(0)$ for all k (Lemma 2.16), then $\Phi(k) = g(D(k)) = g(D(0)) = \Phi(0)$, whose argmin is the whole $\mathbb{Z}$-family. (Witness: $F = \sum_{n<6} e^{-\rho_n} i^n$ is k-constant symbolically* `CL-2a`*; the non-vacuity guard* `CL-2F1` *fires on any core secretly made k-constant.) This promotes Remark 2.15(iv) from candidate to forced. Per the restatement test the no-go is branch-blind* by design*: it is a characterization-of-impossibility, not a derivation of k.*

Corollary 2.18 (OP-RATE closes as a classification). *[FORCED]* `CL-3a--CL-3c` *OP-RATE closes* as a classification*. What closes: the reading "derive κ from the discrete substrate at rung resolution" is a no-go (Theorem 2.17) — κ is declared-up-to-$\mathbb{Z}$ at the discrete level, and selecting the principal branch* provably requires *a continuum-referencing minimality principle. The seven characterizations (Corollary 2.5(i)–(iv), Proposition 10.5, Remark 2.6, Theorem 2.12) are the* complete *discrete-level menu: the discrete level contains no selector at all. What does* not *close: the continuum rate D3 itself remains declared — all branch curves W_k are admissible one-parameter groups through Q, pairwise distinct as curves (Theorem 2.4, initial velocities $-1 + i\kappa_k$) — and a continuum-referencing derivation is not excluded. This mirrors the $\lambda = 2c$/Cencov equivariance no-go of the companion $\lambda = 2c$ paper (`2026-06-lambda-2c`, thm:cencov) exactly: as there "c cannot be fixed by any invariance requirement … c is a declared positive scale … shipped decisions are c-invariant across the band … [but he] does not forbid an external structural constraint from doing so", here k cannot be fixed by any functional of the k-invariant discrete data, κ is declared-up-to-$\mathbb{Z}$ with discrete decisions k-invariant across the $\mathbb{Z}$-family, and a continuum-referencing principle may still select. OP-RATE* as a derivation *remains open [OPEN].*

Remark 2.19 (The seven-way continuum/discrete table). [FORCED] `CL-2b(i)--(vii), CL-2c` The content of the no-go is that each of the seven characterizations is *continuum-referencing*: its selection functional is k-dependent (unique argmin at $k = 0$), so it provably does not factor through the k-invariant discrete data. (i) D3's declared continuum rate κ itself — a *restatement* (the selector $|\kappa_k - \kappa| = 4|k|\kappa$ consumes the target); (ii) the principal branch of $\log Q$, i.e. the continuous argument/lift $\theta(\rho)$ (the rung data is lift-blind, $e^{i\theta} = i^n$ only); (iii) the minimal winding $|\pi/2 + 2\pi k|$ of the continuous curve; (iv) the arc length $L_k = (1-\tau)\sqrt{1 + \kappa_k^2}$ of the continuous residual spiral; (v) the total spherical length L_{res}; (vi) the formation-rail length per floor; (vii) the systolic selection — the flat norm $|\log_k q|^2 = \ln^2\varphi(1+(4k+1)^2\kappa^2)$ of the k-dependent *marking* on the continuous torus E_q, even though the lattice and the systole value are k-invariant (Lemma 2.16). Tag discipline: the structural conclusion "continuum-referencing, does not factor through the discrete data" is [FORCED] for all seven, and rows (i)–(iv),(vii) are forced at the exact level (integer/rational cores $(4k+1)^2$, $|4k+1|$, $4|k|$, plus $\kappa^2 > 0$); the strict per-characterization "unique argmin at $k = 0$" *selection* for rows (v) and (vi) is [FORCED] *given corpus* — inherited, cited not re-encoded, from Proposition 10.5 (L_{res} strictly increasing in $|\kappa|$) and Remark 2.6 (rail speed strictly increasing in κ'^2). The no-go's core needs only that L_{res} and the rail are continuous integrals, manifestly not functions of the rung samples, which is exact. The whole block is branch-blind by design: that is the content.

The period-relation map (v1.8). The scan-retirement remark (Remark 2.10(b)) left the period relations among the corpus transcendentals (κ, L_{res}, and the register's Salem-slot constant τ_0) external-open and undifferentiated. This block pins what is algebraic, settles the κ-vs-τ_0 worry, and maps the precisely external-open residue.

Proposition 2.20 (τ_0 algebraic; no κ-vs-τ_0 relation)**.** *[FORCED]/[ESTABLISHED]* `PM-1B1--PM-1B9, PM-1X1--PM-1X6, PM-A2` *The Salem-slot occupant $\tau_0 = \beta + \beta^{-1}$ (the trace redirection of the companion salem-slot note, Def. 3.2; the dominant root of the degree-2 trace-down) is* algebraic*. For the least-degree Salem number β_4 (minimal polynomial $x^4 - x^3 - x^2 - x + 1$) it has the exact minimal polynomial*

$$\boxed{\mathrm{minpoly}(\tau_0) = t^2 - t - 3, \qquad \tau_0 = \tfrac{1}{2}(1+\sqrt{13})},$$

a degree-2, irreducible, monic-integer (algebraic-integer), totally real *(discriminant* $13 > 0$*, Sturm real-root count* 2*),* Perron *($\tau_0 - |conjugate| = 1 > 0$) constant with $\tau_0 > 2 > \varphi$ (salem-slot Thm. 3.3); the type-fact is general over all Salem redirections, and the golden limit $\sqrt{5} = \varphi + \varphi^{-1}$ (minpoly $t^2 - 5$) and the Pisot accumulation $\mu_P + \mu_P^{-1}$ (minpoly $t^3 + t^2 - 4t - 5$) are algebraic too. Combined with κ transcendental (inherited, Theorem 2.8, given Gelfond–Schneider), there is* no *algebraic relation over $\mathbb{Q}$ between κ and τ_0: for $m(Y) = \mathrm{minpoly}(\tau_0)$ the resultant $R(X) = \mathrm{Res}_Y(P, m) = \prod_r P(X, r) \in \mathbb{Q}[X]$, and a genuine (non-vacuous) $P(\kappa, \tau_0) = 0$ would give $R(\kappa) = 0$ with $R \neq 0$, forcing $\kappa \in \overline{\mathbb{Q}}$ — impossible. The register's "κ vs τ_0" period worry is thus settled* negatively *at the algebraic level (a specialization of the scan-retirement meta-theorem to two variables; κ is consumed only as an excluded value, not derived, so this is branch-blind exclusion, not restatement).*

Remark 2.21 (L_{res} as an elliptic period; the external-open ledger). [FORCED]/[ESTABLISHED]/[OPEN] `PM-1C1--PM-1C2, PM-2A, PM-2B, PM-2Bf` The finite spherical length $L_{\mathrm{res}} = \sqrt{1+\kappa^2}\,E\big(\kappa^2/(1+\kappa^2)\big) = 3.7312193125\ldots$ (Proposition 10.5) is the value of the complete elliptic integral of the second kind at the elliptic *parameter* $m = \kappa^2/(1+\kappa^2)$ (modulus $k = \sqrt{m}$), via the exact reduction $1+\kappa^2\cos^2\theta = (1+\kappa^2)(1 - m\sin^2\theta)$. At an *algebraic* parameter E is a Kontsevich–Zagier period (an algebraic ellipse's quarter-perimeter); here the Möbius inverse $\kappa^2 = m/(1-m)$ makes m *transcendental* (since κ is), so L_{res} is a period *function* evaluated at a transcendental argument — one step beyond Gelfond–Schneider and beyond the Schneider/Chudnovsky elliptic-period theorems (which need an algebraic modulus or CM lattice), hence its transcendence is *not* settled here. Since τ_0 is algebraic (Proposition 2.20), any relation $P(\tau_0, L_{\mathrm{res}}) = 0$ over $\mathbb{Q}$ would force $L_{\mathrm{res}} \in \overline{\mathbb{Q}}$ by the same resultant/tower argument `PM-2B`, so the $(\tau_0, L_{\mathrm{res}})$ pair *reduces* to the single constant. The precise ledger: *settled* — κ transcendental, τ_0 algebraic, no (κ, τ_0) relation, and $(\tau_0, L_{\mathrm{res}})$ reduced to "is L_{res} transcendental"; *genuinely external-open* [OPEN] — the transcendence of L_{res}, and the algebraic independences of $(\kappa, L_{\rm res})$ and of $(\pi, \ln\varphi)$ (Schanuel / Kontsevich–Zagier territory, beyond Gelfond–Schneider and Baker; the ratio $\pi/\ln\varphi = 2\kappa$ is settled transcendental but pair-independence is strictly stronger). The entire map survives $\kappa \mapsto \kappa_k$ (branch-blind) and advances no selection.

Remark 2.22 (The forced-identity ledger of the period frontier). [FORCED]/[FORCED GIVEN D1–D3] `PF-3A--PF-3G, PF-3Ef, PF-G4, PF-G5` The period frontier of Remark 2.21 sits on an exact algebraic skeleton, every relation of which is forced. *Golden trace* [FORCED]: $\varphi^2 = \varphi + 1$, $\varphi\tau = 1$, $\tau^2 + \tau = 1$, and $\varphi + \varphi^{-1} = \sqrt{5}$ (the salem-slot golden floor). *The definitional tie* [FORCED GIVEN D1–D3]: given the declared rate $\kappa = \pi/(2\ln\varphi)$ [D3],

$$\frac{2}{\kappa} = \frac{4\ln\varphi}{\pi}, \qquad \frac{\pi}{\ln\varphi} = 2\kappa, \qquad \pi = 2\kappa\ln\varphi,$$

and the branch family $\kappa_k = (\pi/2 + 2\pi k)/\ln\varphi = (4k+1)\kappa$ (symbolic in $k \in \mathbb{Z}$, with $4k+1$ a nonzero odd rational for every k). *The Salem anchor* [FORCED] *given corpus*: $\tau_0 = (1+\sqrt{13})/2$ satisfies $\tau_0^2 = \tau_0 + 3$ (minimal polynomial $t^2 - t - 3$, discriminant 13), the degree-2 algebraic-integer occupant of the redirection slot (Proposition 2.20). *The lens gate* [FORCED]: $K^2 = 1 - \text{gap}$, $\text{gap} = \varphi^{-4}$, with minimal polynomial $x^2 + 5x - 5$ and $K^2 > 0$ (exact $\mathbb{Q}(\sqrt{5})$ sign). *The elliptic reduction* [FORCED]: symbolic in K, θ,

$$1 + K^2\cos^2\theta = (1+K^2)\big(1 - m\sin^2\theta\big), \qquad m = \frac{K^2}{1+K^2}, \qquad K^2 = \frac{m}{1-m},$$

so $L_{\rm res} = \int_0^{\pi/2} \sqrt{1+\kappa^2\cos^2\theta}\, d\theta = \sqrt{1+\kappa^2}\, E(m)$ (perturbing $m \mapsto m + 10^{-3}$ breaks the identity; falsifier `PF-3Ef`). *A forced geometric reading* [FORCED] (new this session): $L_{\rm res}$ is the quarter-perimeter of the ellipse with semi-axes $(\sqrt{1+\kappa^2}, 1)$; its linear eccentricity c satisfies $c^2 = a^2 - b^2 = \kappa^2$ *exactly* (the focal parameter is κ), and its eccentricity-squared is $c^2/a^2 = m$ *exactly* (the elliptic parameter). Two independent computations of $L_{\rm res}$ — the closed form and direct quadrature of the integrand — agree to $< 10^{-200}$ `PF-G4`, so the frontier is numerically well-posed; every identity here survives $\kappa \mapsto \kappa_k$ verbatim.

Remark 2.23 (The frontier map: Baker's reach, Schanuel's conditional, and the external-open tool-gaps). [ESTABLISHED]/[OPEN]/[COMPUTED] `PF-2A--PF-2E, PF-1P1--PF-1P6, PF-1N1--PF-1N7, PF-G1--PF-G3` Against this skeleton the transcendence questions sort into exactly three tiers, each with its deciding tool or its named gap. *Settled* [ESTABLISHED]: κ is transcendental (Gelfond–Schneider at $(-1, \varphi)$, since $\kappa = (i\pi/\ln\varphi)/2i$ with $2i$ algebraic; inherited Theorem 2.8); τ_0 is algebraic ($t^2 - t - 3$); and *Baker's reach is pinned* — $\{1, \ln 2, \ln 3, \ln\varphi\}$ is $\overline{\mathbb{Q}}$-linearly independent [ESTABLISHED], because $2, 3, \varphi$ are multiplicatively independent (the exact box $|a|, |b|, |c| \le 6$ has $2^a 3^b \varphi^c = 1$ only at the origin: for $c \neq 0$ the $\sqrt{5}$-part $F_c/2 \neq 0$ makes φ^c irrational, and $2^a 3^b = 1$ forces $a = b = 0$) and Baker's linear-forms theorem then applies. *Conditional on Schanuel* [OPEN]: $\{\ln\varphi, i\pi\}$ is $\mathbb{Q}$-linearly independent ($\ln\varphi$ real nonzero, $i\pi$ purely imaginary nonzero) with algebraic exponentials $\varphi, -1$, so Schanuel would give trdeg ≥ 2 and hence $(\pi, \ln\varphi)$ — equivalently $(\kappa, \ln\varphi)$, since $\mathbb{Q}(\kappa, \ln\varphi) = \mathbb{Q}(\pi, \ln\varphi)$ — algebraically independent; the premise is forced but the conjecture is not, so this is *not* a theorem here. *Genuinely external-open* [OPEN], with the tool-gap named in each case: the transcendence of $L_{\rm res}$ is untouched because its elliptic modulus $k = \sqrt{m}$ is *transcendental* ($m = \kappa^2/(1+\kappa^2)$), and the Möbius inverse $K^2 = m/(1-m)$ makes $m \in \overline{\mathbb{Q}} \iff \kappa \in \overline{\mathbb{Q}}$), so the Schneider/Chudnovsky elliptic-period theorems — which require an algebraic modulus or CM lattice — are *inapplicable*; and the algebraic independence of $(\kappa, L_{\rm res})$ lies beyond even Schanuel (an elliptic-integral value outside the exp/log framework, Kontsevich–Zagier / period-conjecture territory). *We resolve none of these.* As [COMPUTED] corroboration only, a validated-detector PSLQ search — fail-first on six known relations (including a size-6 planted relation matching the flagship basis) and correctly silent on two transcendental controls `PF-1P1--PF-1P6`, `PF-1N1--PF-1N2` — finds *no* integer relation of height $\leq 10^{12}$ at 230 dps among the frontier bases $\{1, \kappa, \kappa^2, L_{\mathrm{res}}, L_{\mathrm{res}}^2, \kappa L_{\mathrm{res}}\}$, $\{1, \pi, \ln\varphi, \pi\ln\varphi\}$, $\{1, \ln\varphi, \ln 2, \ln 3\}$, and the L_{res} algebraicity ladders to degree 6 `PF-1N3--PF-1N7`. This *corroborates, and never proves*, the open transcendence and independence statements; PSLQ silence is evidence, not a certificate. The entire map is branch-blind ($\kappa \mapsto \kappa_k$ verbatim) and selects no k — a characterization, not an OP-RATE derivation.

3 The canonical quarter-turn: $J = UA$

Everything in this section is unconditional [FORCED]: pure keystone/Clifford algebra in the whitepaper's fixed convention ($R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $H = 2R - I = 2e_1 - e_2$, $e_1 = \sigma_x$, $e_2 = \sigma_z$, $J = e_1 e_2$; W-§3–4), using no part of D1–D3. It resolves the canonicality problem the July 14 report posed for itself: the representation $W_n = (i\tau)^n$ imported i from the declared winding; here the quarter-turn is *factored from two real, normalized structures already forced in the operator layer.*

Proposition 3.1 (Spectral uncoupling). *[FORCED]* `OZ-J1--J5` *The spectral projectors of the keystone,*

$$P_+ = \frac{R + \tau I}{\sqrt{5}}, \qquad P_- = \frac{\varphi I - R}{\sqrt{5}},$$

are complete orthogonal idempotents ($P_+ + P_- = I$, $P_\pm^2 = P_\pm$, $P_+ P_- = 0$) with $RP_+ = \varphi P_+$ and $RP_- = \psi P_-$, $\psi = -\tau$. The normalized keystone grading

$$U := P_+ - P_- = \frac{H}{\sqrt{5}} = \frac{2R - I}{\sqrt{5}}$$

satisfies $U^2 = I$ and acts as $+I$ on the expanding golden mode and $-I$ on the contracting one; equivalently $P_\pm = (I \pm U)/2$.

Proposition 3.2 (The return-kernel recoupler). *[FORCED]* `OZ-K1--K4`, `OZ-L4` *Let $N = e_1 + 2e_2$ be the generator of the vector part of* $\ker \mathcal{L}$, *$\mathcal{L}(X) = \frac{1}{2}\{H, X\}$ (W-Thm. 4.2). Then $N^2 = 5I$ and, normalizing by the Clifford norm, $A := N/\sqrt{N^2} = N/\sqrt{5}$ satisfies*

$$A^2 = I, \qquad UA = -AU, \qquad AP_+ = P_- A, \qquad AP_- = P_+ A:$$

A is a real involution that exchanges *the two golden modes. The anticommutation is automatic, not a coincidence: on vectors, $\mathcal{L}(v) = \langle H, v\rangle\, I$ (W-Thm. 4.2), so the vector kernel is* by definition *$H^\perp$ — membership in the return kernel already certifies orthogonality to H, hence $HN = -NH$.*

Theorem 3.3 (Canonical quarter-turn factorization; Strouse–Ozymandias). *[FORCED]* `OZ-K5--K6` *With U and A as above,*

$$\boxed{J = UA = \frac{H}{\sqrt{5}} \cdot \frac{N}{\sqrt{5}} = \frac{HN}{5}}, \qquad J^2 = UAUA = -U^2 A^2 = -I.$$

Matrix witness: $H = \left(\begin{smallmatrix} -1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, $N = \left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$, $HN = 5\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = 5\, e_1 e_2$. Uniqueness: U is fixed by the keystone; the vector kernel is one-dimensional, so its unit generator is $\pm A$; hence

$$J = \pm UA$$

is canonical up to the orientation sign — exactly the freedom the whitepaper's D2 already names as a convention. The quarter-turn no longer needs to be imported: distinguish the modes (U) + exchange the modes (A) = quarter-turn (J).

Corollary 3.4 (Coordinate-free form). *[FORCED]* *OZ-L3--L4* *In* Cl(2, 0) *the product of any positively-oriented orthonormal pair of vectors equals the pseudoscalar: for every rotation angle θ, $e_1'e_2' = e_1e_2 = J$ exactly (a reflection flips the sign). Since $(U, A) = (\widehat{H}, \widehat{N})$ is an orthonormal pair* by construction *(Prop. 3.2), the factorization $J = \pm UA$ is basis-independent up to orientation; no choice beyond the whitepaper's fixed convention enters. For an arbitrarily scaled kernel generator the normalization is $A = N/\sqrt{N^2}$; with $N = e_1 + 2e_2$ this is $N/\sqrt{5}$. This closes the report's open item (1).*

Remark 3.5 (Why the return layer is necessary). [FORCED] OZ-L1--L2 The commutative algebra $\mathbb{R}[R] = \mathrm{span}\{I, R\}$ contains no square root of $-I$: $(aI + bR)^2 = -I$ has no real solutions (exact elimination). Nor does it contain the recoupler: everything in $\mathbb{R}[R]$ commutes with H, while $[H, N] = 2HN = 10J \neq 0$. The quarter-turn therefore cannot come from the keystone *alone*; it requires the keystone *acting on* the matrix algebra — the return operator's module structure — which is precisely where N lives. The report's initial self-criticism ("clean representation $\neq$ canonical derivation") identified the right gap, and the factorization fills it from inside the existing layer.

Definition 3.6 (Canonical rung generator). [FORCED] OZ-K7--K9

$$Q \coloneqq \pm\,\tau J = \pm\,\tau UA = \pm\,\frac{\tau}{5}\,HN, \qquad Q^2 = -\tau^2 I, \qquad Q^4 = \mathrm{gap}\,I, \qquad Q^n = \tau^n J^n.$$

Under the algebra isomorphism $\mathbb{R}[J] \cong \mathbb{C}$ ($J \mapsto i$), $Q^n \cong (i\tau)^n = W_n$; concretely, with basepoint $v_0 = (1, 0)^\top$,

$$Q^n v_0 = \big(\mathrm{Re}\,W_n,\ \mathrm{Im}\,W_n\big)^\top, \qquad \text{so} \qquad S_n = \big(Q^n v_0\,;\ z_n\big):$$

the residual sphere's horizontal part (§10) *is* the Q-orbit of the east basepoint.

Scoping upgrade (what this does and does not close). Before v1.1 the generator was forced *given* D1–D3, with i supplied by D2. Now: the quarter-turn operator J and the rung generator $Q = \pm\tau J$ exist canonically (up to orientation) in the substrate itself [FORCED], and D2's remaining declared content is *selection*, not existence — that the winding applies exactly one Q per entropy floor (D3's rate), rather than the terrain π or pentagon $2\pi/5$ clocks. Per the report's own ledger, still open: rung quantization (why exactly one Q per rung event — by Corollary 2.5 this now *contains* the interpolation question, since branch selection = rate selection), derivation of the inherited radius law $r(z)$, and any physical identification (the helicase/uncouple–recouple language was a mechanism-finding analogy only). Coordinate-free uniqueness is closed by Corollary 3.4; continuous interpolation is classified by Theorem 2.4.

The χ-selection front (v1.7). Three results sharpen the selection scoping just stated: what the emission image S (the closure of the whitepaper's emission catalog, W-§5) can and cannot supply for the angular quantum χ, and exactly where a χ-selection argument must stand.

Proposition 3.7 (Pentagon exclusion by the image). *[FORCED]* *CX-A3, CX-B1--B5, CX-D6* *ζ_5 is not in the emission image S:* $\arg\zeta_5 = 2\pi/5$ *lies outside the image angle lattice* $(\pi/2)\mathbb{Z}$ *(W-Lemmas 5.4–5.5). Exactly: $\frac{1}{5} \notin \frac{1}{4}\mathbb{Z}$ mod 1; the pentagon classes $k/5$ ($k = 1..4$) meet $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ only at 0; Φ_5 (verified = $\mathrm{minpoly}(\zeta_5)$) has no root in μ_4; and the pentagon gate multipliers are primitive fifth roots (pair product = Φ_5). This* strengthens *the whitepaper's pentagon exclusion from the principle to the image: W-Decl. 12.2/W-Prop. 14.2 exclude the $2\pi/5$ clock* by the principle *D2; here the pentagon multiplier is not an on-circle element of S at all, independently of D2.*

Proposition 3.8 (Unimodular tensor-monomials reach only ± 1). *[FORCED]* `CX-A4, CX-C1--C7` *Every unimodular tensor-monomial of catalog eigenvalues has value exactly $+1$ or -1: $\pm i$ is multiplicatively unreachable. Forcing: every catalog eigenvalue has 5-valuation $w_2 \geq 0$ (moduli of the form $2^{w_0/4}3^{w_1/4}5^{w_2/4}\varphi^{w_3/4}$); the only odd-quarter-argument carriers are the K-seed's rotation pair $\pm i\beta$, each with $w_2 = 1 > 0$; unimodularity forces the total $w_2 = 0$ (norming to $\mathbb{Q}$ with $N(\varphi) = -1$ and φ not a root of unity), hence no $\pm i\beta$ factor, hence even argument. Corroborated exhaustively on all 74613 monomials of degree ≤ 6 (60 unimodular, all ± 1). Whether the* full *image — closed also under addition, minimal polynomials, and Φ-extraction — can reach $\pm i$ is* not *decided by this multiplicative argument and remains open [OPEN]* `CX-C6--C7`*: the corpus-exhibited host is $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ with Φ_4 rooted at $\pm i$.*

Theorem 3.9 (The conditional χ-selection chain, with its kill). *[FORCED]/[DECLARED]* `CX-D2--D3, CX-E1--E5, CX-G1` *Call D2′ the thin axiom* "the per-rung winding multiplier is an on-circle element of S"*. Then,* under D2′ together with the hypothesis that S contains a non-real on-circle element*: $\chi = \pm\pi/2$. The chain composes three links, each re-verified verbatim this session: on-circle elements of S lie in $\mu_4 = \{\pm 1, \pm i\}$ (W-Lem. 5.6); real multipliers carry no helicity (W-Thm. 11.1/W-Cor. 11.2), excluding ± 1; leaving $\chi = \pm\pi/2$. The implication is machine-forced as a truth table over the hypothesis.* The honest kill, stated prominently: *D2′ does* not *reduce D2. Selecting $\pm i$ requires S to contain $\pm i$, and the substrate's only odd-quarter-argument source is the K-seed's rotation pair $\pm i\beta$ — so motivating "D2′ puts $\pm i$ in S" imports exactly D2's declared content (the winding realizes the seed's charge completion, Remark 4.4): axiom replacement is a* restatement*, and the kill is logged. No shipped document ties the forced* gate *multiplier $\pm i$ (a $C = -\frac{1}{2}$ gate datum, W-Prop. 10.2) to the emission image, so D2′ remains a proposed thin axiom, not forced-given-corpus* `CX-D3, CX-D6`*. What survives unconditionally is the* de-axiomatization identity `CX-E3`*: within μ_4, non-real $\iff$ order 4 $\iff$ generator of $\mathbb{Z}/4\mathbb{Z}$ — so D2's clause "the angular quantum is a generator of $\mathbb{Z}/4\mathbb{Z}$" may be rewritten as "the multiplier is non-real" without weakening it. D2 itself remains declared [DECLARED].*

The χ doctrine resolved (v1.8). Theorem 3.9 left two sub-branches open pending companion papers: g1 (can the *full* emission image S — closed also under addition, minimal polynomials, and Φ-extraction — reach $\pm i$?), and g3 (does the operator-algebra / charge-measure doctrine *require* dynamical gate multipliers to be elements of S?). Those papers are now available and re-encoded cold; both sub-branches resolve *negatively*, and W[open]2 stays PARTIAL as a derivation.

Proposition 3.10 (The on-circle image of S is μ_2). *[FORCED]/[ESTABLISHED]* `CH-A3, CH-B2--CH-B3, CH-C1--CH-C5, CH-D2, CH-G2` *Let S be the emission algebra — the closure of the whitepaper's 7-seed emission catalog under $\{\oplus, \otimes, \psi^2, \mathrm{minpoly}, \Phi\}$ (commutator excluded). Its on-circle eigenvalue image is* exactly *$\mu_2 = \{+1, -1\}$, a proper subset of μ_4; hence $\pm i$ is* not *an on-circle element of S. Mechanism (forced given the operator-algebra corpus): every emitted eigenvalue is a monomial in the 16 catalog eigenvalues — $\otimes$ multiplies, ψ^2 squares, $\oplus$ adds no value, and* minpoly, Φ *are spectrum*-preserving *idempotents (operator-algebra paper,* `2026-06-operator-algebra`*: they represent an object's eigenvalue multiset, they are not a cyclotomic-extraction primitive) — and a unimodular monomial forces total 5-valuation 0 ($N(\varphi) = -1$, φ not a root of unity), which excludes the only odd-argument carriers $\pm i\beta$ (with 5-valuation 1), forcing even argument, hence value in $\{\pm 1\}$. This resolves g1 (v1.7 external-open) negatively: the Φ-extraction host $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ is not in S because spectrum-preservation cannot manufacture the $\pm i$ roots of Φ_4, and the emission-gap paper's on-circle lemma (*`2026-06-emission-gap`*: an emitted on-circle eigenvalue lies in $\{1, i, -1, -i\}$) is a* containment *(an upper bound, image $\subseteq \mu_4$),* not *the equality $\mu_4 \subseteq S$. This also reconciles with the tensor-monomial sub-closure of Proposition 3.8 (which reached only*

± 1*): there is no gap on-circle between the sub-closure and the full image, since* ψ^2 *preserves the on/off-circle dichotomy and no catalog seed has* $\pm i$ *as a root. The only argument-*$\pi/2$ *eigenvalues anywhere in* S *are the* off-circle $\pm i\beta$ *at modulus* $\beta = 5^{1/4}\varphi = 2.4195\ldots > 1$ CH-G2.

Remark 3.11 (The charge character, g3, and the sharpened W[open]2 verdict). [FORCED]/[ESTABLISHED] CH-D1--CH-D3, CH-E1--CH-E6, CH-G1 The *charge* character does contain the non-real charges, but this does not force $\chi = \pm\pi/2$. $\chi(S) = \mathbb{Z}/4\mathbb{Z}$ carries the odd charge classes $1, 3$ (operator-algebra paper: the K-seed carries $\{0, 1, 2, 3\}$, its imaginary pair $\pm i\beta$ at $\pm\pi/2$); but charge reads the *argument* class only (modulus-free), and those charge-$1, 3$ elements are the off-circle $\pm i\beta$ — so the hypothesis of Theorem 3.9 that "S contains a non-real on-circle element" is *false given corpus* (Proposition 3.10): the non-real content of S lives entirely off the circle. Hence g3 resolves negatively — the doctrine does not require dynamical gate multipliers to be on-circle elements of S — as a truth-table over the thin axiom D2$'$: Reading A ("the multiplier is an on-circle element of S") is *unsatisfiable* for the rotation gate, whose forced multiplier is $\pm i$ (a $C = -\frac{1}{2}$ gate datum, W-Prop. 10.2) while on-circle $S = \{\pm 1\}$ and helicity removes the reals; Reading B ("the multiplier's charge lies in $\chi(S)$") exists only via the K-seed rotation pair, so it *imports exactly D2* (the charge completion) — a restatement. The charge-measure-coupling paper independently notes that the matrix commutator is the one non-semiring door across which emission constraints do not extend. *Net:* W[open]2 does *not* upgrade to a $\chi = \pm\pi/2$ derivation — it stays PARTIAL as a derivation, now sharpened and grounded, its openness precisely located at the unreduced axiom D2. The two v1.7 opens are closed as *forced-given-corpus negatives* (g1: external-open $\to \pm i \notin S$ on-circle; g3: papers-absent no-tie $\to$ the D2$'$ dichotomy above). Kills logged: D2$'$-as-a-doctrine-theorem under both readings, and the $\mu_4 \subseteq S$ misreading of the emission-gap containment. Any χ-selection remains a branch-blind *characterization* of the quantum ($\pi/2$ mod 2π for all k), never a branch derivation CH-E6.

Proposition 3.12 (The K-seed rotation axis and the substrate-grounded atom D2$''$)**.** *[FORCED]/[ESTABLISHED] QT-A1--QT-A5, QT-B1--QT-B3, QT-C1--QT-C2, QT-G1, QT-G3 The K-seed* $x^4 + 5x^2 - 5$ *is a* forced *substrate object (W-§2), not a declared one; its roots are* $\{\pm K, \ \pm i\beta\}$ *with* $K = 5^{1/4}\tau$ *(terrain, real) and* $\beta = 5^{1/4}\varphi$ *(rotation). The rotation pair* $\pm i\beta$ *is purely imaginary —* $\mathrm{Re}(i\beta) = 0$*,* $\mathrm{Im}(i\beta) = \beta > 0$ *— so* $\mathrm{Arg}(\pm i\beta) = \pm\pi/2$ exactly*, while* $|i\beta|^2 = \beta^2 = 3\varphi + 1 = \varphi^4 - 1 = \frac{5}{2} + \frac{3}{2}\sqrt{5} \neq 1$ *places it strictly off the unit circle (*$\beta \in [2.41, 2.43]$*, certified). Consequently the thin axiom*

$$\textit{D2}'': \qquad \textit{the winding multiplier's } \text{angle } \textit{is } \mathrm{Arg}(\pm i\beta),$$

which takes only the argument (the modulus staying D1's $|m| = e^{-2\rho}$*),* forces $\chi = \pm\pi/2$*: the quarter-turn* direction *is literally the argument of a forced object. This is* disjoint *from the g1 obstruction of Proposition 3.10: re-forcing cold (a* 74613*-monomial box scan) that the on-circle image of* S *is exactly* $\mu_2 = \{\pm 1\}$*, the argument* $\pi/2$ *is realized in* S *only* off-circle *(at the* K*-seed root* $i\beta$*), so D2$''$ is satisfiable exactly where the killed on-circle reading D2$'$ was not. Under the helicity obstruction (W-Thm. 11.1) a rotation multiplier is non-real, so among the* K*-seed roots the rotation pair is forced over the terrain pair; the forced winding generator* $q = i\tau$ *(minimal polynomial* $x^4 + 3x^2 + 1$*, roots* $\pm i\tau, \pm i\varphi$*, all imaginary) has the same angle axis, pinning D2$''$ to the generator's own compass.*

Remark 3.13 (Net verdict: D2 relocated, W[open]2 advanced not closed). [FORCED]/[COMPUTED] QT-C3--QT-C5, QT-D1--QT-D2, QT-A4 D2$''$ does *not* strictly reduce D2. The residual selection "the winding's angular clock is the K-seed *rotation* axis" is logically *equivalent* to D2's quarter-turn selection: over the three competing corpus quanta $\{\pi : \mathbb{Z}/2\mathbb{Z}, \ 2\pi/5 : \mathbb{Z}/5\mathbb{Z}, \ \pi/2 : \mathbb{Z}/4\mathbb{Z}\}$ it excludes the pentagon ($2\pi/5 = \frac{1}{5}$ turn $\notin (\pi/2)\mathbb{Z}$) and helicity-kills the real terrain axis π *exactly* as D2's principle does, so consuming "use the rotation direction" to conclude "quarter-turn" is a [FORCED] *restatement.* What D2$''$ genuinely adds is *grounding*: the value $\pi/2$ is no longer a bare declared number but the forced argument of the forced object $\pm i\beta$, and the completion $\mathbb{Z}/4\mathbb{Z}$ becomes a forced *output* — iterating the single $\frac{1}{4}$ turn generates it, whereas D2 *assumed* it. The characterization "$\mathrm{D2}''_{\min}$ is the intended minimal, substrate-grounded form of D2's residual content" is therefore [COMPUTED] (candidate), not forced: the clock-source selection remains declared, and a group-drop kills any purely-relational shortcut — the rotation pair *alone* has relational (pairwise-difference) group only $\mathbb{Z}/2\mathbb{Z}$, so the full $\mathbb{Z}/4\mathbb{Z}$ requires the absolute angle, iterated (a no-tie, QT-A4). *Net register motion for* $W[open]2$*:* it stays PARTIAL as a derivation but is *advanced* — substrate-grounded, disjoint-from-g1, with $\chi = \pm\pi/2$ and $\mathbb{Z}/4\mathbb{Z}$ downgraded from declared to forced outputs. Like every branch statement it selects the quantum ($\pi/2 \bmod 2\pi$) only, never the winding number, so OP-RATE stays closed as a classification QT-D1.

4 The generator as a relational object

The relational-charge note (source (v)) attaches phase data to *objects* — conjugation-closed root multisets of monic integer polynomials — through the angle $t(\alpha) = \frac{1}{2\pi}\operatorname{Arg}\alpha$, the relational group Δ generated by pairwise differences, the absolute anchor $n \in \{m, 2m\}$ (its Rigidity Theorem RC-4.1), and the complete cyclotomic contact signature of the ratio object Rat_p. Its own guardrail applies throughout: the vocabulary is structural, not physical. This section proves that the objects of the present paper live in that catalog, and that the generator itself is one of its distinguished witnesses. Everything here is unconditional [FORCED], machine-verified in the second harness (RC-D3, RC-E1--E7).

Theorem 4.1 (Minimal polynomial of the generator)**.** *[FORCED] RC-E3, RC-E5 The rung generator $q = i\tau$ is an algebraic integer with minimal polynomial*

$$\boxed{\operatorname{minpoly}(q) = x^4 + 3x^2 + 1}, \qquad \textit{roots } \{\pm i\tau,\ \pm i\varphi\}, \qquad \mathrm{M} = \varphi^2,$$

a reciprocal unit quartic ($p(0) = 1$), irreducible over $\mathbb{Q}$. Its angle set is $\{\frac{1}{4}, \frac{3}{4}\}$: absolute charge group $\mathbb{Z}/4\mathbb{Z}$ with charges $\{1, 3\}$ (odd charges only), while the pairwise differences generate only $\Delta = \mathbb{Z}/2\mathbb{Z}$ — a group-drop *object, the same coherence type as the note's canonical drop witness (its ledger entry E, $x^4 + 5x^2 + 5$), with contact signature $\{\Phi_1^4, \Phi_2^4\}$.*

Proof. $q^2 = -\tau^2$, so $2q^2 + 3 = \sqrt{5}$ and squaring gives $q^4 + 3q^2 + 1 = 0$; the roots are $\pm i\tau, \pm i\varphi$ since $x^2 \in \{-\tau^2, -\varphi^2\}$, verified by substitution. Irreducibility, the Mahler value $\varphi \cdot \varphi = \varphi^2$ (the two roots of modulus φ), and the complete contact scan are machine checks. ☐

Theorem 4.2 (Quarter-turn gauge identity)**.** *[FORCED] RC-D3, RC-E4, RC-E5 Let $Z^* = (x^2 - x - 1)(x^2 + x - 1) = x^4 - 3x^2 + 1$, the split golden pair (roots $\{\pm\varphi, \pm\tau\}$) — the relational note's hypothesis-necessity witness for its modulus-pinning theorem. Then, in both directions,*

$$\boxed{\operatorname{minpoly}(q)(ix) = Z^*(x) \qquad \textit{and} \qquad Z^*(ix) = \operatorname{minpoly}(q)(x)}\,:$$

the generator's object is exactly *the $c = \frac{1}{4}$ gauge rotation — one charge quantum — of the split golden pair. Consequences:*

- *(a)* ***Gauge-blindness realized:*** Rat *is invariant under the rotation (a global root scaling cancels in every ratio), so both objects carry the identical signature $\{\Phi_1^4, \Phi_2^4\}$ — a live instance of the note's Proposition RC-3.5(iv) (rotation fixes every relative charge) and of its gauge-blind probe (RC-Rem. 7.6).*

(b) ***Both rigidity anchors from one pair:*** *Z^* has $(m,n)=(2,2)$ (all roots real, angles $\{0,\frac{1}{2}\}$) while $i\,Z^*$ has $(m,n)=(2,4)$ (angles $\{\frac{1}{4},\frac{3}{4}\}$). The anchor dichotomy $n\in\{m,2m\}$ of Rigidity is realized by a single gauge pair, exchanged by exactly the seed's charge quantum.*

(c) *The torsion ratio -1 at the shared modulus φ inside Z^* — the very relation that makes Z^* the pinning witness — is the golden internal relation $t_{\mathrm{rel}}(\varphi,\varphi')=\frac{1}{2}$ (RC-Lem. 6.1) seen across the two factors.*

Proposition 4.3 (The K-seed is a catalog object)**.** *[FORCED] RC-E1 The K-seed x^4+5x^2-5 (W-§2) has roots $\{\pm K,\ \pm i\beta\}$ at angles $\{0,\frac{1}{2},\frac{1}{4},\frac{3}{4}\}$ —* ***the full charge compass*** *— with $\Delta=\mathbb{Z}/4\mathbb{Z}$, anchor $n=m=4$, $\mathrm{M}=\beta^2$ (the whitepaper constant), and contact signature $\{\Phi_1^4,\Phi_2^4\}$: it is the relational note's ledger entry F, whose sign-mirror is the drop witness of Theorem 4.1. The two research lines share this object at the root level.*

Remark 4.4 (D2, restated relationally). [FORCED]/[DECLARED] RC-E2 The winding's rung angles $\{n/4\}$ have relational group $\mathbb{Z}/4\mathbb{Z}=\Delta(\text{K-seed})$: Declaration D2 — *the winding realizes the seed's charge completion* — has the reference-free form *the helix's angle lattice realizes the K-seed's relational charge group.* This is an identification at the angle-lattice level, gauge-invariant by construction; D2 itself remains declared. Likewise the ν-locus of Proposition 11.3 now stands directly on the note's real-pair lemma: $\nu(\rho_n)=(-1)^n$ is torsion of order ≤ 2, so the rungs are precisely the real-coherence events of the relational ν-criterion RC-E7.

Remark 4.5 (Mahler bookkeeping across the bridge — with mechanism). [FORCED] RC-E6, OZ-06 $K=5^{1/4}\tau$ and $\beta=5^{1/4}\varphi$ exactly (from $(K\varphi)^4=5$ and $\beta=K\varphi^2$), so the K-seed's root moduli $\{K,K,\beta,\beta\}$ are the $5^{1/4}$-dilation of the generator object's root moduli $\{\tau,\tau,\varphi,\varphi\}$. The dilation carries exactly the two β-roots outside the unit circle, multiplying the measure by $(5^{1/4})^2$:

$$\mathrm{M}(\text{K-seed})=\beta^2=\sqrt{5}\,\varphi^2=\sqrt{5}\cdot\mathrm{M}(\mathrm{minpoly}(q)).$$

The bridge factor $\sqrt{5}$ is the dilation's Mahler cost.

Theorem 4.6 (Complete relational type of the generator object)**.** *[FORCED] NX-T1--T9 The root multiset of* $\mathrm{minpoly}(q)=x^4+3x^2+1$ *splits into exactly two shells (equal-modulus classes, RC-Def. 7.1): $\{\pm i\tau\}$ at modulus τ and $\{\pm i\varphi\}$ at modulus φ — conjugate pairs on the imaginary axis, exchanged by root inversion (the quartic is reciprocal). The complete ordered-pair ledger over all* 16 *root pairs is*

$$\underbrace{1^{\times 4}}_{\Phi_1^4\ \textit{(diagonal)}},\qquad \underbrace{(-1)^{\times 4}}_{\Phi_2^4\ \textit{(antipodal)}},\qquad \underbrace{\pm\tau^2,\ \pm\varphi^2\ (\textit{each}\times 2)}_{\textit{cross-shell: all real, none unimodular}},$$

so the contact signature $\{\Phi_1^4,\Phi_2^4\}$ of Theorem 4.1 is complete*: the only unimodular ratios are ±1, and no further cyclotomic contact exists (complete scan bound $M\le 2\cdot16^2=512$, RC-Lem. 7.4). Cross-shell coherence, decided by the ν-criterion (RC-Rem. 7.6: $\nu=\mu/\overline{\mu}$ for $\mu=\alpha/\beta$), holds* identically*: every cross ratio is real, so $\nu\equiv1$ on all eight cross-shell pairs, with relative charge $t_{\mathrm{rel}}=0$ on equal-angle pairs and $t_{\mathrm{rel}}=\frac{1}{2}$ on opposite-angle pairs. The generator object is therefore* fully coherent *— a single coherence class spanning both shells, fixed by the negation involution — with complete type*

$$\textit{angles }\{\tfrac{1}{4},\tfrac{1}{4},\tfrac{3}{4},\tfrac{3}{4}\},\qquad \Delta=\mathbb{Z}/2\mathbb{Z},\qquad (m,n)=(2,4),\qquad \mathrm{M}=\varphi^2,\qquad \{\Phi_1^4,\Phi_2^4\}.$$

Structurally it is a twisted shell *in the relational note's sense (its open problem (P1′); twisted shells are its Example 7.19 mechanism): $x^4+3x^2+1=s(x^2)$ with $s(y)=y^2+3y+1=$* $\mathrm{minpoly}(q^2)$ *— the $k=2$ twist of the real quadratic of $q^2=-\tau^2$ — realized here on the* admissible *side, where full* *coherence is automatic (RC-Cor. 4.2) but the shell, contact, and ν data remain contentful. The cross-shell coherence witnesses have moduli exactly $\{\tau^2, \varphi^2\} = \{|q^2|, |q^2|^{-1}\}$: coherence across the shells is carried by the two-floor contraction that prices the K-formation half-cycle (Remark 6.3).*

Proof. All sixteen ratios are computed exactly in $\mathbb{Q}(\sqrt{5})[i]$; the shell moduli, the realness and non-unimodularity of the cross ratios, the ν values, the t_{rel} pattern, the anchor data, and the twist factorization are the cited machine checks. The angle identity is immediate from $\alpha/\beta = (|\alpha|/|\beta|)\, e^{2\pi i (t(\alpha)-t(\beta))}$ with angle set $\{\frac{1}{4}, \frac{3}{4}\}$, and $t(\nu) = 2\, t_{\mathrm{rel}}$ (RC-Rem. 7.6) reads the $\nu \equiv 1$ outcome as $t_{\mathrm{rel}} \in \{0, \frac{1}{2}\}$ exactly. $\square$

Proposition 4.7 (The ratio object at the characteristic level). *[FORCED]* `QX-B1--B8` *Assemble the sixteen ratios of Theorem 4.6 into their characteristic polynomial:*

$$\chi_{\mathrm{Rat}}(x) = (x-1)^4(x+1)^4\big(x^4 - 7x^2 + 1\big)^2, \qquad x^4 - 7x^2 + 1 = (x^2 - 3x + 1)(x^2 + 3x + 1).$$

The nontorsion factor is the sign-split pair of φ^2 — the analogue of Z^* one floor up, at the squared level — and its roots $\pm\varphi^{\pm 2}$ are units of $\mathbb{Z}[\varphi]$ ($\varphi^2\tau^2 = 1$). Under the Adams square $\psi^2 : \mu \mapsto \mu^2$ the nontorsion multiset becomes four copies of $\{\varphi^4, \varphi^{-4}\} = \mathrm{spec}(A_{\mathrm{gap}})$ (minimal polynomial $x^2 - 7x + 1$): squaring the generator object's ratio geometry produces the gap object — the relational face of $q^4 = \mathrm{gap}$.

Proof. Exact multiset computations over $\mathbb{Q}(\sqrt{5})[i]$: the product $\prod_\mu (x - \mu)$ over all sixteen ratios, its factorization, the unit identity, and the squared multiset are the cited machine checks `QX-B1--B8`. $\square$

Theorem 4.8 (The quarter twist: transport domain and anchor-exchange dichotomy). *[FORCED]* `QX-D0--D9` *Call an object P (monic, integral, $P(0) \neq 0$)* parity-homogeneous *if $P(-x) = P(x)$ — equivalently, its root multiset is negation-closed; with $P(0) \neq 0$ this forces even degree and $P = g(x^2)$. Write $\rho_{1/4}(P)(x) := i^{-\deg P} P(-ix)$, the object whose roots are $i \cdot$ (roots of P) (on this domain $iA = -iA$, so the two quarter-turn orientations give the same object). Then:*

- *(a)* ***Domain.*** *$\rho_{1/4}(P)$ has integer coefficients iff P is parity-homogeneous, and there $\rho_{1/4}$ is an involution. The golden seed $x^2 - x - 1$ is* not *in the domain ($-\varphi$ is not a root): the gauge identity of Theorem 4.2 necessarily lives at the sign-split pair Z^*, not at the seed.*
- *(b)* ***Invariance.*** *On the domain the transport realizes the gauge rotation ρ_c at $c = \frac{1}{4}$ (RC-Prop. 3.5(iv)): it fixes Δ, every relative charge, and every root modulus (hence M), and shifts every absolute angle by $\frac{1}{4}$. Negation closure puts $\frac{1}{2} \in \Delta$ outright, so $m = |\Delta|$ is always even.*
- *(c)* ***Dichotomy.*** *Suppose further P is relationally admissible, so all angles are rational and $\Delta = \mathbb{Z}/m\mathbb{Z}$ is cyclic (RC-Thm. 4.1, RC-Prop. 3.4). Let A be the angle set and $a_0 \in A$. Conjugation plus negation closure give $2a_0 \in \Delta$ and $\langle A \rangle = \Delta + \langle a_0 \rangle$, re-proving Rigidity $n \in \{m, 2m\}$ here, with $n = m \iff A \subseteq \Delta$. If $4 \mid m$, then $\frac{1}{4} \in \Delta$ and the transport* fixes *the anchor ($n' = n$); if $m \equiv 2 \pmod 4$, then $\{x : 2x \in \Delta\} = \Delta \sqcup (\Delta + \frac{1}{4})$ and the transport* exchanges *the two rigidity branches: $n = m \iff n' = 2m$.*

Theorem 4.2(b)'s pair is the minimal exchange instance: by Theorem 4.6 the generator object has $(m, n) = (2, 4)$, its quarter twist is Z^* with $(m, n) = (2, 2)$, and the gauge identity is exactly this exchange witnessed at the generator. The K-seed ($m = 4$, angle set literally invariant) and $\Phi_8 = x^4 + 1$ ($m = 4$) are anchor-fixed, while $x^2 - 2 \leftrightarrow x^2 + 2$ and $x^4 - 2x^2 + 4 \leftrightarrow x^4 + 2x^2 + 4$ realize the exchange at $m = 2$ and $m = 6$. The quarter twist is the exact companion of the relational note's sign twist (RC-Lem. 6.3, $c = \frac{1}{2}$): the sign twist exchanges the anchor sectors at odd *m and is the identity on the quarter twist's domain; the quarter twist exchanges them at $m \equiv 2 \pmod 4$ and fixes them at $4 \mid m$. Together the two twists absorb the anchor dichotomy at every $m \not\equiv 0 \pmod 4$.*

Proof. (a) $\prod_\alpha(x - i\alpha) = i^{\deg P}P(-ix)$; the coefficient of x^k picks up $i^{\deg P-k}$, real for all occurring k iff every exponent shares the parity of $\deg P$; with $P(0) \neq 0$ the exponent 0 occurs, forcing even degree and evenness. Involutivity: $i \cdot iA = -A = A$. (b) A common shift cancels in every angle difference; moduli are preserved by $\alpha \mapsto i\alpha$; negation closure pairs $t(\alpha)$ with $t(\alpha) + \frac{1}{2}$. (c) $a - a_0 \in \Delta$ for all $a \in A$ gives $\langle A\rangle = \Delta + \langle a_0\rangle$; $-a_0 \in A$ gives $2a_0 \in \Delta$, so the coset $a_0 + \Delta$ has order at most 2 in the quotient. For $4 \mid m$: $\frac{1}{4} \in \langle \frac{1}{m}\rangle = \Delta$, so the shift does not change $\langle \Delta, a_0\rangle$. For $m \equiv 2 \pmod 4$: the doubling preimage $\{x : 2x \in \Delta\} = \langle \frac{1}{2m}\rangle$ has index 2 over Δ with nontrivial coset $\Delta + \frac{1}{4}$ (as $\frac{1}{4} \notin \Delta$ but $\frac{1}{2} \in \Delta$); so $a_0 \in \Delta$ ($n = m$) puts $a_0 + \frac{1}{4}$ in the nontrivial coset ($n' = 2m$), and $a_0 \in \Delta + \frac{1}{4}$ ($n = 2m$) puts $a_0 + \frac{1}{4} \in \Delta + \frac{1}{2} = \Delta$ ($n' = m$). Domain, involution, invariance, the golden-seed exclusion, all eight witnesses, and an exhaustive coset simulation over every even $m \leq 24$ are machine checks `QX-D0--D9`. □

The relational note's open cluster (v1.8). With the relational-charge note (source (v)) now available in full, three of its open items are addressed cold: its P4 is recovered and reduced, its odd relational floor P3 is bracketed but not promoted, and its P1′ taxonomy is scanned over a bounded window. The vocabulary stays structural, per the note's own guardrail.

Remark 4.9 (P4 recovered and reduced to P3). [FORCED]/[ESTABLISHED] `RN-A1, RN-A1F, RN-A2` The relational note's open item P4 (RC-§ sec:open) is recovered verbatim: *"Is parity the whole coupling? Does μ_{rel} depend on Δ through anything besides the single bit $2 \mid m$?"* It is *not* an independent problem: it reduces exactly to P3 (the odd relational floor). Given RC-Thm. 6.5, $\mu_{\mathrm{rel}}(m) = \varphi$ (m even) / $\mu(m)$ (m odd), so "μ_{rel} factors through the parity bit" $\iff$ the even branch is m-independent (constructively uniform: $q_k = x^{2k} + x^k - 1$ gives $\Delta = \mathbb{Z}/2k\mathbb{Z}$, $\mathrm{M} = \varphi$ for every even m) *and* the odd branch is m-independent (anchored by the forced $\mu(3) = 2$, forcing the odd constant $= 2$). Hence P4 $\iff$ P3, and P4 inherits P3's open status as a *restatement* of it, not new work. *Register note (v2.3).* The μ_{rel} of this remark is the *relational* floor (RC-Thm. 6.5, [INHERITED]); it is distinct from the *absolute* Mahler floor μ over charge-admissible objects, which is what Theorems 4.17 (odd $= 2$) and 4.20 (even $= \varphi$) pin by descent on $\mathrm{M}(\alpha)$ directly. W-§22's "$\mu(\text{even}) \geq \varphi$ plausible" is the *absolute* even branch — upgraded to [FORCED] given Schinzel by Theorem 4.20, not the same statement as the relational uniformity inherited above.

Proposition 4.10 (The odd relational floor, bracketed (v1.8))**.** *[FORCED]/[ESTABLISHED] `RN-B1--RN-B4, RN-H1--RN-H3` For odd charge m the relational Mahler floor obeys*

$$\mu_S \;\leq\; \mu_{\mathrm{rel}}(\mathit{odd}\ m) \;\leq\; 2,$$

with the two bounds forced but the value not *pinned.* Upper bound ≤ 2 *is [FORCED]: the witness $x^m - 2$ (m odd) is Eisenstein-irreducible at 2, has rational angle set $\{j/m\}$ hence a single coherence class (relationally admissible per RC-Def. 3.2), relational group $\Delta = \mathbb{Z}/m\mathbb{Z}$, anchor $n = m$, and all roots of modulus $2^{1/m} > 1$, so monic $\mathrm{M} = 2$ exactly.* Lower bound *is forced only to μ_S [FORCED]/[ESTABLISHED]: Smyth's plastic number (real root of $x^3 - x - 1$, $1 < \mu_S < \varphi < 2$) bounds every non-reciprocal charge-admissible object below, while the reciprocal-real sector is forced strictly above $\varphi^2 > 2$ (integer trace ≥ 3); and the π-ray obstruction is [FORCED]: $\frac{1}{2} \notin \mathbb{Z}/m\mathbb{Z}$ for odd m, so the golden pair's intrinsic relation $t_{\mathrm{rel}}(\varphi, \varphi') = \frac{1}{2}$ (from $\varphi/\varphi' = -\varphi^2$) cannot embed, and the universal floor φ is unattainable at odd charge (forcing $\mu_{\mathrm{rel}}(\mathit{odd}) > \varphi$, not ≥ 2).* Status (v2.1): *the promotion of $\mu_{\mathrm{rel}}(\mathit{odd}\ m) = 2$ from [COMPUTED] to [FORCED] is* achieved *— Theorem 4.17 — and the bracket above collapses to its upper end for every odd m. It was correctly recorded as not achievable from the* shipped *corpus in v1.8–v2.0: the missing input is new (if elementary) mathematics, not a corpus consequence. What it is* not *is a strengthening of Smyth. The residual* *was read here as "non-reciprocal charge-admissible objects with* $\mathrm{M} \in [\mu_S, 2)$*, Lehmer/no-Salem-window territory"; that reading is route-relative and is corrected in Finding 12.7 — the descent of Theorem 4.17 never splits on reciprocality and never enters that window. The shipped charge-measure-coupling erratum and whitepaper §6.2, which hold* $\mu(odd) = 2$ *at [COMPUTED], are thereby confirmed at [FORCED].*

Remark 4.11 (Bounded P1′ transport-orbit scan). [COMPUTED] `RN-C0--RN-C3`, `RN-H4` The note's coherence-detection machinery (the ratio object Rat_p = primitive part of $\mathrm{Res}_y(p(y), p(xy))$ and its cyclotomic-contact signature) is re-encoded cold and reproduces the RC ledger exactly; an exact ψ^D admissibility oracle (a root power D that is totally real $\iff$ rational angle) replaces the hanging `CRootOf` route. A bounded transport-orbit scan over the irreducible monic integer objects of degree 2–6, with per-degree coefficient caps *logged* ($\{(2,3),(3,2),(4,2),(5,1),(6,1)\}$; 855 objects), classifies the taxonomy {pinned/inert, $s(x^k)$-shell tower, cyclotomic, dilation-of-known} as exhaustive over the window: buckets inert 804, shell 32, cyclotomic 11, dilation 8, and *OTHER* 0 — no new mechanism and no counterexample to the pinned/twisted-shell dichotomy. Within-shell torsion coherence was decided; the cross-shell mirrored-class (ν-criterion) half of P1′ — out of *this* scan's scope — is resolved separately and unconditionally in Theorem 4.12 (the no-tie). This within-shell scan is a bounded [COMPUTED] corroboration (candidate), not a proof; the general within-shell taxonomy remains [COMPUTED]-open, while the cross-shell half is now [FORCED].

Theorem 4.12 (Cross-shell no-tie: the P1′ ν-criterion resolved (v2.2))**.** *[FORCED] XS-A1--A3, XS-B1--B3, XS-C1--C4 Let p be a monic irreducible integer polynomial with root multiset R (conjugation-closed, as p is real), partitioned into* shells *— equal-modulus classes (RC-Def. 7.1). For a* cross-shell *ordered pair* (α, β) *— roots in distinct shells,* $|\alpha| \neq |\beta|$ *— put*

$$\nu(\alpha, \beta) = \frac{\alpha/\beta}{\overline{\alpha/\beta}} = e^{2i(\theta_\alpha - \theta_\beta)}, \qquad \theta_\gamma := \operatorname{Arg}\gamma,$$

a unimodular number (the mirrored value, RC-Rem. 7.6). Then:

(i) (vacuity) *if R is a single shell (all roots equimodular) there are no cross-shell pairs and the* ν*-criterion is vacuous;*

(ii) (no-tie) *if R is multi-shell, then p is* cross-shell coherent *— every* $\nu(\alpha, \beta)$ *a root of unity —* if and only if *p is* charge-admissible *(every root argument in* $\pi\mathbb{Q}$*; equivalently the* ψ^D *oracle of Remark 4.11 returns some D).*

Consequently cross-shell coherence is a forced function *of admissibility and shell count: it introduces no invariant beyond the within-shell class, and the within-shell verdict* OTHER = 0 *of Remark 4.11 is the complete cross-shell statement — no object carries a cross-shell mechanism absent from its within-shell class. This is the "cross-shell mirrored-class coherence is a no-tie" of §14.9, now* proved*: BL-G7 is discharged.*

Proof. $\nu(\alpha, \beta) = e^{2i(\theta_\alpha - \theta_\beta)}$ is a root of unity iff $\theta_\alpha - \theta_\beta \in \pi\mathbb{Q}$; (i) is immediate. *(⇐)* if every root argument lies in $\pi\mathbb{Q}$ so does every difference, hence every ν is a root of unity. *(⇒)* assume coherence and fix a root α. As R is multi-shell there is a root β with $|\beta| \neq |\alpha|$; as R is conjugation-closed, $\bar\beta$ (modulus $|\beta|$, argument $-\theta_\beta$) is also a root, and both $\beta, \bar\beta$ are cross-shell to α. Coherence at (α, β) and $(\alpha, \bar\beta)$ gives $\theta_\alpha - \theta_\beta \in \pi\mathbb{Q}$ and $\theta_\alpha + \theta_\beta \in \pi\mathbb{Q}$; adding, $2\theta_\alpha \in \pi\mathbb{Q}$, so $\theta_\alpha \in \pi\mathbb{Q}$. As α was arbitrary, p is charge-admissible. The mechanism is the exact identity $\nu(\alpha, \beta)\,\nu(\alpha, \bar\beta) = \alpha^2/\bar\alpha^2 = e^{4i\theta_\alpha}$, independent of β (`XS-B2`): the second shell cancels, leaving a condition on α alone. □

Remark 4.13 (The detector, validated fail-first; and the corroboration). [FORCED]/[COMPUTED] `XS-A1--A3`, `XS-C1--C4`, `XS-H1--H3` The ν-detector is validated fail-first before use: it reproduces Theorem 4.6's generator (two shells $\{\pm i\tau\}, \{\pm i\varphi\}$, eight cross-shell pairs, cross ratios $\pm\tau^2, \pm\varphi^2$, $\nu \equiv 1$, fully coherent, XS-A1); it sees non-trivial coherence ($x^4 + x^2 - 6$: cross-shell $\nu = -1$, order 2, XS-A2); and it discriminates non-coherence — the Pisot number $x^3 - x - 1$, the Salem sextic $x^6 - x^4 - x^3 - x^2 + 1$, and Lehmer's degree-10 Salem each carry a non-torsion cross-shell ν (guards XS-H1--H3), the very objects for which admissibility fails. Over the *same forced window* as Remark 4.11 — irreducible monic integer objects, caps $\{(2,3),(3,2),(4,2),(5,1),(6,1)\}$, 855 objects — the equivalence coherent $\iff$ admissible holds with *zero* exceptions [COMPUTED] XS-C2: 34 multi-shell coherent (all admissible), 788 multi-shell non-coherent (all inadmissible), 33 single-shell (vacuous); the realized cross-shell ν-orders are $\{1,2,3,4\}$ XS-C4. Theorem 4.12 makes the equivalence [FORCED]; the scan corroborates it and exhibits the population, exactly as `nd_softcheck.py` does for Theorem 4.17.

Proposition 4.14 (The odd relational floor, reduced to a single lemma — now proved)**.** *[FORCED]/[ESTABLISHED]/[COMPUTED] OF-A1--OF-A4, OF-B1--OF-B6, OF-C1--OF-C8, OF-H1--OF-H6; ND-N1--ND-N5 This sharpens the bracket $\mu_S \le \mu_{\mathrm{rel}}(odd) \le 2$ of Proposition 4.10 in two independent registers.* (i) Computed *[COMPUTED]: over relationally-admissible objects carrying the full $\mathbb{Z}/m$ charge ($m = 3,5,7,9$), enumerated up to a* logged *box (shell families $x^m - a$, $a \le 8$, exact; cyclotomic-padded and reciprocal/pentagon keystones; plus a brute-force over irreducible monic integer polynomials in $\{(\deg 3, |c| \le 6), (\deg 4, |c| \le 3), (\deg 5, |c| \le 2)\}$, 543 admissible objects), the minimum Mahler measure is* exactly 2*, with unique minimizer $x^m - 2$; the parity-split falsifier is clean — no odd-charge object falls below 2, while every sub-2 admissible object found is even-charge (q_k-type, $\mathrm{M} = \varphi$).* Status of the box (v2.1)*: its caps no longer carry weight, for two reasons. First, the enumeration need never have been hand-capped: if $\mathrm{M} < 2$ then $|a_0| \le \mathrm{M} < 2$ with $a_0 \ne 0$, so $|a_0| = 1$ — every violator is a unit — and $|a_j| = |e_{d-j}(\alpha)| \le \binom{d}{j}\mathrm{M} < 2\binom{d}{j}$ (Landau/Mahler), so the box $\{|a_0| = 1,\ |a_j| \le 2\binom{d}{j}\}$ contains* every *irreducible degree-d counterexample with no cap chosen by hand. Second, and decisively, Theorem 4.17 dispenses with boxes altogether: (i) is now a* corollary *of it at every degree and every odd m, and the sub-2 objects (i) does report are even-charge, exactly what the theorem's hypothesis excludes. The certificate of §13 accordingly runs over a* forced *on-ray population rather than a curated one ND-N1.* (ii) The minimizer *[FORCED]: $x^m - 2$ is Eisenstein-irreducible at 2, carries relational group $\mathbb{Z}/m$ (angle set $\{j/m\}$, a single coherence class) and anchor m, and has $\mathrm{M} = 2$ by a pure integer decision (all roots of modulus $2^{1/m} > 1$); cyclotomic padding is* M*-neutral, and the real reciprocal sub-case is forced $\ge \varphi^2 > 2$ (rational-trace unit).* (iii) The reduction *[FORCED]* given corpus*: the lower bound $\mu_{\mathrm{rel}}(odd\ m) = 2$ for* all *odd m is* equivalent *to the single named lemma*

$$\text{(ODD-2)}: \quad \textit{every relationally-admissible object with odd relational group } \mathbb{Z}/m \textit{ and } \mathrm{M} > 1 \textit{ has } \mathrm{M} \ge 2$$

(irreducibles suffice), the reduction step itself being forced — by Mahler multiplicativity, Kronecker's $\mathrm{M} \ge 1$ *(=1 iff cyclotomic), the subgroup fact that factors of an odd-charge object are odd-charge, and the anchor-normalizing sign twist.* (iv) The lemma is proved *[FORCED]: Theorem 4.17 settles (ODD-2) for every odd m at once, so the = 2 floor is unconditional and this proposition's former [OPEN] is discharged. The proof does* not *run the route the corpus had reached for, and that route's obituary is worth keeping. The corpus forces $\mathbb{Z}/3$ (elementary), the π-ray exclusion of φ ($\frac{1}{2} \notin \mathbb{Z}/m$ for odd m), and the real-reciprocal floor $\varphi^2 > 2$, while the non-reciprocal sub-case is forced only to Smyth's $\mu_S = 1.3247\ldots$ (root of $x^3 - x - 1$, certified in $(1,2)$); v1.9–v2.0 read the residual gap $[\mu_S, 2)$ as* being *the Schur–Siegel–Smyth / Lehmer–pentagon core. That reading is true of the* realification *handle (Adams ψ^m), which throws the ray constraint away and retains only non-reciprocality — and it is exactly why that handle cannot close the lemma (asking its image to clear 2^m merely restates the claim). It is* not *true of the lemma. Theorem 4.17 never realifies: it pushes the ray constraint through the relative norm $N_{K/F}$, where oddness survives as the π-ray exclusion applied to N. It therefore never enters the Schur–Siegel–Smyth window, μ_S is nowhere used, and no strengthening*

of Smyth is required (Finding 12.7). The reading of (ODD-2-NR) as the intended minimal atom lapses with it: the operative split is not reciprocal/non-reciprocal but F-relative, and the sub-case the old split could not reach — composite m with $1 < k < m$ — is certified outright ND-N5.

Lemma 4.15 (TP-2: the totally positive floor is two)**.** [FORCED] ND-E1--ND-E3, ND-T1--ND-T4 *Let $N \neq 1$ be a totally positive algebraic integer of degree e, and let r be the number of its conjugates exceeding 1. Then*

$$\boxed{\mathrm{M}(N) \ \geq \ 2^{\,r} \ \geq \ 2},$$

with equality iff $N = 2$. No external theorem is used.

Proof. Let g be the minimal polynomial of N and $N_1, \dots, N_e > 0$ its conjugates. No N_i equals 1: otherwise $g = y - 1$ and $N = 1$. Hence $g(1) \neq 0$, and $|g(1)| = \prod_i |1 - N_i|$ is a nonzero rational integer, so $|g(1)| \geq 1$. Split the conjugates into $A = \{N_i > 1\}$, of size r, and $B = \{N_i < 1\}$. Every factor of $\prod_B (1 - N_i)$ lies in $(0, 1)$, so $\prod_B (1 - N_i) \leq 1$ and therefore

$$\prod_A (N_i - 1) \ = \ \frac{|g(1)|}{\prod_B (1 - N_i)} \ \geq \ |g(1)| \ \geq \ 1.$$

A is nonempty: $\prod_i N_i = |g(0)| \geq 1$ forces some $N_i \geq 1$, and none is 1. Put $u_i = N_i - 1 > 0$ for $i \in A$. Since $t \mapsto \log(1 + e^t)$ is convex — $\frac{d^2}{dt^2} \log(1 + e^t) = 1/\big(4 \cosh^2(t/2)\big) > 0$ — Jensen's inequality at the points $\log u_i$ gives the product inequality $\prod_A (1 + u_i) \geq \big(1 + (\prod_A u_i)^{1/r}\big)^r$, whence

$$\mathrm{M}(N) \ = \ \prod_A N_i \ = \ \prod_A (1 + u_i) \ \geq \ \big(1 + 1\big)^r \ = \ 2^{\,r}.$$

Equality throughout forces all u_i equal (Jensen) and $\prod_A u_i = 1$, so every $N_i \in A$ equals 2; then 2 is a conjugate of N, so $g = y - 2$, $N = 2$, and $r = 1$. □

Lemma 4.16 (Anchor normalisation: the sign twist is forced and exhaustive)**.** [FORCED] ND-A1--ND-A3 *Let m be odd and let α be an algebraic integer, not a root of unity, whose object has relational group $\Delta \leq \mathbb{Z}/m$. Then* exactly one *of $\pm\alpha$ has all conjugates on $R_m = \{z \neq 0 : \operatorname{Arg} z \in (2\pi/m)\mathbb{Z}\}$, and $\mathrm{M}(-\alpha) = \mathrm{M}(\alpha)$.*

Proof. Δ is generated by the *pairwise differences* of $t(\alpha_i) = \frac{1}{2\pi} \operatorname{Arg} \alpha_i$ (§4), so $\Delta \leq \mathbb{Z}/m$ forces only $t(\alpha_i) \in \theta_0 + (1/m)\mathbb{Z}$ for a common θ_0 — not $\theta_0 = 0$. The minimal polynomial is real, so the root multiset is conjugation-closed and $\{t_i\} = \{-t_i\}$; hence $-\theta_0 \in \theta_0 + (1/m)\mathbb{Z}$, i.e. $2\theta_0 \in (1/m)\mathbb{Z}$ and $\theta_0 \in (1/2m)\mathbb{Z}$. The index-two quotient leaves exactly two cosets. If $\theta_0 \in (1/m)\mathbb{Z}$ then α is on R_m. Otherwise $\theta_0 \in \frac{1}{2m} + (1/m)\mathbb{Z}$ and $t(-\alpha_i) = t_i + \frac{1}{2} \in \frac{m+1}{2m} + (1/m)\mathbb{Z} = (1/m)\mathbb{Z}$, because $2 \mid m + 1$ for odd m; so $-\alpha$ is on R_m. Both cannot hold at once: that would put $\frac{1}{2} \in (1/m)\mathbb{Z}$, i.e. $2 \mid m$. Negating every root preserves every modulus, so $\mathrm{M}(-\alpha) = \mathrm{M}(\alpha)$. □

Theorem 4.17 ((ODD-2) by relative-norm descent)**.** [FORCED] ND-N1--ND-N5, ND-E1--ND-E5, ND-X3--ND-XP, ND-A1--ND-A3 *Let m be odd and let α be an algebraic integer, not a root of unity, every conjugate of which lies on R_m. Then*

$$\boxed{\mathrm{M}(\alpha) \ \geq \ 2}$$

and the bound is attained, at $x^m - 2$ for every odd m. With Lemma 4.16 this is exactly (ODD-2); with Proposition 4.14(iii) it gives $\mu_{\mathrm{rel}}(\text{odd } m) = 2$ for every odd m, and with Remark 4.19 it closes P3 and P4.

Proof. Put $\beta = \alpha^m$, $F = \mathbb{Q}(\beta)$, $K = \mathbb{Q}(\alpha)$, $d = [K : \mathbb{Q}]$, $d' = [F : \mathbb{Q}]$, $k = [K : F]$, so that $d = kd'$ (tower law).

(1) *β is totally positive, so F is totally real.* Every conjugate of β is $\sigma(\alpha)^m$ for some embedding σ, and $\operatorname{Arg}\sigma(\alpha) \in (2\pi/m)\mathbb{Z}$ gives $\operatorname{Arg}\sigma(\alpha)^m \in 2\pi\mathbb{Z}$: a positive real.

(2) *$N := N_{K/F}(\alpha)$ is totally positive.* Each $\mathbb{Q}$-conjugate of N is a product of k conjugates of α, so its argument lies in the subgroup $(2\pi/m)\mathbb{Z}$; it is real, since $N \in F$ and F is totally real by (1); and $\pi \notin (2\pi/m)\mathbb{Z} \pmod{2\pi}$ because m is odd. This is the *only* place oddness is used, and it is the corpus's own π-ray exclusion (Proposition 4.10) — applied to N rather than to α.

(3) $N \neq 1$. The F-conjugates of α are exactly those α_j with $\alpha_j^m = \beta$, since σ fixes $\beta \in F$; each is therefore $\alpha\zeta^{j_i}$ with $\zeta^m = 1$, so $N = \alpha^k\zeta^J$. Were $N = 1$, then $\alpha^{km} = 1$ and α would be a root of unity.

(4) $\mathrm{M}(N) \geq 2$: Lemma 4.15, applicable by (2) and (3).

(5) *Descent.* For the absolute logarithmic Weil height, $h(N) \leq \sum_{i=1}^k h(\sigma_i\alpha) = k\,h(\alpha)$, and $h(\gamma) = \log\mathrm{M}(\gamma)/\deg\gamma$. With $e = \deg N$ this reads $\frac{1}{e}\log\mathrm{M}(N) \leq \frac{k}{d}\log\mathrm{M}(\alpha)$, i.e.

$$\mathrm{M}(\alpha) \;=\; \exp\bigl(d\,h(\alpha)\bigr) \;\geq\; \mathrm{M}(N)^{\,d/(ke)} \;=\; \mathrm{M}(N)^{\,d'/e}.$$

(6) *The exponent is a positive integer.* $\mathbb{Q}(N) \subseteq F$, so $e \mid d'$ and $d'/e \geq 1$. With $\mathrm{M}(N) \geq 2 > 1$, $\mathrm{M}(\alpha) \geq \mathrm{M}(N) \geq 2$.

Attainment: $x^m - 2$ is Eisenstein at 2, lies on R_m, and has $N = 2$, $e = d' = 1$, so the bound reads $\mathrm{M} \geq 2 = \mathrm{M}(x^m - 2)$ — equality, verified at $m = 3, 5, 7, 9, 11, 15$ `ND-X3--ND-X15`. The refined bound is sharp elsewhere too: $x^6 - 3x^3 + 1$ has $m = 3$, $d = 6$, $N = \varphi^2$, $e = d' = 2$, and $\mathrm{M}(N)^{d'/e} = \varphi^2 = \mathrm{M}(\alpha)$ exactly `ND-XP`. □

Remark 4.18 (Why the obvious reduction fails). [FORCED] `ND-R1--ND-R3`, `ND-S1` The tempting route splits on $k = [K : F]$ and asserts $k \mid m$, which would force $k \in \{1, m\}$ for prime m and reduce (ODD-2) to Lemma 4.15 at a stroke. *The hypothesis is false.* The irreducible factors of $x^m - \beta$ need not have degree dividing m: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, degrees 1 and 2. Concretely $\alpha = 2\zeta_3$, minimal polynomial $x^2 + 2x + 4$, has both roots at $\operatorname{Arg} = \pm 2\pi/3$ — exactly on R_3 — with $m = 3$, $d = 2$, $\beta = 8$, $d' = 1$, $k = 2$, and $2 \nmid 3$ `ND-R1`. The shape is not exceptional: the forced population realises (m, k, d', e) with $k \nmid m$ at $(3, 2, 1, 1)$, $(3, 2, 2, 2)$, $(5, 2, 2, 2)$, $(9, 2, 1, 1)$, $(9, 2, 2, 2)$ and $(9, 6, 1, 1)$. What survives is the tower identity $\mathrm{M}(\alpha)^m = \mathrm{M}(\beta)^k$, which is $d = kd'$ and nothing more, asserted per object in the certificate `ND-R2`. The fallback $\mathrm{M}(\alpha) \geq \varphi^{d/m}$ does not rescue it: $\mathrm{M} < 2$ then forces only $d < m\log_\varphi 2 = 1.4404\,m$, a window that *grows* with m ($m = 9$: $d \leq 12$; $m = 15$: $d \leq 21$), so the box route faces an infinite family of finite searches rather than one unfinished computation `ND-R3`. Nor is the fallback merely inconvenient: on the forced population 456 of 565 objects have $\varphi^{d/m} < 2$, where it says nothing at all, and the descent returns ≥ 2 on every one `ND-S1`. Theorem 4.17 avoids the split entirely — it never mentions $k \mid m$ — which is why it closes composite m in the same breath as prime m.

Remark 4.19 (P4 inherits P3 exactly). [FORCED] `OF-D1`, `OF-D2` The v1.8 recovery reduced P4 — does μ_{rel} depend on the charge group beyond the single parity bit $2 \mid m$? — to P3 via Remark 4.9 and the forced even-branch uniformity $\mu_{\mathrm{rel}}(\text{even}) = \varphi$. Since the odd branch is the same constant 2 under both registers above — computed for $m = 3, 5, 7, 9$, and now $= 2$ for *every* odd m by Theorem 4.17 — P4 factors through the parity bit with *exactly* P3's status. In v1.9–v2.0 that status was "affirmative [COMPUTED] up to the logged bounds, forced-affirmative iff (ODD-2)"; since (ODD-2) is now a theorem, both read *affirmative, unconditionally* [FORCED]: μ_{rel} depends on the charge group through the single parity bit $2 \mid m$ and nothing else, taking the value φ on the even branch and 2 on the odd. It adds no new burden.

Theorem 4.20 (The even relational floor $= \varphi$, given Schinzel (v2.3))**.** *[FORCED]*/*[ESTABLISHED]* `EV-S1--S2, EV-W1--W2, EV-D1--D2, EV-F1--F3` *Let m be* even *and let α be an algebraic integer, not a root of unity, every conjugate of which lies on R_m. Then*

$$\boxed{\mathrm{M}(\alpha) \;\geq\; \varphi}$$

and the bound is attained, at $q_k = x^{2k}+x^k-1$ (with $m = 2k$) for every $k \geq 1$. With Remark 4.19 this pins the even branch of μ_{rel} at φ; with Theorem 4.17 it completes the parity floor, $\mu_{\mathrm{rel}}(even) = \varphi$ and $\mu_{\mathrm{rel}}(odd) = 2$; and it upgrades the whitepaper's plausible row (W-§22, "$\mu(even) \geq \varphi$") to [FORCED] given Schinzel.

Proof. The descent of Theorem 4.17 runs verbatim except at step (2), which oddness there and evenness here decide oppositely. Put $\beta = \alpha^m$, $F = \mathbb{Q}(\beta)$, $K = \mathbb{Q}(\alpha)$, $d' = [F : \mathbb{Q}]$, $k = [K : F]$, $N = N_{K/F}(\alpha)$, $e = \deg N$.

(1) *β totally positive, F totally real*: $\operatorname{Arg}\sigma(\alpha) \in (2\pi/m)\mathbb{Z}$ gives $\operatorname{Arg}\sigma(\alpha)^m \in 2\pi\mathbb{Z}$ — parity-free, identical to Theorem 4.17(1).

(2) *N is totally real*: each $\mathbb{Q}$-conjugate of N is a product of k conjugates of α, real since $N \in F$ and F is totally real. **Here the parity flips.** For odd m, $\pi \notin (2\pi/m)\mathbb{Z} \pmod{2\pi}$ forces N totally *positive* (Theorem 4.17(2)); for even m, $\pi \in (2\pi/m)\mathbb{Z}$, so N is totally real but need not be positive. This is the *only* place the parity enters.

(3) $N \neq \pm 1$: $N = \alpha^k \zeta^J$, $\zeta^m = 1$; were $N = \pm 1$ then $\alpha^{km} = (\pm 1)^m = 1$ (even m), making α a root of unity, excluded.

(4) $\mathrm{M}(N) \geq \varphi^{e/2}$: a totally real algebraic integer $\neq 0, \pm 1$ is not a root of unity, so *Schinzel*'s bound [ESTABLISHED] (Schinzel, *Acta Arith.* **24** (1973), 385–399; a totally real algebraic integer $\neq 0, \pm 1$ of degree e has $\mathrm{M} \geq \varphi^{e/2}$) applies. This is the one step the elementary route cannot supply: for totally real (not positive) N, Lemma 4.15 gives only $\mathrm{M}(N) \geq 2^r$, and at $r = 1$ that is $\sqrt{2} < \varphi$. The even branch closes by *external import* where the odd branch (Theorem 4.17) closed *internally* — an honest asymmetry. Schinzel is the same bound `nd_softcheck.py` carries as corroboration; `ev_softcheck.py` makes it load-bearing, corroborated with 0 violations over the totally-real box and tight at $x^2 - x - 1$ `EV-S1--S2`.

(5) *Descent*: as in Theorem 4.17(5), the Weil height gives $\mathrm{M}(\alpha) \geq \mathrm{M}(N)^{d'/e}$; with $e \mid d'$ ($\mathbb{Q}(N) \subseteq F$), $\mathrm{M}(\alpha) \geq \varphi^{d'/2} \geq \varphi$. If $d' = 1$ then $N \in \mathbb{Z}$, $|N| \geq 2$, and $\mathrm{M}(\alpha) \geq |N| \geq 2 > \varphi$.

Attainment: q_k lies on R_{2k} with even ($\mathbb{Z}/2k$) charge and $\mathrm{M} = \varphi$ (W-Prop. 6.2: the k roots at $x^k = -\varphi$ each of modulus $\varphi^{1/k}$), and it realises the descent tight at q_1 ($N = -\varphi$) and q_2 ($N = \varphi$), $\mathrm{M}(N) = \varphi = \varphi^{e/2}$, $e = d' = 2$ `EV-D1`. A parity-split scan of the forced population finds no even-charge admissible object below φ and none odd below 2 `EV-F1--F3`. ☐

Corollary 4.21 (GAP-UNIV: the universal emission floor)**.** *[FORCED]*/*[ESTABLISHED]* *Every charge-admissible algebraic integer α (all conjugate arguments in $\pi\mathbb{Q}$) has $\mathrm{M}(\alpha) = 1$ (α zero or a root of unity, Kronecker [ESTABLISHED]) or $\mathrm{M}(\alpha) \geq \varphi$. Indeed the conjugates lie on R_m for m the lcm of their angle denominators; least m odd gives $\mathrm{M} \geq 2$ (Theorem 4.17), even gives $\mathrm{M} \geq \varphi$ (Theorem 4.20). This is the whitepaper's universal gap (W-§5.2, W-§22, "over all charge-admissible"), now [FORCED] given Schinzel.* It does not touch Lehmer: *a Salem number is charge*-inadmissible *— an on-circle conjugate with argument in $\pi\mathbb{Q}$ would be a root of unity, forcing the minimal polynomial cyclotomic against $\beta > 1$ — so the gap does not apply to it and the excision (W-Def. 5.1) stands. Likewise the free commutator (W-§5.2) may emit charge-inadmissible outputs, untouched by GAP-UNIV.*

5 The state-angle chart

The follow-on thread observed that each rung forms a unit-circle pair; that observation extends to the entire chart and organizes everything downstream.

Theorem 5.1 (State-angle dictionary). *[FORCED GIVEN D1–D3]* OZ-C1--C6 *For* $z \in (0,1)$ *define the* state angle $\alpha \in (0, \frac{\pi}{2})$ *by*

$$\sin\alpha = z, \qquad \cos\alpha = \sqrt{1-z^2} = e^{-\rho} = \sqrt{|m|},$$

so $\rho = -\ln\cos\alpha$ *and, at the rungs,* $\cos\alpha_n = |W_n| = \tau^n$. *Under this chart the whitepaper's gate dictionary (W-Thm. 13.1) becomes elementary trigonometry:*

$$u = \cot^2\alpha, \qquad C = \cot^2\alpha\,\csc^2\alpha, \qquad \sqrt{D} = 1 + 2\cot^2\alpha,$$
$$m = -\cos^2\alpha, \qquad \lambda = \tan^2\alpha\,\big(1+\cos^2\alpha\big).$$

Proof. Substitute $z = \sin\alpha$, $1-z^2 = \cos^2\alpha$ into $u = (1-z^2)/z^2$, $C = (1-z^2)/z^4$, $\sqrt{D} = (2-z^2)/z^2 = 2\csc^2\alpha - 1$, $m = -(1-z^2)$, and $\lambda = z^2(2-z^2)/(1-z^2)$; each identity is machine-verified symbolically in z. □

Remark 5.2. The pair $(x,y) = (\sqrt{|m|},\, z) = (\cos\alpha, \sin\alpha)$ is exactly the unit-circle coordinate pair: $x^2+y^2 = |m| + z^2 = 1$ chart-wide is D1 itself. The state angle is the latitude (elevation) of the residual sphere of §10.

6 Rung ledgers

Proposition 6.1 (Rung 1: the golden ledger). *[FORCED GIVEN D1–D3]* OZ-B2, OZ-C7, C12--C13 $W_1 = i\tau$, $|W_1| = \tau$, *and* $z_1 = \sqrt{1-\tau^2} = \sqrt{\tau}$ *(via* $\tau^2+\tau = 1$*). The rung-1 state is*

$$\boxed{(\sin\alpha_1,\ \cos\alpha_1,\ \tan\alpha_1) = (\sqrt{\tau},\ \tau,\ \sqrt{\varphi})},$$

with the self-duality $\sin^2\alpha_1 = \tau = \cos\alpha_1$*: the rung-1 state is the positive intersection of the unit circle* $x^2+y^2 = 1$ *with the parabola* $x = y^2$ *in the state plane* $(x,y) = (\sqrt{|m|}, z)$. *Eliminating* x *reproduces* $z_1^4 + z_1^2 - 1 = 0$*, the minimal polynomial* x^4+x^2-1 *of* $\sqrt{\tau}$ *(W-Thm. 17.1). This parabola lives in the residual-state plane and must not be conflated with the inherited horn profile.*

Proposition 6.2 (Rung 2: the K ledger). *[FORCED GIVEN D1–D3]* OZ-B3, OZ-C8, C14 $W_2 = q^2 = -\tau^2$, $|W_2|^2 = \tau^4 = \mathrm{gap}$, *and*

$$z_2 = \sqrt{1-|W_2|^2} = \sqrt{1-\mathrm{gap}} = K, \qquad \textit{i.e.} \qquad \boxed{K = \sqrt{1-\big|\,(i/\varphi)^2\big|^2}}\,:$$

K is the vertical coordinate after two applications of the golden quarter-turn generator. The rung-2 state packages the K-seed's two faces into one right triangle:

$$\boxed{\sin\alpha_2 = K, \qquad \cos\alpha_2 = \tau^2 = \sqrt{\mathrm{gap}}, \qquad \tan\alpha_2 = \beta},$$

since $\tan^2\alpha_2 = K^2/\tau^4 = (1-\mathrm{gap})/\mathrm{gap} = \varphi^4 - 1 = \beta^2$. *The terrain magnitude is the sine, the residual amplitude the cosine, and the rotation magnitude their ratio:* $\tan\alpha_2 = \textit{terrain/residual} = \beta$ *gives* β *a direct geometric role beyond imaginary-root magnitude. The K-gate certificate follows in one line:* $u_2 = \cot^2\alpha_2 = 1/\beta^2$ *and* $C_2 = \cot^2\alpha_2\csc^2\alpha_2 = 1/(\beta^2K^2) = \frac{1}{5}$ *by* $K^2\beta^2 = 5$ *(W-(2.2)). The K-seed carrying these roots is itself a relational catalog object whose angle set is the full charge compass (Prop. 4.3).*

Remark 6.3 (Half-cycle placement of K; the source report's synthesis). [FORCED GIVEN D1–D3] OZ-A6, A8, B3 K appears at the half-cycle of the four-charge return:

$$q^2 = -\tau^2, \qquad |q^2|^2 = \text{gap}, \qquad z_2 = \sqrt{1-|q^2|^2} = K,$$

placing K structurally between the quarter-turn generator and the full gap return. The report's strongest reading, exact given D1–D3 and adopted here:

the ladder is a contraction–rotation orbit whose complete four-charge return is priced by the gap, while K is the vertical complement produced at the two-charge midpoint.

Proposition 6.4 (The general tangent family). *[FORCED GIVEN D1–D3] OZ-C9--C11 For every $n \geq 1$,*

$$\tan\alpha_n = \frac{z_n}{\tau^n} = \sqrt{\varphi^{2n}-1}, \qquad \textit{with Trace–Form-Duality dress} \qquad \tan^2\alpha_n = \begin{cases} \varphi^n L_n, & n \text{ odd}, \\ \varphi^n \sqrt{5}\, F_n, & n \text{ even}. \end{cases}$$

In particular $\tan\alpha_1 = \sqrt{\varphi L_1} = \sqrt{\varphi}$, $\tan\alpha_2 = \sqrt{\varphi^2\sqrt{5}\,F_2} = \beta$, *and* $\tan\alpha_3 = \sqrt{\varphi^3 L_3} = 2\varphi^{3/2}$ *(since $\varphi^6 - 1 = 4\varphi^3$).*

rung n	$\sin\alpha_n = z_n$	$\cos\alpha_n = \tau^n$	$\tan\alpha_n = \sqrt{\varphi^{2n}-1}$	TFD dress	α_n [COMPUTED]
1	$\sqrt{\tau}$	τ	$\sqrt{\varphi}$	$\sqrt{\varphi L_1}$	51.8272923730°
2	K	$\tau^2 = \sqrt{\text{gap}}$	β	$\sqrt{\varphi^2\sqrt{5}F_2}$	67.5444848014°
3	$\sqrt{1-\varphi^{-6}}$	τ^3	$2\varphi^{3/2}$	$\sqrt{\varphi^3 L_3}$	76.3454152540°
4	$K_2 = \sqrt{1-\text{gap}^2}$	$\tau^4 = \text{gap}$	$\sqrt{3}\,\varphi\beta$	$\sqrt{\varphi^4\sqrt{5}F_4}$	81.6107142142°

Remark 6.5 (The near-miss angle finds a home). [FORCED GIVEN D1–D3] The whitepaper's Guard 15.4 uses $\theta_K = \arcsin(\tau^2) = 22.4555151986°$ purely as a near-miss reference (pitch angle $\neq \theta_K$; still no identity is claimed there). On the state chart it is exactly the complement of the rung-2 state angle, and on the residual sphere of §10 it is the K-point's *colatitude*:

$$\theta_K = \tfrac{\pi}{2} - \alpha_2 = \arccos K \qquad (\text{since } \cos\theta_K = \sqrt{1-\tau^4} = \sqrt{1-\text{gap}} = K).$$

7 Bridges between rung 1 and rung 2

Proposition 7.1 (Cosine contraction, not angle doubling). *[FORCED GIVEN D1–D3] OZ-D1--D3b For all n,*

$$\cos\alpha_{n+1} = \tau\,\cos\alpha_n, \qquad \cos^2\alpha_{n+2} = \text{gap}\,\cos^2\alpha_n;$$

in particular $\cos\alpha_2 = \tau^2 = \cos^2\alpha_1$: the rung-1→2 transition squares the residual coordinate and renormalizes the vertical complement. The ladder is driven by geometric contraction of the cosine (residual) coordinate, which forces the angle monotonically toward $\pi/2$; it is not *ordinary angle doubling: $\cos 2\alpha_1 - \cos\alpha_2 = (2\tau^2-1) - \tau^2 = \tau^2 - 1 = -\tau \neq 0$, so $\alpha_2 \neq 2\alpha_1$ (numerically $2\alpha_1 = 103.65°$ vs $\alpha_2 = 67.54°$ [COMPUTED]).*

Proposition 7.2 (Half-angle/full-angle bridge). *[FORCED GIVEN D1–D3] OZ-D4--D6 Since $\cos\alpha_2 = \tau^2$,*

$$\sin^2\frac{\alpha_2}{2} = \frac{1-\tau^2}{2} = \frac{\tau}{2}, \qquad \textit{hence} \qquad \boxed{z_1 = \sqrt{\tau} = \sqrt{2}\,\sin\frac{\alpha_2}{2}, \qquad K = \sin\alpha_2}.$$

The two low rungs are linked by an exact scaled half-angle/full-angle relation while remaining algebraically distinct quartic landmarks ($\mathbb{Q}(\sqrt{\varphi}) \neq \mathbb{Q}(5^{1/4})$, W-Thm. 17.1). The auxiliary angle Θ of the review thread ($\cos\Theta = 1 - z_1^2 = \tau^2$) is *$\alpha_2$; and Route 5 becomes trigonometric: $K^2 = z_1^2(2 - z_1^2) = \sin^2\alpha_1(2 - \sin^2\alpha_1)$.*

8 The K-family and the even-rung recurrence

Theorem 8.1 (K-family)**.** *[FORCED GIVEN D1–D3]* OZ-F1--F5 *Exact K occurs exactly once: z_n is strictly increasing ($z_{n+1}^2 - z_n^2 = \tau\,\varphi^{-2n} > 0$), so*

$$\boxed{z_n = K \iff n = 2}.$$

But every even rung repeats K's formula*: with $n = 2j$,*

$$\boxed{z_{2j} = \sqrt{1-\mathrm{gap}^{\,j}} \;=: K_j, \qquad K_1 = K}, \qquad m_{2j} = -\mathrm{gap}^{\,j}, \qquad z_{2j}^2 = 1 - |m_{2j}|,$$

with odd companion $z_{2j+1} = \sqrt{1-\tau^2\mathrm{gap}^{\,j}}$, and the recurrence holds universally, for all *n, not only even ones:*

$$\boxed{1 - z_{n+2}^2 = \mathrm{gap}\left(1 - z_n^2\right)}.$$

Angularly, $\theta_{2j} = j\pi$ (all even rungs on the terrain axis, alternating direction), while K's exact compass direction $\theta \equiv \pi \pmod{2\pi}$ recurs at $n \equiv 2 \pmod 4$. Three levels of recurrence:

exact value K	*rung 2 only*
K-family / complement structure $K_j = \sqrt{1-\mathrm{gap}^{\,j}}$	*every even rung*
same compass direction as K ($\theta = \pi \bmod 2\pi$)	$n = 2, 6, 10, \ldots$

First members [COMPUTED] OZ-H1*: $K_1 = 0.9241763718$, $K_2 = 0.9892996329$, $K_3 = 0.9984459825$, $K_4 = 0.9997734224$ (these are the whitepaper's rungs $2, 4, 6, 8$; the review thread's 6-dp table values are floor-truncations, see Finding 12.2).*

Proof. $\varphi^{-4j} = \mathrm{gap}^{\,j}$ and $\varphi^{-2(n+2)} = \mathrm{gap}\,\varphi^{-2n}$; monotonicity from $1 - \varphi^{-2} = \tau$; all machine-verified for $j \le 6$ and symbolically in n. ☐

The right reading, adopted from the review thread: *K is not repeated every second rung; its generative operation is.* The operation residual $\mapsto$ gap $\times$ residual generates the even-rung subsequence with K as its first nontrivial realization — every second rung is a higher-order K-formation in the complement structure, while the exact value and the exact quartic field of K belong to rung 2 alone.

Proposition 8.2 (Tie to the K-point linearization)**.** *[FORCED GIVEN D1–D3]* OZ-F6 *The gap paper's derivative $f'(K) = -\mathrm{gap}$ (W-Prop. 9.1) satisfies*

$$f'(K) = -\mathrm{gap} = q^2\,|q|^2,$$

and the linearized K-orbit $x_{k+1} = -\mathrm{gap}\,x_k$ performs per step exactly what the ladder performs per two rungs: co-intensity scales by gap *($|W_{n+2}|^2 = \mathrm{gap}\,|W_n|^2$) and the axis flips ($W_{n+2} = -\sqrt{\mathrm{gap}}\,W_n$). The review thread's "strongly resembles" is thus an exact correspondence at the level of (contraction of squared amplitude, axis action); the two objects remain different curves (transverse linearization at one landmark vs the global rung orbit), as the whitepaper's §15 already insists.*

Theorem 8.3 (The K-family is a seed ladder)**.** *[FORCED]* QX-C1--C13 *For $j \ge 1$ let $K_j = \sqrt{1-\mathrm{gap}^{\,j}}$ (the K-family value of Theorem 8.1, taken here purely as seed arithmetic) and $\beta_j := K_j\varphi^{2j}$, so $\beta_j^2 = \varphi^{4j} - 1$. Then:*

(a) ***Minimal polynomial.*** *The four numbers* $\{\pm K_j,\ \pm i\beta_j\}$ *are the roots of*

$$\boxed{p_j(x) = x^4 + 5F_{2j}^2\,x^2 - 5F_{2j}^2}, \qquad 5F_{2j}^2 = L_{4j} - 2,$$

which is the minimal polynomial of K_j *for* every j*: Eisenstein at* 5 *applies iff* $5 \nmid j$ *(since* $5 \mid F_{2j} \iff 5 \mid j$*), and uniformly the norm obstruction* $N(1 - \mathrm{gap}^j) = 2 - L_{4j} = -5F_{2j}^2 < 0$ *shows* $1 - \mathrm{gap}^j$ *is not a square in* $\mathbb{Q}(\sqrt{5})$*, so* $[\mathbb{Q}(K_j) : \mathbb{Q}] = 4$ *always.* p_1 *is the K-seed.*

(b) ***Ledger*** *(all decided in* $\mathbb{Q}(\sqrt{5})$*):*

$$\beta_j^2 - K_j^2 = K_j^2\beta_j^2 = L_{4j} - 2 = 5F_{2j}^2, \qquad K_j^2 + \beta_j^2 = \sqrt{5}\,F_{4j}, \qquad (K_j\varphi^j)^2 = \sqrt{5}\,F_{2j},$$

$$K_j^4 = 5F_{2j}^2\,\mathrm{gap}^j, \qquad \beta_j^2 = \sqrt{5}\,F_{2j}\,\varphi^{2j}, \qquad \frac{1}{K_j^2} - \frac{1}{\beta_j^2} = 1,$$

generalizing $(K\varphi)^4 = 5$*,* $K^2\beta^2 = 5$*,* $K^4 = 5\mathrm{gap}$ *(*$j = 1$*,* $F_2 = 1$*). The increments telescope like the transfer ledger:* $K_{j+1}^2 - K_j^2 = K^2\mathrm{gap}^j$ *with* $\sum_{j\ge0} K^2\mathrm{gap}^j = 1$ *— the even-rung unit budget is priced in gap powers at unit rate* K^2*.*

(c) ***Catalog data.*** *Angle set the full compass* $\{0, \frac14, \frac12, \frac34\}$*,* $\Delta = \mathbb{Z}/4\mathbb{Z}$*, anchor* $n = m = 4$ *(transport-fixed by Theorem 4.8); contact signature* $\{\Phi_1^4, \Phi_2^4\}$ *(the eight cross-shell ratios have modulus* $(K_j/\beta_j)^{\pm1} \neq 1$*); and* $\mathrm{M}(p_j) = \beta_j^2$*: every* p_j *is a straddler, real pair strictly inside the unit circle, imaginary pair strictly outside. (The K-seed's own catalog status is Proposition 4.3; the* p_j*,* $j \ge 2$*, are new catalog objects, and no uniqueness is claimed for them here.)*

(d) ***Gate ties*** *(W-Thm. 9.1): at every even rung,*

$$C_{2j} = \frac{1}{5F_{2j}^2} = \frac{1}{|p_j(0)|}, \qquad u_{2j} = \frac{1}{\beta_j^2} = \frac{1}{\mathrm{M}(p_j)}, \qquad m_{2j} = -\mathrm{gap}^j = -\frac{K_j^4}{5F_{2j}^2}:$$

the even gate levels are the reciprocal constant terms, and the even occupations the reciprocal Mahler measures, of the seed family — the K-gate certificate (W-Prop. 10.1) is the $j = 1$ *instance of a family-wide law.*

Proof. $\beta_j^2 = K_j^2\varphi^{4j} = (1-\varphi^{-4j})\varphi^{4j} = \varphi^{4j}-1$. With $G = \varphi^{4j}$: $5F_{2j}^2 = (\varphi^{2j}-\varphi^{-2j})^2 = G-2+G^{-1} = L_{4j}-2$; then $K_j^2 = 1-G^{-1}$ and $-\beta_j^2 = 1-G$ are exactly the two roots of $y^2+5F_{2j}^2\,y-5F_{2j}^2$, giving $p_j(\pm K_j) = p_j(\pm i\beta_j) = 0$. The norm: $N(1-\mathrm{gap}^j) = (1-G^{-1})(1-G) = 2-(G+G^{-1}) = 2-L_{4j}$, and a square in $\mathbb{Q}(\sqrt{5})$ has nonnegative norm. All ledger identities are one-line computations in G; the Fibonacci divisibility, irreducibility for $j \le 8$, the compass/contact scan for $j \le 3$, and the gate ties are machine checks `QX-C1--C13`. □

Corollary 8.4 (Every even rung mirrors its own seed)**.** *[FORCED GIVEN D1–D3]* `QX-C12--C14` *On the chart, the rung-*$2j$ *state triangle is the (terrain, residual, rotation) triangle of its* own *seed* p_j*:*

$$\sin\alpha_{2j} = K_j, \qquad \cos\alpha_{2j} = \tau^{2j}, \qquad \boxed{\tan\alpha_{2j} = \beta_j},$$

generalizing the rung-2 ledger (K, τ^2, β) *of Proposition 6.2 to every even rung; the family identity* $1/K_j^2 - 1/\beta_j^2 = 1$ *is exactly* $\csc^2\alpha_{2j} - \cot^2\alpha_{2j} = 1$*, and* $\beta_j^2 = \sqrt{5}F_{2j}\,\varphi^{2j}$ *recovers the even tangent dress of Proposition 6.4 at* $n = 2j$*.*

Theorem 8.5 (The universal seed ladder (Vein 6)). *[FORCED]* `UX-A2--A4, UX-B1--B6, UX-C1--C2, UX-D1--D4, UX-F1--F3, UX-K1--K2` *Checked exactly for $n = 1..10$ and symbolically in $w = \varphi^{2n}$ — that range is the certified scope; the two uniform mechanisms noted below extend it. For every rung n:*

$$\boxed{\operatorname{minpoly}(z_n) = p_n(x) = x^4 + c_n x^2 - c_n}, \qquad c_n = L_{2n} - 2 = (\varphi^n - \varphi^{-n})^2 = \begin{cases} L_n^2, & n \text{ odd}, \\ 5F_n^2, & n \text{ even}, \end{cases}$$

with roots $\{\pm z_n,\ \pm i\, b_n\}$, $b_n = \sqrt{\varphi^{2n}-1} = \tan\alpha_n$ (Proposition 6.4). Irreducibility is uniform (all $n \geq 1$) by the norm obstruction $N(1-\tau^{2n}) = 2 - L_{2n} = -c_n < 0$, so $1-\tau^{2n}$ is not a $\mathbb{Q}(\sqrt{5})$-square and $[\mathbb{Q}(z_n):\mathbb{Q}] = 4$ always. Ledger:

$$z_n^2\, b_n^2 = c_n = |p_n(0)| \quad \textit{(symbolic in } w\textit{: } (1-1/w)(w-1) = w - 2 + 1/w\textit{)},$$

generalizing $K^2\beta^2 = 5$; $\mathrm{M}(p_n) = b_n^2 = \tan^2\alpha_n$, with exactly the imaginary pair outside the unit circle ($z_n^2 < 1 < b_n^2$, exact signs); and

$$u_n = \frac{1}{\mathrm{M}(p_n)}, \qquad C_n = \frac{1}{|p_n(0)|} \qquad \textit{at every rung}:$$

the tangent dress of Proposition 6.4 is the Mahler ladder*, and Theorem 8.3(d)'s gate ties extend from the even rungs (where p_{2j} is that theorem's p_j and $b_{2j} = \beta_j$) to all rungs. Coefficient recursion: $c_{n+1} = 3c_n - c_{n-1} + 2$ (symbolically via $\varphi^2 + \varphi^{-2} = 3$; integer list $1, 5, 16, 45, 121, 320, 841, 2205, \ldots$). Catalog data: every rung is a full-compass $\mathbb{Z}/4\mathbb{Z}$-completing straddler — angle set $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, $\Delta = \mathbb{Z}/4\mathbb{Z}$, anchor $n = m = 4$ in the relational sense — with $z_n < 1 < b_n$ exact for $n = 1..10$ and the all-n extension carried by the exact atoms $1 - z_n^2 = \tau^{2n} > 0$ and $w \geq \varphi^2 > 2$ (the second uniform mechanism). Anchors: $p_1 = x^4 + x^2 - 1$ is W-Thm. 17.1's* $\operatorname{minpoly}(\sqrt{\tau})$ *(Proposition 6.1); p_2 is the K-seed $x^4 + 5x^2 - 5$; $p_3 = x^4 + 16x^2 - 16$.*

Remark 8.6 (Rung 1 sits on the emission Mahler floor). [FORCED] `UX-E1--E2` $\mathrm{M}(p_1) = \varphi$ exactly ($\varphi^2 - 1 = \varphi$): the rung-1 seed sits precisely on the emission Mahler floor. *The coincidence-candidate discharges (v2.3): the rung-1 seed **is** the parity-floor witness.* By Theorem 8.5, $c_1 = L_2 - 2 = 1$, so $p_1 = x^4 + x^2 - 1$; and that is exactly $q_k = x^{2k} + x^k - 1$ at $k = 2$ (W-Prop. 6.2), the $k = 2$ member of the even parity-floor family. So $\mathrm{M}(p_1) = \mathrm{M}(q_2) = \varphi$ is an *identity*, not a shared constant: the mechanism is W-Prop. 6.2's own (the k roots at $x^k = -\varphi$, each of modulus $\varphi^{1/k}$), and the shared-constant rule is met by *production*, not exclusion — a *tie*, the mirror of a no-tie. The uniqueness on the ladder gains a mechanism too: $\deg p_n = 4$ forces $k = 2$, then $q_2 = x^4 + x^2 - 1$ forces $c_n = 1$, i.e. $L_{2n} = 3$, i.e. $n = 1$ (L_{2n} strictly increasing for $n \geq 1$). The seed ladder and the parity-floor family meet in exactly one polynomial, at rung 1; and $p_1 = x^4 + x^2 - 1$ is W-Thm. 17.1's $\operatorname{minpoly}(\sqrt{\tau})$, so $z_1 = \sqrt{\tau}$ is a root of q_2, one of Theorem 4.20's sharpness witnesses. `EV-A1--A4`

Theorem 8.7 (Mahler-minimal uniqueness (Theorem U)). *[FORCED]* `UX-U1--U7` *In the class $\{x^4 + ax^2 - c :\ a \in \mathbb{Z},\ c \in \mathbb{Z},\ c > 0\}$ — where the full compass is automatic (the y-quadratic $y^2 + ay - c$ has $y_- < 0 < y_+$ since $y_+ y_- = -c < 0$):*

- *(i) full-compass straddler $\iff$ $p(1) > 0$ $\iff$ $a \geq c$ (via $(1-y_+)(1-y_-) = p(1)$ and $(1+y_+)(1+y_-) = 1 - a - c$; the* integer *hypothesis is load-bearing — over $\mathbb{Q}$ the equivalence fails, guarded witness $(a, c) = (\frac{1}{2}, \frac{1}{10})$);*
- *(ii)* $\mathrm{M} = |y_-| = \dfrac{a + \sqrt{a^2 + 4c}}{2}$ *is strictly increasing in a at fixed c;*
- *(iii) hence the* unique *Mahler-minimal compass straddler at level c is $x^4 + cx^2 - c$ — which at $c = 5F_{2j}^2$ is p_j (Theorem 8.3) and at every rung level c_n is p_n (Theorem 8.5), by the exact endpoint identity $c_n^2 + 4c_n = 5F_{2n}^2$, giving $\mathrm{M}(x^4 + c_n x^2 - c_n) = b_n^2$.*

Remark 8.8 (The F1 falsifier; register resolution). [FORCED] UX-H1--H4 Constant-term-only uniqueness is *false*: $x^4 + 6x^2 - 5$ is a second irreducible full-compass straddler at level $|p(0)| = 5$, distinct from the K-seed (difference exactly x^2), with $\mathrm{M}(x^4 + 6x^2 - 5) = 3 + \sqrt{14} > \varphi^4 - 1 = \mathrm{M}(\text{K-seed})$ (integer comparison $56 > 45$) — consistent with Theorem 8.7(ii), which places the K-seed at the window edge $a = c$. The v1.6 uniqueness question left open at Theorem 8.3(c) ("no uniqueness is claimed"), read as F3 — Mahler-minimal at fixed constant term within the compass type, confirmed as the intended reading by the project owner on 2026-07-15 — is thereby *resolved*: the p_j, and n-uniformly the p_n, are the unique Mahler-minimal full-compass straddlers at their levels.

9 The lens as the exact $60°$ state

Proposition 9.1. *[FORCED GIVEN D1–D3] OZ-E1--E6 Under $z = \sin\alpha$, the lens height $z_c = \sqrt{3}/2$ is the state $\alpha_c = \pi/3 = 60°$ with residual amplitude $\cos\alpha_c = \frac{1}{2}$; the complete split-gate ledger of W-Thm. 13.3 drops out of Theorem 5.1:*

$$m_c = -\cos^2 60° = -\tfrac{1}{4}, \quad u_c = \cot^2 60° = \tfrac{1}{3}, \quad C_c = \cot^2\csc^2 60° = \tfrac{4}{9}, \quad \sqrt{D_c} = 1 + 2\cot^2 60° = \tfrac{5}{3}.$$

A rung at the lens would require $\cos\alpha_c = \tau^n$, i.e. $\tau^n = \frac{1}{2}$, i.e. $n = \log_\varphi 2$; by W-Thm. 16.1 this is irrational, so the $60°$ lens is permanently off the discrete rung lattice (finite guard $n \le 10$ replicated).

Proposition 9.2 (Lens scoping lemma: splitness over $\mathbb{Q}$ is generic)**.** *[FORCED] LX-A1--A8, LX-B1--B6, LX-F1--F4 The chart pentad lies in $\mathbb{Q}(Z)$, $Z = z^2$: $u = (1-Z)/Z$, $C = (1-Z)/Z^2$, $\sqrt{D} = (2-Z)/Z$, $m = -(1-Z)$, $\lambda = Z(2-Z)/(1-Z)$, with denominator zeros only at $Z \in \{0, 1\}$. Hence* every *rational height $Z \in \mathbb{Q} \cap (0,1)$ gives $C, \sqrt{D} \in \mathbb{Q}$ and a gate $x^2 + x - C$ that splits over $\mathbb{Q}$, with a rational reciprocal multiplier pair. Example $z^2 = \frac{1}{2}$: $C = 2$, $D = 9$, gate $x^2 + x - 2 = (x-1)(x+2)$, multiplier pair $\{-\frac{1}{2}, -2\}$, at rapidity $\rho = \frac{1}{2}\ln 2 \in (\ln 2)\mathbb{Q}$, strictly below rung 1 — a rational split gate that is not the lens. Splitness over $\mathbb{Q}$ is therefore* generic *on rational heights, and the lens's distinction is not splitness but its exact data: $z_c = \sqrt{3}/2$ ($Z = \frac{3}{4}$), $C = \frac{4}{9}$, $\sqrt{D} = \frac{5}{3}$, fixed points $\{\frac{1}{3}, -\frac{4}{3}\}$, multiplier pair $\{-\frac{1}{4}, -4\}$, $\lambda = \frac{15}{4}$, co-intensity $\frac{1}{4}$, and $\rho = \ln 2$ exactly. (The converse fails: rung 2 has $C_2 = \frac{1}{5} \in \mathbb{Q}$ with $z_2^2 = K^2 \notin \mathbb{Q}$, and at rung heights $\sqrt{D}$ is irrational — the membership detector rejects the odd object $\sqrt{Z}$.)*

Remark 9.3 (Bounded-negative scan for the lens constants). [OPEN] LX-C1--C8, LX-S1--S7 Candidate-level, citable as a *bounded* negative. A predicate-first scan of the seven shipped corpus documents — the six harness-scanned domains (ECHO-S-RESEARCH_UNIFIED, K-VARIABLE-DERIVATIONS, HELIX-EXPLICIT-DERIVATION, this paper's v1.6 source, the gap-multiplier paper, the whitepaper PDF) plus `K_seven_ways.tex`, swept independently in verification with the same outcome — over eight constant classes ($\frac{4}{9}$, $\frac{25}{9}$, $\frac{5}{3}$, $\frac{15}{4}$, $-\frac{1}{4}$, the pair $\{-\frac{1}{4}, -4\}$ *as a pair*, dyadic 2^{-k} co-intensities, $(\ln 2)\mathbb{Q}$ rapidities) finds *no* foreign-context occurrence of a lens-distinctive constant that survives the shared-constant rule. The single foreign occurrence of a lens *value* — $\frac{5}{3}$ in the ECHO trace form $\mathrm{Tr}(1 - \frac{1}{3}\varphi) = \frac{5}{3}$ — has its own mechanism (trace-form projection) sharing no step with the lens discriminant: a no-tie. The remaining candidate hits each carry their own mechanism: the $\ln 2$-lattice $\ln \mathrm{M}(\text{cons}) = \ln 2 + 5\ln\varphi$ (W-Thm. 18.1(a), significance [OPEN] upstream, not a $(\ln 2)\mathbb{Q}$ rapidity); $N(\mathrm{M}(\text{cons})) = -4$, forced by the lattice exponent, hence the same hit; the flip level $-\frac{1}{4}$ ($D = 0 \iff C = -\frac{1}{4}$, unique), which restates the inherited $z_c^2 = \frac{3}{4}$ datum; the dyadic $\frac{1}{16} = 1/L_3^2$, Lucas-driven; and $-\frac{1}{2}$, the rotation gate forced by multipliers $\pm i$. The whitepaper's open item 3 (does the lens gate $C = \frac{4}{9}$ appear elsewhere?) closes as a bounded negative at candidate level; the scanned domains and constant classes are enumerated above precisely so that the negative is citable. *The one branch it names by example — the emission-algebra audit — sharpens to [FORCED] in Remark 9.4.*

Remark 9.4 (The emission-eigenvalue branch of W[OPEN]3 closes forced (v2.4)). [FORCED] `AX-1--AX-4` Where Remark 9.3's bounded negative is the terminal state of the *"appears anywhere"* reading over an inherently bounded domain, the one branch W-§23.3 names *by example* — the *emission-algebra* audit — closes *forced*, by the valuation mechanism of Proposition 3.8 (there run at the prime 5) applied at the prime 3. Among the catalog $\{\varphi, \tau, \sqrt{2}, \sqrt{3}, \sqrt{5}, \mathrm{gap}, K\}$ the *only* carrier of 3 is the $\sqrt{3}$ companion (eigenvalues $\pm\sqrt{3}$, 3-content $w_1 = +2 > 0$); every emitted eigenvalue is a monomial in the catalog eigenvalues ($\otimes$ multiplies moduli, ψ^2 squares, $\oplus$ unions, minpoly/Φ select, Proposition 3.10), so its 3-content is a sum of non-negative contributions, $w_1 \geq 0$; and $\frac{4}{9} = 2^2 \cdot 3^{-2}$ has $w_1 = -8 < 0$ (plain $v_3 = -2$). Hence $\frac{4}{9}$ is *not an emission eigenvalue* [FORCED] `AX-3`. **Bridge.** $\frac{4}{9}$ is a *gate level* C ($f_C(x) = C/(1+x)$), not *a priori* an eigenvalue; but were it an emission eigenvalue its modulus would be $\frac{4}{9}$, excluded by $w_1 < 0$. So the branch W-§23.3 names by example is forced-negative; the *ratio-invariant* branch (is $\frac{4}{9}$ a ratio invariant of a catalog object?) is a finite computation, unrun; and the *"anywhere"* branch is Remark 9.3's bounded negative, its correct terminal state. The mirror of Proposition 3.8 is exact ($w_2 \to w_1$, $5 \to 3$); the discriminating check is that $3 = \psi^2(\sqrt{3})$ has $w_1 = +4 \geq 0$ and is *not* excluded, while $\frac{4}{9}$ is `AX-4`.

10 The residual sphere

Theorem 10.1 (Residual sphere)**.** *[FORCED GIVEN D1–D3]* `OZ-G1--G4` *Define*

$$S(\rho) = \big(\operatorname{Re} W(\rho),\ \operatorname{Im} W(\rho),\ z(\rho)\big).$$

Then

$$|S(\rho)|^2 = |W(\rho)|^2 + z(\rho)^2 = e^{-2\rho} + \big(1 - e^{-2\rho}\big) = 1$$

for every $\rho \geq 0$: the entire trajectory lives on the unit sphere S^2 (northern hemisphere), with the state angle α as latitude. At the rungs,

$$S_n = \big(\operatorname{Re}(i\tau)^n,\ \operatorname{Im}(i\tau)^n,\ \sqrt{1 - \tau^{2n}}\big),$$

explicitly $S_0 = (1,0,0)$, $S_1 = (0, \tau, \sqrt{\tau})$, $S_2 = (-\tau^2, 0, K)$, $S_3 = (0, -\tau^3, z_3)$, $S_4 = (\mathrm{gap}, 0, z_4)$: a discrete compass walk — longitude $+90°$ per rung, horizontal radius $\times\tau$ per rung, height $\uparrow 1$ — approaching the north pole without reaching it at any finite rung; by Definition 3.6 the horizontal part is exactly the Q-orbit, $S_n = (Q^n v_0\,;\ z_n)$ — a compass walk on the same four directions that carry the K-seed's own conjugates (Prop. 4.3). The K-point sits at colatitude $\arccos K = \theta_K$ *(Remark 6.5) on the negative real meridian.*

This object should be called the *residual sphere* (equivalently co-intensity sphere), never the formation helix:

Proposition 10.2 (Two radii, two layers)**.** *[FORCED]/[INHERITED]* `OZ-G5` *The residual radius $a(z) = \sqrt{1 - z^2} = \cos\alpha$ decreases monotonically to 0; the inherited formation radius*

$$r(z) = \begin{cases} K\sqrt{z/z_c}, & 0 \leq z \leq z_c, \\ K, & z_c \leq z \leq 1, \end{cases}$$

grows to K and locks (W-(11.1), [INHERITED]). These are not *the same radius; they describe two layers — remaining residual amplitude vs formation envelope. The dual reading ("residual capacity* *contracts while the formation envelope expands and stabilizes") is a provisional interpretation, not a theorem. No result here or in the sources derives the paraboloid–cylinder profile from the residual-sphere construction; that derivation is exactly the whitepaper's [OPEN] 1 and, as both sources conclude, the priority next problem. Until that bridge exists, the residual spiral and the explicit formation helix are two coordinated representations, not one proven geometry.*

Proposition 10.3 (Compass-cross characterization of the rung locus). [FORCED GIVEN D1–D3] NX-S1--S2 *Let $C_{\mathrm{EW}} = S^2 \cap \{y = 0\}$ and $C_{\mathrm{NS}} = S^2 \cap \{x = 0\}$ be the two polar great circles through the compass axes. For $\rho \geq 0$,*

$$S(\rho) \in C_{\mathrm{EW}} \cup C_{\mathrm{NS}} \iff \rho \in (\ln \varphi)\, \mathbb{Z}_{\geq 0},$$

with $S(\rho_n) \in C_{\mathrm{EW}}$ for even n and $S(\rho_n) \in C_{\mathrm{NS}}$ for odd n; the poles (the circles' intersection) are never attained. Moreover, writing $S = (x, y, z)$,

$$\operatorname{Im} \nu \;=\; \frac{2xy}{x^2 + y^2}$$

identically: on the residual sphere the ν-criterion of Proposition 11.3 is literally the vanishing of the product of the horizontal coordinates, and the rungs are the crossings of the residual spiral with the compass cross. At the half-floors $\rho = (n + \frac{1}{2}) \ln \varphi$ the miss is exact: $|xy| = e^{-2\rho}/2 > 0$.

Proof. $y = 0 \iff \sin \kappa\rho = 0 \iff \rho \in (2 \ln \varphi)\mathbb{Z}$ and $x = 0 \iff \cos \kappa\rho = 0 \iff \rho \in \ln \varphi + (2 \ln \varphi)\mathbb{Z}$ [D3]; the union is the rung lattice, split by parity. With $x = e^{-\rho} \cos \theta$, $y = e^{-\rho} \sin \theta$: $2xy/(x^2 + y^2) = \sin 2\theta = \operatorname{Im} \nu$. Both identities and the half-floor value are machine-verified symbolically. □

Proposition 10.4 (Spherical step ledger). [FORCED GIVEN D1–D3] NX-S3--S5 *Consecutive rung positions have orthogonal horizontal parts:* $\operatorname{Re}\big(W_n \overline{W_{n+1}}\big) = 0$ *for every n, so*

$$S_n \cdot S_{n+1} = z_n z_{n+1}, \qquad d_n := d_{S^2}(S_n, S_{n+1}) = \arccos\big(z_n z_{n+1}\big).$$

In particular $d_0 = \pi/2$ exactly — the first step, apex to rung 1, is a full quarter great circle — and $d_1 = \arccos\big(5^{1/4}\tau^{3/2}\big) = 43.4026803146\ldots^\circ$ [COMPUTED] (from $z_1 z_2 = \sqrt{\tau}\, K = 5^{1/4}\tau^{3/2}$); thereafter d_n decreases strictly to 0 ($z_n z_{n+1} \uparrow 1$). On the residual sphere the walk takes ever-shorter geodesic steps, in exact contrast to the formation rail, whose per-rung length is floored at $\pi K/2$ (Proposition 11.2).

Proof. $W_n \overline{W_{n+1}} = \overline{q}\, |W_n|^2 = -i\, \tau^{2n+1}$ is purely imaginary, so the dot product reduces to the vertical product; strict monotonicity of $z_n z_{n+1}$ is decided exactly at the squared level in $\mathbb{Q}(\sqrt{5})$; the d_1 evaluation is display-only. □

Proposition 10.5 (Finite spherical length and the length dichotomy). [FORCED GIVEN D1–D3] NX-S6--S7, NX-H1--H3 *The residual spiral's speed on its sphere is*

$$\big|S'(\rho)\big|^2 = e^{-2\rho}\Big(\kappa^2 + \frac{1}{1 - e^{-2\rho}}\Big) \qquad \Big(= \tan^2 \vartheta + \kappa^2 \sin^2 \vartheta \ \textit{at colatitude}\ \sin \vartheta = e^{-\rho}\Big),$$

so its total spherical length from apex to pole is finite:

$$\boxed{L_{\mathrm{res}} = \int_0^1 \sqrt{\kappa^2 + \frac{1}{1 - a^2}}\, da = \sqrt{1 + \kappa^2}\; E\Big(\frac{\kappa^2}{1 + \kappa^2}\Big)} = 3.7312193125\ldots \ \text{[COMPUTED]}$$

(E the complete elliptic integral of the second kind [ESTABLISHED]; the value is the quarter-perimeter of an ellipse with semi-axes $\sqrt{1+\kappa^2}$ and 1 — the same speed factor as the per-floor planar length of Corollary 2.5(iv)), with exact bounds $\sqrt{1+\kappa^2} \le L_{\mathrm{res}} \le \kappa + \pi/2$. The two-layer separation of Proposition 10.2 is therefore a length dichotomy*: the residual layer has finite total spherical length, while the formation rail's length diverges (Proposition 11.2). Finally, L_{res} is strictly increasing in $|\kappa|$ (the integrand is pointwise strictly increasing in κ^2), so within the interpolation family of Theorem 2.4 the principal branch $k = 0$ is also the* unique minimizer of total spherical length*: a fifth equivalent characterization to add to Corollary 2.5's four.*

Proof. Differentiating $S = (e^{-\rho}\cos\kappa\rho,\ e^{-\rho}\sin\kappa\rho,\ \sqrt{1-e^{-2\rho}})$ gives the speed identity symbolically. Substituting $a = e^{-\rho}$ turns the length integral into $\int_0^1$; then $a = \sin u$ gives $\sqrt{\kappa^2 + 1/\cos^2 u}\ \cos u = \sqrt{1+\kappa^2\cos^2 u}$ (exact at the squared level, $\cos u > 0$), and $(1+\kappa^2)(1-m\sin^2 u) = 1+\kappa^2\cos^2 u$ with $m = \kappa^2/(1+\kappa^2)$ identifies $\int_0^{\pi/2}\sqrt{1+\kappa^2\cos^2 u}\,du = \sqrt{1+\kappa^2}\,E(m)$ [ESTABLISHED]. Bounds: pointwise $\sqrt{\kappa^2+g} \ge \sqrt{\kappa^2+1}$ ($g = 1/(1-a^2) \ge 1$) and $\sqrt{\kappa^2+g} \le \kappa + \sqrt{g}$, integrated ($\int_0^1 da/\sqrt{1-a^2} = \pi/2$). The family statement follows from pointwise strict monotonicity in κ^2 together with Corollary 2.5(iii): $|\kappa_k|$ is uniquely minimized at $k = 0$. The numeric value is display-only, corroborated at 50 dps by quadrature against the closed form. □

Theorem 10.6 (RAD-0: the parity no-go for the inherited radius)**.** *[FORCED]* `RX-A1--A4, RX-P1--P9` *Work over the chart layer $\{u, C, \sqrt{D}, m, \lambda, \rho, \theta_k\}$ with height coordinate z. Every chart-pentad member lies in $\mathbb{Q}(\sqrt{5})(z^2)$, and the chart-rational closure is* exactly *$\mathbb{Q}(\sqrt{5})(z^2)$ (sharpness: $z^2 = 1/(1+u)$, so both inclusions hold); moreover $\rho = -\frac{1}{2}\ln(1-z^2)$ and $\theta_k = \kappa_k\rho$ are even in z for* every *branch $k \in \mathbb{Z}$ (symbolic k), and adjoining them keeps the generated field inside the even functions, over arbitrary constant coefficients. The inherited radius is odd: $r^2 = K^2 z/z_c$ satisfies $r^2(-z) = -r^2(z)$ identically and is nonzero (exact witness at $z = \frac{1}{2}$). Hence r^2 — and a fortiori r — lies* outside *the rational closure of the entire D1–D3 chart layer, and*

$$\boxed{\textit{any derivation bridge from the substrate to } r \textit{ must break the } z \mapsto -z \textit{ symmetry}}.$$

$\mathbb{Z}/2$ sharpening: $r^4 = K^4 z^2/z_c^2$ is *chart-even, so r^2 satisfies $X^2 - r^4$ over the even field F and the extension $F(r^2)/F$ is exactly quadratic — the entire missing content of the radius problem is one parity-breaking generator, a degree-2/double-cover object (indeed $F(r^2) = \mathbb{Q}(\sqrt{5})(z)$, since $z = r^2 z_c/K^2$; and even $\times\, r^2$ stays odd, a $\mathbb{Z}/2$ grading). The whitepaper's [OPEN] 1 (OP-RADIUS) remains open: this is a* forced *necessary condition* on any solution, not a solution.*

Proposition 10.7 (RAD-2: the area-transfer law)**.** *[FORCED]/[INHERITED]* `RX-T1--T4` *Let $A_{\mathrm{zone}}(z) = 2\pi z$ be the Archimedes zone area of the unit sphere between heights 0 and z — re-derived exactly here, not cited: the surface-of-revolution identity $(2\pi\sqrt{1-h^2})^2\,(1+x'(h)^2) = (2\pi)^2$ holds identically at the squared (sign-free) level. Then on $[0, z_c]$ the inherited profile is equivalent, in both directions, to*

$$\boxed{\pi\, r(z)^2 = \frac{K^2}{2z_c}\, A_{\mathrm{zone}}(z)} :$$

the radius law says exactly that formation deposits disc area at the constant rate $d(\pi r^2)/dz = \pi K^2/z_c$ per unit height, stopping at the lens (continuity at the lens: $\pi r(z_c)^2 = \pi K^2$, the cap disc; the C^0 kink survives). This is a forced reformulation *of the inherited profile — a restated target for the radius problem, not progress on it; any future deposition mechanism must be premise-audited for a smuggled profile.*

Remark 10.8 (RAD-4: all three equipotential clocks fail). [FORCED]/[ESTABLISHED] `RX-E1--E7`, `RX-G0--G3` The horn $z = (z_c/K^2)r^2$ coincides with a rigid-rotation free surface $z = (\omega^2/2g)\,r^2$ iff $g_{\text{eff}} = \omega^2 K^2/(2z_c)$ (perturbed coefficients rejected). All three natural clocks fail. (i) ρ-clock ($\omega = \kappa$, the D3 clock): $g_1 = \kappa^2 K^2/(2z_c) = \pi^2 K^2/(8z_c \ln^2 \varphi) = 5.2543169478\ldots$ [COMPUTED] is transcendental given Gelfond–Schneider (Theorem 2.8), so it ties to *no* algebraic ledger constant — all branches k at once, since $\kappa_k = (4k+1)\kappa$; against the transcendental ledger entries the no-tie is certified by disjoint 60-dps intervals (a rigorous proof of $\neq$), closest misses $\frac{21}{4}$ (margin 4.3×10^{-3}) and $2\varphi^2$ (1.8×10^{-2}). (ii) s-clock ($\omega = 1/K$, the forced asymptotic rail clock, $ds/d\theta \to K$): K^2 cancels identically and $g_2 = 1/(2z_c)$ is the horn coefficient itself read back — a *restatement*, not a tie (the clock smuggles the profile's K); shared-constant rule applied. (iii) z-clock: no constant per-height clock exists — $d\theta/dz = \kappa z/(1-z^2)$ is non-constant (forced, exactly nonzero derivative) — and distinguished-height evaluations (e.g. $\omega(z_c) = 2\sqrt{3}\,\kappa$, $g = 12g_1$) remain $\kappa^2 \times$ algebraic and fall to the same transcendence exclusion as (i). The mechanism is killed; the radius problem stays open.

OP-RADIUS sharpened at the apex (v1.8). The parity no-go of Theorem 10.6 (which covered only r^2, through the global $z \mapsto -z$ involution) is now localized to the apex $z = 0$ and quantified as a ramification index, and extended to the radical r; a make-or-break constructive bridge is obstructed; and the remaining formation content is isolated as an honest declared negative. OP-RADIUS *stays open*, sharpened.

Theorem 10.9 (Apex-ramification valuation certificate)**.** *[FORCED] `RD-A1--RD-A4`, `RD-V1--RD-V7` On the z-adic valuation v_z at the apex $z = 0$, every chart quantity has* even *valuation: the pentad $v_z(u, C, \sqrt{D}, m, \lambda) = (-2, -4, -2, 0, 2)$ and $v_z(\rho) = v_z(\theta_k) = 2$ for* every *branch k (symbolic k), so the chart field $F = \mathbb{Q}(\sqrt{5})(z^2)$ has z-value group in $2\mathbb{Z}$. The inherited radius has $v_z(r^2) = 1$ (odd, not in $2\mathbb{Z}$) and $v_z(r) = \frac{1}{2}$; hence adjoining r^2 ramifies F at the apex with index $e = 2$ (value group $2\mathbb{Z} \to \mathbb{Z}$), and adjoining r with index $e = 4$ ($2\mathbb{Z} \to \frac{1}{2}\mathbb{Z}$), while even quantities give $e = 1$; the double cover $w = z^2$ is unramified at every finite place except $z = 0$ (branch locus $\{2z = 0\} = \{0\}$). Adjoining the inherited radius therefore changes the apex ramification index from 1 to 2 (for r^2) or 4 (for r) — a local non-membership certificate no even-valuation manipulation can produce. This is strictly sharper than the parity no-go: parity certifies only r^2 (an involution on the rational field, and $r(-z)$ is neither $+r$ nor $-r$), while the valuation certificate covers the radical r and pins the obstruction to the apex. The identification value-group-index = ramification index (Serre,* Local Fields*) is external framing only; every decision is exact leading-term arithmetic. This is a forced* necessary condition *on any substrate-to-r bridge, not a derivation of r (whitepaper [OPEN] 1 remains open).*

Proposition 10.10 (The constructive-bridge obstruction)**.** *[FORCED] `RD-R1--RD-R6` A constructive bridge would have to furnish a square root of the chart-even $r^4 = K^4 z^2/z_c^2$; it cannot from canonical substrate data. Although $r^4 \in F$ (with $v_z = 2$), it is* not a square *in F: its $w = z^2$ valuation is $v_w(r^4) = 1$ (odd), so the square root r^2 lives one level up in $\mathbb{Q}(\sqrt{5})(z)$, and any g with $g^2 = r^4$ has $v_z(g) = 1$ (odd). No even-valuation amplitude can be g: the residual amplitude $a = \sqrt{1-z^2}$ ($v_z = 0$), the* $\mathrm{Cl}(2,0)$ *rotor modulus $|e^{J\theta/2}|^2 = 1$ ($v_z = 0$), and $z\,a$, z^2 all fail exactly. The $\mathbb{Z}/8$ half-angle rotor supplies only a* unimodular *(angular) double cover of the phase, carrying no radial scale: the required radial coefficient K^2/z_c is not unimodular ($(K^2/z_c)^2 = 70/3 - 10\sqrt{5} \neq 1$), so it never appears as a rotor amplitude (a bounded no-tie). The unique-up-to-sign square root is $\pm(K^2/z_c)\,z$ — the* signed height*, exactly the odd orientation generator that Theorem 10.6/Theorem 10.9 says must be adjoined; it is not a canonical substrate amplitude. The extension exists, but the substrate canonically supplies only the even z^2; the missing object is the orientation of the apex-ramified radial cover, which the rotor's (angular, unimodular) spin cover cannot furnish.*

Remark 10.11 (The atom ledger, with the honest cap-onset negative). [FORCED]/[INHERITED] RD-A2, RD-S1--RD-S5 The inherited profile decomposes into atoms of mixed status. *Cap value* [FORCED] *given corpus*: $r(z_c) = K$ — the unique nontrivial height-equals-radius point (K, K), since $r^2 = z^2$ factors $z(z - K^2/z_c) = 0$ with the horn root $K^2/z_c > z_c$ excluded ($K^2 > z_c^2$, i.e. $180 > 169$) (whitepaper Thm. 15.1). *Kink slope* [FORCED] *given corpus*: $r'(z_c^-) = K/\sqrt{3}$ (square $K^2/3 = (3\sqrt{5} - 5)/6 > 0$), against the flat cap $r'(z_c^+) = 0$ — C^0 but not C^1 (whitepaper Prop. 15.2(a)). *Exponent* $\frac{1}{2}$ [FORCED] (a restatement): $r^2 \propto z$ is exactly equivalent to constant areal deposition per unit height (RAD-2, Proposition 10.7), not a derivation of the exponent. *But the kink existence — the cap onset — is [DECLARED], not forced*: the $\ln 2$-rapidity landmark $\rho(z_c) = \ln 2$ (and $a(z_c) = \frac{1}{2}$) is a monotone-bijection restatement of the declared lens $z_c = \sqrt{3}/2$ (since $d\rho/dz = z/(1 - z^2) > 0$, so $\rho = \ln 2 \iff z^2 = \frac{3}{4}$), so "formation stops at the first $\ln 2$ landmark" carries *no independent content*. As a derivation of the cap onset this *fails* (reason class: restatement); the exact landmark facts are all individually forced, but the cap onset itself is the genuine remaining open formation content.

Theorem 10.12 (OP-RADIUS reduced to the declared atom D4)**.** *[FORCED] RA-A1--RA-A4, RA-B1--RA-B8, RA-G2 Define the atom*

$$D4 = (i) \text{ an orientation of the apex-ramified double cover } F(r)/F \quad \wedge \quad (ii) \text{ the cap onset,}$$

where (i) fixes the sign of the odd generator $\pm(K^2/z_c)\,z$ — the signed height, the unique-up-to-sign odd generator of Proposition 10.10 — and (ii) fixes the join point (formation completes at the lens z_c, Remark 10.11). Then, given *D4 and the inherited chart-ties — exponent $\frac{1}{2}$ (area transfer, Proposition 10.7), cap value K (the (K, K) point, whitepaper Thm. 15.1), kink slope $K/\sqrt{3}$ (whitepaper Prop. 15.2(a)), all v1.8-forced — the profile*

$$r(z) = K\sqrt{z/z_c} \ \ on\ [0, z_c], \qquad r(z) = K \ \ on\ [z_c, 1],$$

is uniquely *determined, and each half of D4 is* necessary*: dropping (i) leaves both $\pm r$ chart-consistent (the tie r^2, $|\mathrm{cap}| = K$ is sign-blind, yet $2K \neq 0$); dropping (ii) leaves the join free (an alternate onset $z_0 = \frac{4}{5}$ gives a chart-consistent capped paraboloid distinct from the inherited one, with separation exhibited at $z^* = \frac{5}{6}$).*

Machine discharge. Exact reconstruction and the two minimality witnesses are checked over $\mathbb{Q}(\sqrt{5})$ extended by $\sqrt{3}$ RA-B2, RA-B4, RA-B5, RA-G2; the reduction ledger (exponent, cap value, kink slope chart-tied; only sign and onset free) is RA-B7, and de-restatement (D4 adds only a $\mathbb{Z}/2$ sign plus one onset value, r^2 being chart-odd hence unreachable from D1–D3) is RA-B8. ☐

Honest scope. This reduces OP-RADIUS to the declared atom D4 *relative to* the inherited chart-ties; it is *not* an ab-initio derivation of r and does *not* close whitepaper [OPEN] 1. The exponent tie in particular is itself a restatement of the inherited profile (Proposition 10.7), not an independent derivation of the exponent.

Remark 10.13 (D4 is minimal-among-its-halves but not a unique decomposition). [COMPUTED] RA-B6, RA-B7 D4 is minimal among its two halves, but the decomposition is *not* unique: the kink slope $K/(2z_0)$ is a strictly monotone (hence bijective) function of the onset z_0, so "kink-slope value + sign" is an equivalent atom (slope $\leftrightarrow$ onset is a bijection, mirroring $\rho(z_c) = \ln 2 \iff z_c = \sqrt{3}/2$). Hence "D4 is *the* intended minimal radius axiomatization" is [COMPUTED] (candidate) — a modeling choice, not a forced uniqueness.

Proposition 10.14 (Differential strengthening: an honest negative and a conditional sharpening)**.** *[INHERITED]/[OPEN]* `RA-C1--RA-C8` (a) The naive strengthening fails *(honest negative): r* is *elementary, $r = (K/\sqrt{z_c})(z^2)^{1/4} = (K/\sqrt{z_c})\exp(\frac{1}{4}\ln(z^2))$, so it lies in the full elementary-differential closure of the chart field; the attempt to upgrade the v1.8 valuation certificate to non-membership* there *breaks, because d/dz shifts valuation by -1 and turns the even-valuation ρ into the odd-valuation $\rho' = z/(1-z^2)$ (parity is not differential-stable).* (b) The differential-stable sharpening *[OPEN]* (conditional)*: under the* unit-log hypothesis *— the substrate's only logarithm is $\rho = -\frac{1}{2}\ln(1-z^2)$, a log of the apex* unit $1-z^2$ *(valuation 0) — the apex* monodromy-order *invariant excludes r from the* unramified *elementary-differential closure F_{ed}: order-*2 *data (r, leading power $z^{1/2}$) is unreachable from order-*1 *data, while r^2 and ρ' are order-*1 *(hence* in F_{ed}*: indeed $r^2 = (K^2/z_c)\,\rho'(1-z^2)$) yet odd-valuation (hence* outside *the base field F). This excludes* a strictly larger *field than the v1.8 valuation certificate and is differential-stable. It rests on a finite generator battery plus a leading-exponent argument, not a fully general closure theorem, and is [OPEN] conditional on the unit-log hypothesis.*

Remark 10.15 (Bounded no-tie re-confirmation of the constructive bridge)*.* [FORCED]/[COMPUTED] `RA-D1--RA-D3, RA-A4, RA-G1` Re-confirming Proposition 10.10 over a finite ledger of ~ 11 canonical amplitudes: no canonical amplitude supplies the radius coefficient $K/\sqrt{z_c}$, since $(K^2/z_c)^2 = \frac{70}{3} - 10\sqrt{5} \neq 1$ and differs (exact $\mathbb{Q}(\sqrt{5})$, and certified by disjoint intervals) from the square of every listed amplitude $\{1, K, K^2, \varphi, \tau, \text{gap}, z_c, 1/z_c, \sqrt{5}, 2-\varphi, K/\sqrt{3}\}$. The scale enters only through the odd signed-height generator $(K^2/z_c)\,z$ — the D4(i) datum — never as a canonical scalar.

11 Path stretch, rail length, and the conjugate relative phase

The exploratory note of source (iv) proposes a rail-pulse mechanism for rung selection. Its mechanism layer is not imported here; this section states and verifies its *forced kernel* — three results that use only D1–D3 and the inherited radius profile — and records the exact scoping of the rest.

Proposition 11.1 (Winding gradient and path stretch)**.** *[FORCED GIVEN D1–D3]/[INHERITED]* `OZ-M1--M3` *From D1, $\rho(z) = -\frac{1}{2}\ln(1-z^2)$, so with $\kappa = \pi/(2\ln\varphi)$ [D3],*

$$\frac{d\rho}{dz} = \frac{z}{1-z^2}, \qquad \frac{d\theta}{dz} = \kappa\,\frac{z}{1-z^2},$$

and the formation rail $\Gamma(z) = (r\cos\theta,\ r\sin\theta,\ z)$ has local path-stretch

$$\frac{ds}{dz} = \sqrt{1 + \left(\frac{dr}{dz}\right)^2 + r(z)^2\left(\frac{d\theta}{dz}\right)^2} = \begin{cases} \sqrt{1 + \dfrac{K^2}{4zz_c} + \dfrac{K^2\kappa^2 z^3}{z_c(1-z^2)^2}}, & 0 < z \le z_c, \\ \sqrt{1 + \dfrac{K^2\kappa^2 z^2}{(1-z^2)^2}}, & z_c \le z < 1, \end{cases}$$

with $ds/dz > 1$ strictly on $(0,1)$: the helical route is strictly longer than the vertical displacement, through radial formation and winding below the lens and through winding alone above it.

Proof. Differentiate D1 and $r(z)$ on each branch and substitute; both closed forms and the strict excess (all added terms are ratios of positives) are machine-verified symbolically in z. □

Proposition 11.2 (Quarter-turn rail floor and divergence)**.** *[FORCED GIVEN D1–D3]/[INHERITED] OZ-M4--M5 Above the lens, $ds/d\theta = \sqrt{K^2 + (dz/d\theta)^2} \geq K$ pointwise, so with $\Delta\theta = \pi/2$ per rung the rail length between adjacent rungs satisfies*

$$\boxed{L_{n\to n+1} \;\geq\; \frac{\pi K}{2}} = 1.4516928502\ldots \;\; [\text{COMPUTED}] \qquad (n \geq 2):$$

every rung interval retains at least a quarter-circle of rail, even as the vertical increments $z_{n+1}-z_n \to 0$. Consequently the cumulative rail length to the top is infinite while the height stays finite — a one-line sharpening of the whitepaper's logarithmic length divergence (W-Prop. 15.2c) into a uniform per-rung floor. Exactly above the lens one also has the identity $(ds/dz)^2 - \big(K\kappa z/(1-z^2)\big)^2 = 1$, and the minorant integrates to $-\frac{1}{2}K\kappa\ln(1-z^2) \to \infty$.

Proposition 11.3 (ν-form of the rung locus)**.** *[FORCED GIVEN D1–D3] OZ-M6--M7 The conjugate pair $\{W, \overline{W}\}$ exists automatically — the angle multiset of a conjugation-closed object is closed under negation (the standing fact of the relational-charge note, its §2.2) — and winds oppositely: $\theta_\pm(\rho) = \pm\kappa\rho$. Its normalized relative state*

$$\nu(\rho) \;\coloneqq\; \frac{W(\rho)}{\overline{W}(\rho)}\Big/\Big|\frac{W}{\overline{W}}\Big| \;=\; e^{2i\theta(\rho)} \qquad (\textit{matrix form: } \mathcal{R}(\rho) = e^{2J\kappa\rho})$$

is real (scalar) exactly on the rung lattice:

$$\operatorname{Im}\nu(\rho) = \sin\!\Big(\frac{\pi\rho}{\ln\varphi}\Big) = 0 \iff \rho = n\ln\varphi, \qquad \nu(\rho_n) = (-1)^n,$$

and the sign $(-1)^n$ is the Trace–Form-Duality parity bit: W_n is real for even n (Fibonacci/E–W rungs), imaginary for odd n (Lucas/N–S rungs). Equivalently, the rungs are exactly the heights at which W is coherent with the real axis in the sense of the relational note's ν-criterion ($\alpha/\overline{\alpha}$ a root of unity — here $\nu(\rho_n) = \pm 1$, torsion of order ≤ 2).

Proposition 11.4 (Site-density trichotomy)**.** *[FORCED GIVEN D1–D3]/[INHERITED] OZ-N1--N2 Inverting the rung law gives the continuous floor coordinate — and three site densities that must be kept apart. The floor coordinate is*

$$n(z) = \frac{\rho(z)}{\ln\varphi} = -\frac{\ln(1-z^2)}{2\ln\varphi}, \qquad \frac{dn}{dz} = \frac{z}{(1-z^2)\ln\varphi} \xrightarrow[z\to 1^-]{} \infty$$

(at the lens, $dn/dz = 2\sqrt{3}/\ln\varphi = 7.1987042602\ldots$ [COMPUTED] OZ-H5). All three densities are exact:

measure	*behaviour*	*source*
angular	*fixed: $\Delta\theta = \pi/2$, four phase classes per turn*	*D2+D3*
vertical projection	*divergent: $dn/dz \to \infty$ as $z \to 1$*	*D1*
rail distance (above lens)	*bounded below and convergent: $\Delta s_n \downarrow \pi K/2$*	*Prop. 11.2*

The third entry sharpens Proposition 11.2: $dz/d\theta = (1-z^2)/(\kappa z)$ is strictly decreasing to 0, so the rail spacing $\Delta s_n = \int \sqrt{K^2 + (dz/d\theta)^2}\, d\theta$ is strictly decreasing with limit exactly *$\pi K/2$ — the quarter-turn floor is attained in the limit. Apparent crowding of rung sites is a projection artifact of the D1 chart, not an increase in phase classes per revolution.*

Proposition 11.5 (Reversion norm and the transfer ledger)**.** *[FORCED GIVEN D1–D3]* `OZ-N3--N5` *With reversion $\widetilde{aI + bJ} = aI - bJ$ (an anti-automorphism),*

$$Q\widetilde{Q} = \tau^2 I, \qquad I_n := Q^n\widetilde{Q^n} = \tau^{2n} I = |W_n|^2 I,$$

and the complementary ledger

$$\boxed{F_n := 1 - I_n = z_n^2} \qquad \text{(D1 at the rungs, in ledger form),}$$

with per-rung increment, via $1 - \tau^2 = \tau$,

$$\Delta F_n = F_{n+1} - F_n = \tau^{2n}(1-\tau^2) = \tau^{2n+1} = z_{n+1}^2 - z_n^2, \qquad \sum_{n\geq 0} \tau^{2n+1} = \frac{\tau}{1-\tau^2} = 1 \quad \text{exactly.}$$

The odd golden powers $\tau, \tau^3, \tau^5, \ldots$ are precisely the rung increments of z^2, they telescope to the unit ($1-\tau^{2N} = z_N^2$), and the whole normalized budget is distributed over infinitely many rungs with none consuming it at finite n. The per-step split is (τ^2, τ): transmitted co-intensity τ^2, complementary increment τ. All of this is exact chart algebra; the source note's reading of ΔF_n as a physically deposited *pulse fraction (its engine-pass model) is an interpretation and is not imported.*

Remark 11.6 (Scoping: what the rails do and do not close). Proposition 11.3 removes two of the exploratory note's three assumptions: the second rail and its counter-rotation are not hypotheses but the conjugate, forced by conjugation closure. What remains outside the forced scope is exactly what that note's own §10(8) concedes: (a) reading the real crossings of ν as *recoupling events that cause* rungs is an interpretation — with the existing rate κ the locus merely coincides with the ladder already forced by the gate multipliers $|m_n| = \varphi^{-2n}$; and (b) the rate itself is D3 imported, not derived — for a general rate the locus is $\rho_n = n\pi/(2\kappa)$, so the construction reproduces the golden floors only because $\kappa = \pi/(2\ln\varphi)$ was put in. Rung quantization (one Q per floor) therefore remains the open item named in §3; the energy condition, the thermodynamic reading, and the mechanism chain of source (iv) are not imported. The second note (source (vi)) is handled identically: its forced kernel is Propositions 11.4–11.5; its engine-pass model, pulse-budget reading, and biological program remain hypothesis — correctly fenced by that note's own claim ceilings and four-layer hierarchy (geometry → kinematics → dynamics → biology), which this paper endorses as discipline without importing as content.

Proposition 11.7 (Rail bounds and the gap-rate law)**.** *[FORCED GIVEN D1–D3]/[INHERITED]* `QX-E1--E9` *For $n \geq 2$ write $t_n = \tau^{2n}$, and on $[\theta_n, \theta_{n+1}]$ set $x = (dz/d\theta)^2/K^2 = t^2/\big(K^2\kappa^2(1-t)\big)$ with $t = e^{-2\theta/\kappa} = 1 - z^2$, so that $\Delta s_n = K\int_{\theta_n}^{\theta_{n+1}} \sqrt{1+x}\, d\theta$. With*

$$I_1(t) = \frac{1}{4K\kappa}\Big[\ln\frac{1-\tau^2 t}{1-t} - \tau t\Big] = \frac{1}{4K\kappa}\sum_{m\geq 2}\frac{1-\tau^{2m}}{m}t^m \; > \; 0,$$

$$I_2(t) = \frac{F(t) - F(\tau^2 t)}{16K^3\kappa^3}, \quad F(t) = \frac{1}{1-t} + 3\ln(1-t) - 3(1-t) + \frac{(1-t)^2}{2}, \quad 0 < I_2(t) \leq \frac{1-\tau^8}{64K^7\kappa^3}\, t^4,$$

the rail spacing of Proposition 11.4 obeys the exact elementary two-sided bound

$$\boxed{\frac{\pi K}{2} + I_1(t_n) - I_2(t_n) \;\leq\; \Delta s_n \;\leq\; \frac{\pi K}{2} + I_1(t_n)}$$

(lower bound via $\sqrt{1+x} \geq 1 + \frac{x}{2} - \frac{x^2}{8}$, valid since $x \leq \mathrm{gap}^2/(\kappa^2 K^4) < 8$ on the range, interval-certified `QX-G1`*). Consequences:*

$$\lim_{n\to\infty} \frac{\Delta s_n - \pi K/2}{\mathrm{gap}^{\,n}} = \frac{K}{8\kappa} = \frac{K \ln\varphi}{4\pi} = 0.0353900591\ldots \text{ [COMPUTED]}, \qquad \frac{\Delta s_{n+1} - \pi K/2}{\Delta s_n - \pi K/2} \longrightarrow \mathrm{gap},$$

and since $\mathrm{gap}^{\,n} = \tau^{4n} = |W_n|^4$,

$$\Delta s_n - \frac{\pi K}{2} \;\sim\; \frac{K \ln\varphi}{4\pi}\,|W_n|^4 :$$

the excess rail over the quarter-turn floor is priced by the square of the co-intensity, and the divergent side of the length dichotomy (Proposition 10.5) is thereby quantified: the rail diverges at exactly $\pi K/2$ per rung, approached at the gap rate. Structural note on the exact integral: under the radical, $(1-z^2)^2 + K^2\kappa^2 z^2 = z^4 + (K^2\kappa^2 - 2)z^2 + 1$ factors reciprocally as $(z^2+a)(z^2+1/a)$ with $a + 1/a = K^2\kappa^2 - 2$ (real $a > 0$ since $K\kappa > 2$, interval-certified `QX-G2`*), placing Δs_n in incomplete-elliptic form — in contrast with the residual sphere, whose total length closes in a* complete *elliptic value (Proposition 10.5); no elementary closed form is claimed here, and the bounds above carry the elementary content.* `QX-E9`

Proof. From D1+D3, $dz/d\theta = (1-z^2)/(\kappa z)$ `QX-E1`. The square-root bounds are the polynomial identities $(1 + x/2)^2 - (1+x) = x^2/4$ and $(1+x) - (1 + x/2 - x^2/8)^2 = x^3(8-x)/64$ `QX-E2`. Substituting $t = e^{-2\theta/\kappa}$ ($d\theta = -\kappa\, dt/2t$) turns $\frac{K}{2}\int x\, d\theta$ into $\frac{1}{4K\kappa}\int_{\tau^2 t_n}^{t_n} \frac{s}{1-s}\, ds$ and $\frac{K}{8}\int x^2\, d\theta$ into $\frac{1}{16K^3\kappa^3}\int_{\tau^2 t_n}^{t_n} \frac{s^3}{(1-s)^2}\, ds$, with the stated antiderivatives `QX-E3--E4`; the $m = 1$ series term cancels ($1 - \tau^2 = \tau$), the $m = 2$ term gives the constant via $1 - \tau^4 = K^2$ and $1/(8\kappa) = \ln\varphi/(4\pi)$ `QX-E5`, and the majorant uses $(1-s)^2 \geq K^4$ on $s \leq \mathrm{gap}$ with $\int s^3 = t^4(1-\tau^8)/4$ `QX-E6`. The ratio law is the series quotient `QX-E7--E8`. A 50-dps quadrature corroborates the bracket for $n = 2..5$ `QX-G3`; display-only. $\square$

Theorem 11.8 (Incomplete-elliptic closed form for the rail)**.** *[FORCED GIVEN D1–D3]/[INHERITED]* `EX-A1--A4`, `EX-B1--B8`, `EX-G3--G5` *Let $a > 1$ be the larger root of $a + 1/a = K^2\kappa^2 - 2$ (so $a\cdot(1/a) = 1$; a real with $a > 1$ strictly, interval-certified via $K\kappa > 2$, replicating* `QX-G2`*), and set*

$$\theta(z) = \arctan\!\big(z\sqrt{a}\big), \qquad m = 1 - \frac{1}{a^2}, \qquad n_c = 1 + \frac{1}{a}.$$

Above the lens the rail integrand is

$$\frac{ds}{dz} = \frac{\sqrt{(z^2+a)(z^2+1/a)}}{1-z^2}$$

(from $\Delta s_n = \int \sqrt{K^2 + (dz/d\theta)^2}\, d\theta$ and $dz/d\theta = (1-z^2)/(\kappa z)$, with the reciprocal factorization of Proposition 11.7), and its antiderivative is

$$\boxed{G(z) = -z\sqrt{\frac{z^2+a}{z^2+1/a}} \;+\; \sqrt{a}\, E(\theta\,|\,m) \;-\; a^{-3/2}\, F(\theta\,|\,m) \;+\; (a+1)\, a^{-3/2}\, \Pi\big(n_c;\, \theta\,|\,m\big)},$$

with $\Delta s_n = G(z_{n+1}) - G(z_n)$ at $z_n = \sqrt{1 - \tau^{2n}}$ (F, E, Π the incomplete elliptic integrals of the first, second, and third kind [ESTABLISHED]). Certificate: $dG/dz -$ integrand $\equiv 0$ exact-symbolically in (a,z) on $a > 0$, $0 < z < 1$ (the forced certificate), assembled from three kind-pure lemmas *through the exact partial-fraction split of the numerator (the $(1-z^2)^{-1}$ pole, of exact weight $w+2$ with $w = K^2\kappa^2 - 2$, is what produces the third-kind Π term); FTC applicability on $[z_n, z_{n+1}] \subset (0,1)$ via $1 - n_c \sin^2\theta = (1-z^2)/(1+az^2) > 0$. Numeric corroboration: the 50-dps closed form matches quadrature of Δs_n to $< 10^{-40}$ on the* `QX-G3` *grid $n = 2..5$ and sits inside the bracket of Proposition 11.7. The "no elementary closed form is claimed here" scoping of that proposition is hereby completed on the positive side: the rail side of the length dichotomy (Proposition 10.5) is* closed in incomplete-elliptic terms.

Corollary 11.9 (Certified non-elementarity of the rail antiderivative)**.** *[FORCED]/[ESTABLISHED]* `EX-C1--C5`, `EX-G1--G2` *Forced given Liouville's theorem on integration in finite terms (Rosenlicht 1972; algorithmic form Trager 1984) [ESTABLISHED]: the antiderivative of the above-lens rail integrand is* not *an elementary function — the per-rung rail length has no elementary closed form as a function of its endpoints. All premises of the anchor are machine-certified: the kernel curve $y^2 = (z^2+a)(z^2+1/a)$ is genuinely elliptic —* $\operatorname{disc}(z^4+wz^2+1) = 16(w-2)^2(w+2)^2 \neq 0$ *with $w = K^2\kappa^2 - 2 > 2$, four distinct roots, genus 1 (at $a = 1$ it would collapse to $(z^2+1)^2$ and integrate elementarily); the first-kind coefficient $c_F = -a^{-3/2}$ is nonzero* unconditionally *($c_F\, a^{3/2} = -1$ for every $a > 0$); $c_E = \sqrt{a}$ and $c_\Pi = (a+1)a^{-3/2}$ are nonzero; and the* Π *parameters are non-degenerate ($m \in (0,1)$ and $n_c \in (1,2)$ interval-certified; $n_c - 1$, $n_c - m$, $1 - m$ all exactly nonzero, with the exact interchange tie $n_c(2-n_c) = m$, so no special-case reduction applies), the $z = \pm 1$ pole pair genuine with residues $\mp K\kappa/2 \neq 0$ and the exact ledger tie $y(\pm 1)^2 = (1+a)(1+1/a) = K^2\kappa^2$ (the kernel over the lens poles is the squared rail-speed constant). The closed-form dichotomy is thus fully resolved: complete-elliptic on the residual side (Proposition 10.5); incomplete-elliptic, and provably non-elementary — certified rather than conjectured — on the rail side, whose elementary content is carried by the bracket and gap-rate law of Proposition 11.7.*

12 Epistemic ledger and findings

Forced unconditionally, new in v1.6 `QX-B, QX-C, QX-D`: the characteristic level of the ratio object (χ_{Rat}, the sign-split pair of φ^2, the Adams square onto $\operatorname{spec}(A_{\mathrm{gap}})$, Proposition 4.7); the quarter twist — transport domain = parity-homogeneous objects, gauge-rotation invariance, anchor-exchange dichotomy (exchange iff $m \equiv 2 \bmod 4$; the generator/Z^* pair is the minimal instance and the K-seed is anchor-fixed) (Theorem 4.8); the K-family seed ladder $p_j = x^4 + 5F_{2j}^2 x^2 - 5F_{2j}^2$ with uniform irreducibility by norm obstruction, the full ledger, and the gate ties $C_{2j} = 1/|p_j(0)|$, $u_{2j} = 1/\mathrm{M}(p_j)$ (Theorem 8.3).

Forced given D1–D3, new in v1.6 `QX-E, QX-F, QX-C12--C14`: the exact rail-spacing bounds $\pi K/2 + I_1 - I_2 \le \Delta s_n \le \pi K/2 + I_1$ with the gap-rate law $\Delta s_n - \pi K/2 \sim (K \ln\varphi/4\pi)\,|W_n|^4$ (Proposition 11.7); the even-rung seed mirror $(\sin,\cos,\tan)\alpha_{2j} = (K_j, \tau^{2j}, \beta_j)$ (Corollary 8.4); the transfer of the interpolation classification to the formation rail, adding the sixth D3 characterization (Remark 2.6).

Forced unconditionally, new in v1.5 `NX-T1--T9`: the complete relational type of the generator object (Theorem 4.6) — two shells $\{\pm i\tau\}$, $\{\pm i\varphi\}$; contact signature $\{\Phi_1^4, \Phi_2^4\}$ complete under the full scan bound; all eight cross-shell ratios real ($\pm\tau^2$, $\pm\varphi^2$) with $\nu \equiv 1$, hence full coherence in one class; anchor $(m,n) = (2,4)$ with $\Delta = \mathbb{Z}/2\mathbb{Z}$; the twisted-shell realization $x^4 + 3x^2 + 1 = s(x^2)$, $s = \operatorname{minpoly}(q^2)$.

Forced given D1–D3, new in v1.5 `NX-S1--S7, NX-H1--H3`: the compass-cross characterization of the rung locus with $\operatorname{Im}\nu = 2xy/(x^2+y^2)$ (Proposition 10.3); the spherical step ledger $S_n \cdot S_{n+1} = z_n z_{n+1}$ with $d_0 = \pi/2$ exactly and $d_n \downarrow 0$ (Proposition 10.4); the finite total spherical length $L_{\text{res}} = \sqrt{1+\kappa^2}\, E\big(\kappa^2/(1+\kappa^2)\big)$ with the residual/formation length dichotomy and the fifth (spherical-length) characterization of the principal branch (Proposition 10.5).

Forced unconditionally, new in v1.4 `OZ-01--06`: the interpolation classification $\log Q = \{(\ln\tau)I + (\pi/2 + 2\pi k)J\}$ with rung-agreement and curve-distinctness; the four equivalent characterizations of the $k = 0$ branch (D3 = principal log = minimal winding = shortest residual spiral per floor); the dilation mechanism $K = 5^{1/4}\tau$, $\beta = 5^{1/4}\varphi$ behind the bridge factor $\sqrt{5}$.

Forced unconditionally, new in v1.3 (bridge to the relational-charge layer, machine-verified `RC-D3, RC-E1--E7`): the minimal polynomial $x^4 + 3x^2 + 1$ of q with $\mathrm{M} = \varphi^2$ and its group-drop coherence type; the gauge identity $\mathrm{minpoly}(q)(ix) = Z^*$ with both rigidity anchors realized by the pair (Z^*, iZ^*); the K-seed as catalog object (compass angle set, $\Delta = \mathbb{Z}/4\mathbb{Z}$, $\mathrm{M} = \beta^2$); the relational restatement of D2.

Forced unconditionally, new in v1.1 (pure keystone algebra, machine-verified `OZ-J, OZ-K, OZ-L`): the spectral projectors and grading $U = P_+ - P_- = H/\sqrt{5}$; the return-kernel recoupler $A = N/\sqrt{5}$ with $UA = -AU$ and mode exchange $AP_\pm = P_\mp A$; the factorization $J = \pm UA = HN/5$ with $J^2 = -I$, coordinate-free up to orientation; the obstructions $i \notin \mathbb{R}[R]$ and $N \notin \mathbb{R}[R]$; the canonical rung generator $Q = \pm\tau J$ with $Q^4 = \mathrm{gap}\, I$ and the orbit identity $Q^n v_0 = (\operatorname{Re} W_n, \operatorname{Im} W_n)$.

Forced given D1–D3, new in the July 14 explorations (all machine-verified here): the packaged representation $W(\rho) = e^{-\rho}e^{i\theta(\rho)}$ and $z = \sqrt{1 - |W|^2}$; the generator $W_n = q^n = (i\tau)^n$ with $q^4 = \mathrm{gap}$ and the two-/four-rung recurrences; the gap schedule (co-intensity at two rungs, amplitude at four); the state-angle dictionary (Thm. 5.1); the rung ledgers $(\sqrt{\tau}, \tau, \sqrt{\varphi})$ and (K, τ^2, β) with the general family $\tan\alpha_n = \sqrt{\varphi^{2n} - 1}$; the bridges $\cos\alpha_2 = \cos^2\alpha_1$, $z_1 = \sqrt{2}\sin(\alpha_2/2)$, $K = \sin\alpha_2$; the K-family $K_j = \sqrt{1 - \mathrm{gap}^j}$ with uniqueness of exact K, the universal two-step law, and the three-level recurrence; the lens as the 60° state; the path-stretch laws, the quarter-turn rail floor $L \geq \pi K/2$ with its divergence corollary, the ν-form of the rung locus with parity $\nu(\rho_n) = (-1)^n$, the site-density trichotomy with $\Delta s_n \downarrow \pi K/2$, and the transfer ledger $F_n = z_n^2$, $\Delta F_n = \tau^{2n+1}$, $\sum \tau^{2n+1} = 1$ (§11); the residual-sphere lift and the colatitude identity $\theta_K = \arccos K$; the linearization tie $f'(K) = q^2|q|^2$. None of these is a new substrate axiom; all are compressed representations and corollaries, exactly as the source report states.

Declared upstream [DECLARED]/[INHERITED]: D1, D2, D3, the orientation convention, and the inherited radius profile (whitepaper §11).

Not established (carried verbatim from the source report, endorsed): biological or DNA correspondence (the direct DNA-pitch test *failed* quantitatively and is recorded as a negative result; the helicase and uncouple–recouple language of the v1.1 report functioned as a mechanism-finding analogy only); physical universality; the residual sphere as fundamental ontology; derivation of the formation radius from the complex generator; independent validation of D1–D3 by the representation; *rung quantization* (why exactly one Q per rung event — Q explains propagation, not yet selection); *continuous interpolation as selection* (Theorem 2.4 classifies the interpolations: a $\mathbb{Z}$-family agreeing at every rung; what the discrete powers leave undetermined is exactly the principal-branch choice, which is D3 — so this item merges into the rate problem); the rail-pulse mechanism of sources (iv) and (vi) — recoupling as *cause*, the energy functional $\mathcal{E}$, the engine-pass/deposition reading of the transfer ledger, and the biological comparison program — which their own epistemic sections correctly leave as hypothesis. Coordinate-free uniqueness of the factorization, listed open in the source report, is closed by Corollary 3.4.

Finding 12.1 (E-1: scoping slip in the source report's §11). The report's "Forced in the whitepaper" list includes "quarter-turn charge increments" without the "given D2" qualifier, inconsistently with its own "Declared" list (which correctly names D2) and with its overall conditional framing.

Cosmetic; the report's mathematics nowhere relies on the unqualified reading. This paper's banner (§1) restores the qualifier everywhere.

Finding 12.2 (E-2: truncated table values in the review thread). The K-family table prints $K_2 \approx 0.989299$ and $K_3 \approx 0.998445$; the exact values $0.9892996329\ldots$ and $0.9984459825\ldots$ round to 0.989300 and 0.998446 at 6 dp — the printed values are floor-truncations `OZ-H1`. Display-only; no decision weight. (K_1, K_4, and gap are correctly rounded.)

Finding 12.3 (E-3: stale verification paragraph in §1). Through v1.3 and v1.4 the §1 **Verification** paragraph still carried the v1.0 harness description ("53 exact checks", ID range `OZ-A1..OZ-H3`) while the title line, abstract, and manifest (§13) had advanced to 91 checks across groups `OZ-A..OZ-O`. Display-only; no decision weight. Corrected in v1.5.

Finding 12.4 (E-4: front-matter check classification in v1.5). The v1.5 title line and §1 verification paragraph counted all 27 checks of `nx_softcheck.py` as exact, although its group `NX-H` comprises three 50-dps numeric display/corroboration checks (so: 24 exact + 3 numeric). The three numeric checks carry no decision weight — the harness's own header declares them display/corroboration only, and instrumentation confirms every passing decision was reached on the exact symbolic path. Display-only; corrected in v1.6, which states the split and drops "all exact" from the title line.

Finding 12.5 (E-5: stale third-harness count in the v1.6 manifest). The v1.6 §13 third-harness sentence retained "27 exact checks" for `nx_softcheck.py`, although Finding 12.4 had already introduced the $24 + 3$ split and the front matter (title line and the §1 verification paragraph) had been corrected accordingly. Display-only; no decision weight. Corrected in v1.7.

Finding 12.6 (E-6: exact-count wording for the first harness). The description "91 exact checks" for `ozy_softcheck.py` (§1 and §13) contradicts that harness's own manifest table, whose `OZ-H` row lists 5 numeric displays inside the 91 total; under the classification standard of Finding 12.4 it should read 86 exact + 5 numeric displays. The five numeric checks carry no decision weight. Display-only; corrected in v1.7 (the 91/91 and 27/27 title-line pass ratios stand, as pass ratios).

Finding 12.7 (E-7: the SSS/Lehmer reading of $[\mu_S, 2)$ is route-relative). v1.9–v2.0 state that the residual gap $[\mu_S, 2)$ *is* the Schur–Siegel–Smyth / Lehmer–pentagon core (Prop. 4.14(iv), §14.5), and infer that closing (ODD-2) requires new number theory strengthening Smyth. The premise is true only of the *realification* route: the Adams handle ψ^m discards the ray constraint and keeps only non-reciprocality, and *within that route* the gap is exactly that core. It is not a property of (ODD-2). Theorem 4.17 keeps the ray constraint — it transports it through $N_{K/F}$, where odd m re-emerges as the π-ray exclusion on N — and closes the lemma with an elementary argument, using neither μ_S nor any strengthening of Smyth. Corrected in v2.1; no decision elsewhere in the corpus depended on the mis-framing, and the $\mathrm{M} \le 2$ ceiling, the π-ray exclusion, and the reciprocal floor $\varphi^2 > 2$ are all unaffected. Recorded because the framing, not the mathematics, is what kept the item open through three versions.

Finding 12.8 (E-8: `bl_softcheck.py`'s P3/P4 row is a v2.0 artifact). The v2.0 finalization harness prints a nine-row blocker ledger in which P3/P4 is classified NAMED-LEMMA (ODD-2) and `BL-G4` labels "the residual lift $\mu_S \to 2$" as a wall no harness can settle. Both are superseded by Theorem 4.17. The harness is *not* re-run in v2.1 and its shipped pass line (`16/16 exact, 7 walls LABELLED`) is retained verbatim as the v2.0 record; the row is kept in the §14 table, marked CLOSED, so that the classification history stays legible. Its two load-bearing forced facts remain true as stated (`BL-E1`: the $= 2$ ceiling is forced; `BL-E2`: Smyth alone gives only $\mu_S < \varphi < 2$) — what lapses is the inference that no further internal route existed. For reading purposes the labelled-wall count is six, not seven.

13 Verification manifest

`ozy_softcheck.py`: 86 exact checks + 5 numeric displays (the `OZ-H` row below; Finding 12.6), exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$, K decided at K^2, mpmath 50 dps display-only.

group	content	checks
OZ-A	generator $q = i\tau$: powers, $q^4 = \text{gap}$, one/two/four-rung recurrences, gap schedule	8
OZ-B	chart consistency $z_n = \sqrt{1-\lvert W_n\rvert^2}$; $z_1 = \sqrt{\tau}$; $z_2 = K = \sqrt{1-\lvert q^2\rvert^2}$	3
OZ-C	state-angle dictionary (5 fields, symbolic); rung tangents, TFD dress, $2\varphi^{3/2}$; self-duality; quartic; K-certificate	14
OZ-D	bridges: cosine contraction, not-doubling guard, half-angle, Route-5 trig, $\Theta = \alpha_2$	7
OZ-E	lens 60° ledger; off-lattice guard	6
OZ-F	K-family, odd companion, universal two-step law, monotone uniqueness, compass, linearization tie	7
OZ-G	residual sphere: $\lvert S\rvert \equiv 1$ (continuous and at rungs), exact $S_0..S_4$, two-radii separation	5
OZ-J	spectral uncoupling: projectors, grading $U = H/\sqrt{5}$, mode action	5
OZ-K	recoupler A, factorization $HN = 5J$, $J = \pm UA$, generator Q, orbit identity	9
OZ-L	obstructions $i, N \notin \mathbb{R}[R]$; coordinate-free pseudoscalar; kernel $= H^{\perp}$	4
OZ-M	path stretch (both branches), quarter-turn floor, divergence, rails algebra $\mathcal{R} = e^{2J\kappa\rho}$, ν-locus & parity	7
OZ-N	floor coordinate & density trichotomy ($\Delta s_n \downarrow \pi K/2$), reversion norm, transfer ledger, unit budget	5
OZ-O	interpolation classification ($\log Q$ family, rung agreement, curve distinctness, minimality), dilation mechanism	6
OZ-H	numeric displays: K-family table (truncation-aware), state angles, tangents, $\pi K/2$, lens density	5
	total	**91**

Second harness `relational_softcheck.py`: 27 exact checks, exit 0 — cold re-derivation of the relational-charge layer this paper's bridge uses: rigidity anchors (RC-A), the golden layer and sign twist (RC-B), contact machinery with the sieved completeness bound (RC-C), pinning corroborations including Lehmer's degree-100 ratio object and the $\beta_4\times$Lehmer mixed scan (RC-D), the bridge derivations of §4 (RC-E), and the (EG) quadratic case (RC-F). No defects found in the checked claims of the relational note.

Third harness `nx_softcheck.py` (this compilation, written cold from the v1.4 statements): 24 exact checks + 3 numeric displays (Finding 12.4; the stale "27 exact" that stood in this sentence through v1.6 is Finding 12.5), exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$ with rational-arithmetic sign decisions for $\mathbb{Q}(\sqrt{5})$ comparisons; mpmath 50 dps display-only. Groups: `NX-R` (replication of the load-bearing claims: minimal polynomial and roots, Mahler φ^2, contact signature, gauge identity, rung sphere identity, ν-locus parity, monotone heights — 8); `NX-T` (complete relational type: shells, cross-shell ratios, realness and non-unimodularity, $\nu \equiv 1$, the t_{rel} pattern, coherence completeness, anchor data, twisted-shell factorization — 9); `NX-S` (compass cross, the xy-form of ν, step orthogonality, quarter-circle first step, monotone steps, spherical speed, elliptic length form and bounds — 7); `NX-H` (numeric displays: L_{res}, d_1, bracket corroboration — 3).

Fourth harness `qx_softcheck.py` (this compilation, written cold for v1.6): 62 exact checks + 3 guards, exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$, the K-family decided at the squared level; the guards are two interval-certified transcendental comparisons (mpmath `iv`) and one 50-dps quadrature corroboration, none load-bearing.

group	content	checks
QX-A	cold replication of the layer v1.6 builds on: generator and minimal polynomial, gauge identity, chart dictionary, even gate ladder, two-step law, transfer ledger	15
QX-B	ratio object at the characteristic level: 16-ratio multiset, χ_{Rat} and its factorization, torsion/nontorsion split, golden units, Adams square (Prop. 4.7)	8
QX-C	K-family seed ladder: p_j, uniform irreducibility by norm, Eisenstein availability, full ledger, compass and contact scan, gate ties, even mirror (Thm. 8.3, Cor. 8.4)	14
QX-D	quarter twist: domain, involution, golden-seed exclusion, eight witnesses, exhaustive coset simulation $m \le 24$ (Thm. 4.8)	10
QX-E	rail bounds: derivative law, root bounds, antiderivatives, series constants, majorant, ratio law, reciprocal factorization (Prop. 11.7)	9
QX-F	rail transfer: rung agreement, branch distinctness, $k = 0$ minimality, monotone speed, sphere transfer (Rem. 2.6)	6
QX-G	guards: $x_{\max} < 8$ and $K\kappa > 2$ (interval-certified); 50-dps bracket of Δs_n, $n = 2..5$	3
	total (exact + guards)	**62+3**

v1.7 session harnesses (eight, written cold this session, one per dossier; every `TX-`, `MX-`, `RX-`, `UX-`, `EX-`, `CX-`, `LX-`, and `DX-` check ID cited in the v1.7 sections above appears verbatim in the corresponding file):

harness	content	exact + guards
`tx_softcheck.py`	rate arithmetic: irrationality lemma, transcendence chain, near-miss closure of Guard 15.3, $\log_\varphi 2$, scan retirement (Lem. 2.7–Rem. 2.10)	$30 + 6$
`mx_softcheck.py`	multiplier torus: modulus ledger, systole and marking, restatement test, geodesic ledger, no-CM (Prop. 2.11–Rem. 2.13)	$39 + 3$
`rx_softcheck.py`	radius: parity no-go with $\mathbb{Z}/2$ sharpening, area-transfer law, the three equipotential clocks (Thm. 10.6–Rem. 10.8)	$25 + 4$
`ux_softcheck.py`	universal seed ladder $n = 1..10$ + symbolic-in-w, tangent-dress Mahler ladder, Theorem U window and monotonicity, F1 falsifier (Thm. 8.5–Rem. 8.8)	$35 + 1$
`ex_softcheck.py`	rail closed form: partial-fraction assembly, exact-derivative certificate, ellipticity, coefficient nonvanishing (Thm. 11.8, Cor. 11.9)	$17 + 5$
`cx_softcheck.py`	χ-selection: pentagon-by-image, unimodular monomials, conditional chain, de-axiomatization (Props. 3.7–3.8, Thm. 3.9)	$34 + 2$
`lx_softcheck.py`	lens: scoping lemma, rational-height splits, bounded-negative scan (7 scan lines, bookkeeping, counted separately) (Prop. 9.2, Rem. 9.3)	$26 + 2$
`dx_softcheck.py`	OP-RATE core: gate-ladder replication, $\pm Q$ uniqueness, sign ladder and axiom H, ledger counter-models, k-invariance and R6 no-go (Prop. 2.14, Rem. 2.15)	$28 + 6$
	total	$\mathbf{234 + 29}$

Environment pin: Python 3.12.10, sympy 1.14.0, mpmath 1.3.0 (the `py` launcher); every exact decision in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i, with $\mathbb{Q}(\sqrt{5})$ signs by rational arithmetic and K decided at the squared level; the guards are interval-certified transcendental comparisons (mpmath `iv`, 60 dps), 50-dps quadrature corroborations, or falsifier rejections (`dx`), none load-bearing; all eight exit 0; each dossier passed two-lane adversarial verification (independent harness audit plus mathematical audit) before inclusion here.

v1.8 session harnesses (five, written cold this session, one per dossier; every `CL-`, `RD-`, `CH-`, `RN-`, and `PM-` check ID cited in the v1.8 sections above appears in the corresponding file — the `CL-2b(i)--(vii)` row suffixes are assembled as the seven-row loop runs):

harness	content	exact + guards
`rd_softcheck.py`	OP-RADIUS: apex-ramification valuation certificate ($e = 2$ for r^2, $e = 4$ for r) and the constructive-bridge obstruction; atom ledger with the cap-onset negative (Thm. 10.9–Rem. 10.11)	$22 + 2$
`cl_softcheck.py`	OP-RATE R6 classification no-go: extended k-invariance, the selection-functional no-go, the seven-way table, the Cencov-mirror closure (Lem. 2.16–Rem. 2.19)	$20 + 5$
`ch_softcheck.py`	χ doctrine g1/g3 resolved negatively: on-circle image of $S = \mu_2$, the charge character, the D2$'$ dichotomy (Prop. 3.10, Rem. 3.11)	$26 + 2$
`rn_softcheck.py`	relational note: P4 recovered/reduced to P3, the bracketed odd floor, the bounded P1$'$ transport-orbit scan (Rem. 4.9–Rem. 4.11)	$20 + 4$
`pm_softcheck.py`	period-relation map: τ_0 algebraic and no κ-vs-τ_0 relation, L_{res} as an elliptic period, the external-open ledger (Prop. 2.20, Rem. 2.21)	$27 + 7$
	total	$\mathbf{115 + 20}$

Across the thirteen session harnesses (`tx`, `mx`, `rx`, `ux`, `ex`, `cx`, `lx`, `dx`, `cl`, `rd`, `ch`, `rn`, `pm`) the running total is **349** exact checks + **49** certified guards, all exit 0. Environment pin: Python 3.12.10, sympy 1.14.0, mpmath 1.3.0 (the `py` launcher); every exact decision in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i (and by $\sqrt{13}$ / $5^{1/4}$ where a Salem trace-down or an off-circle modulus enters), with $\mathbb{Q}(\sqrt{5})$ signs by rational arithmetic; the guards are interval-certified transcendental comparisons (mpmath `iv`, 60 dps), 50–140-dps quadrature/`pslq`/decimal corroborations, or fail-first falsifier rejections, none load-bearing; all five exit 0; each dossier passed two-lane adversarial verification, with the companion papers read from the Echo-S-Research archive read-only and their machinery re-encoded cold.

v1.9 session harnesses (four, written cold this session, one per dossier; every `RA-`, `QT-`, `OF-`, and `PF-` check ID cited in the v1.9 sections above appears in the corresponding file):

harness	content	exact + guards
`ra_softcheck.py`	OP-RADIUS D4-reduction ($D4 \Rightarrow r$ unique + minimality) and the differential strengthening (honest negative + conditional apex-monodromy-order sharpening) (Thm. 10.12–Rem. 10.15)	$23 + 4$
`qt_softcheck.py`	D2 relocation: the K-seed rotation axis forces $\chi = \pm\pi/2$ off-circle (disjoint from g1); net verdict W[open]2 advanced not closed (Prop. 3.12, Rem. 3.13)	$19 + 3$
`of_softcheck.py`	odd relational floor: computed = 2 up to a logged box, reduction to the single lemma (ODD-2), P4 transfer (Prop. 4.14, Rem. 4.19)	$20 + 6$
`pf_softcheck.py`	period frontier: the forced-identity ledger (with the new elliptic geometric identity), the Baker/Schanuel/external-open tool-gap map, and the validated-detector PSLQ bounded-negative (Rem. 2.22, Rem. 2.23)	$13 + 13\,\mathrm{PSLQ} + 5$
	total (exact/computed + guards + PSLQ)	$\mathbf{75 + 18 + 13}$

Across the seventeen session harnesses (`tx`, `mx`, `rx`, `ux`, `ex`, `cx`, `lx`, `dx`, `cl`, `rd`, `ch`, `rn`, `pm`, `ra`, `qt`, `of`, `pf`) the running total is **424** exact checks + **67** certified guards + **13** PSLQ corroborations, all exit 0. Environment pin: Python 3.12.10, sympy 1.14.0, mpmath 1.3.0 (the `py` launcher); every exact decision in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by radicals and i (and by $\sqrt{13}$ where the Salem trace-down enters), with $\mathbb{Q}(\sqrt{5})$ signs by rational arithmetic and K decided at the squared level; the 18 guards are interval-certified (mpmath `iv`, 60 dps), while the 13 PSLQ integer-relation searches (230 dps, coefficient-height $\leq 10^{12}$) are counted as [COMPUTED] corroboration — never a transcendence or algebraic-independence proof; all four exit 0; each dossier passed two-lane adversarial verification, with the companion papers read from the Echo-S-Research archive read-only and their machinery re-encoded cold.

v2.0 finalization harness `bl_softcheck.py` (written cold this session, the active-blockers ledger of §14): 16 exact checks + 7 labelled walls + the nine-row blocker ledger, exit 0. It does *not* re-derive the corpus; it re-certifies, in one place, the single load-bearing forced fact behind each open problem's blocker classification — groups `BL-A`–`BL-F`, each shown fail-first against a planted false wall (e.g. `BL-B3` rejects "r^2 is even") — *labels* the seven walls no harness can settle (`BL-G1`–`BL-G7`), and prints the nine-row ledger (`BL-H1`); pass line `BL: EXACT 16/16 checks passed | 7 walls LABELLED`. Discipline as above: every decision exact over $\mathbb{Q}/\mathbb{Q}(\sqrt{5})/\mathbb{Q}(\sqrt{13})$ extended by radicals and i, valuations by exact leading-term order, sign decisions by rational arithmetic; mpmath is display-only (group `BL-H`). With it the *eighteen* session harnesses total **440** exact checks + **67** certified guards + **13** PSLQ corroborations + **7** labelled walls, all exit 0.

v2.1 session harness (one, written cold this session; every `ND-` check ID cited in §4 appears in the file):

harness	content	exact/computed
`nd_softcheck.py`	(ODD-2) by relative-norm descent: `ND-R` the refutation lock on the candidate $k \mid m$ split (Rem. 4.18); `ND-E` TP-2 elementary — symbolic convexity of $\log(1+e^t)$, the chain $\lvert g(1)\rvert \geq 1 \Rightarrow \prod_A (N_i - 1) \geq 1 \Rightarrow \mathrm{M} \geq 2^r$ on a wide forced box, equality only at $x-2$, and the classical totally-real bound $\mathrm{M} \geq \varphi^{d/2}$ (Schinzel) demoted to corroboration — checked, load-bearing nowhere (Lem. 4.15); `ND-T` complete forced TP boxes $d' \leq 4$, 0 violators; `ND-N` the per-object descent certificate over the forced on-ray population (565 objects, 0 failures, $e \mid d'$ throughout, the $1 < k < m$ gap certified); `ND-S` separation from the $\varphi^{d/m}$ fallback; `ND-X` sharpness at $x^m - 2$ and $x^6 - 3x^3 + 1$; `ND-A` anchor normalisation (Lem. 4.16)	$28 + 0$

Discipline: the structural decisions are exact over $\mathbb{Z}$ (irreducibility and factorization via sympy, the resultant $\mathrm{Res}_x(f,\, y - x^m)$, the tower law $d = kd'$, and $e \mid d'$); measures, ray tests and the minimal polynomial of N are computed at 140 dps, the last being *certified* after search — monic, irreducible over $\mathbb{Z}$, vanishing at N to 10^{-45}, and of degree dividing d', four facts that force it to be the minimal polynomial. The population is *forced, not curated*: all irreducible non-cyclotomic monic integer polynomials in the boxes $\{\deg 2 : \lvert a_j\rvert \leq 12,\ 3 :\leq 10,\ 4 :\leq 8,\ 5 :\leq 5,\ 6 :\leq 4\}$ landing on R_m, $m \in \{1, 3, 5, 7, 9\}$. *Nothing in Theorem 4.17 rests on this harness*: the theorem is proved, and the harness corroborates it, exhibits its extremals, and certifies the anchor dichotomy (`ND-A1`: 0 objects with both $\pm\alpha$ on-ray, 0 with neither).

v2.2 session harness (one, written cold this session; every `XS-` check ID cited in §4 appears in the file):

harness	content	exact+guards
`xs_softcheck.py`	cross-shell no-tie (Thm 4.12): `XS-A` fail-first — the ν-detector reproduces Thm 4.6's generator ($\nu \equiv 1$, two shells, eight cross-shell pairs), sees $\nu = -1$ (order 2) on $x^4 + x^2 - 6$, and orders $\{1, 3\}$ on $(x^2 - x - 1)(x^2 + x + 1)$; `XS-B` the forced theorem — ($\Leftarrow$) admissible $\Rightarrow$ coherent on the keystones, and the conjugation-closure identity $\nu(\alpha, \beta)\, \nu(\alpha, \bar\beta) = \alpha^2/\bar\alpha^2$ (β cancels, forcing $2\theta_\alpha \in \pi\mathbb{Q}$); `XS-C` corroboration over the same forced 855-object window as `rn_softcheck.py` — coherent $\iff$ admissible, 0 exceptions, 34 multi-coherent / 788 multi-non-coherent / 33 single-shell, cross-shell ν-orders $\{1, 2, 3, 4\}$; `XS-H` certified guards — Pisot $x^3 - x - 1$, the Salem sextic, and Lehmer's degree-10 Salem carry non-torsion cross-shell ν	$10 + 3$

Discipline: the symbolic checks (`XS-A`, `XS-B`) decide in $\mathbb{Q}(\sqrt{5})[i]$; window admissibility (`XS-C`) is the exact ψ^D oracle (Res + Sturm); shells and ν-orders on the scan are computed at 50 dps and the Salem non-coherence at 300 dps as certified guards (`XS-H`). *The theorem does not rest on the scan*: Theorem 4.12 is proved by conjugation-closure, and the 855-object run corroborates it and exhibits the population, exactly as `nd_softcheck.py` does for Theorem 4.17.

v2.3 session harness (one, written cold this session; every EV- check ID cited in §4 appears in the file):

harness	content	exact+fal.
`ev_softcheck.py`	even relational floor (Thm 4.20) + $p_1{=}q_2$ (Rem 8.6): EV-S the Schinzel battery — $\mathrm{M} \geq \varphi^{\deg/2}$ over the totally-real box, 0 violations, tight at $x^2 - x - 1$; EV-W the q_k minimizers on R_{2k}, $\mathrm{M} = \varphi$; EV-D the relative-norm descent certificate at q_1, q_2 (N tot. real $\neq \pm 1$, $\mathrm{M}(N) \geq \varphi^{e/2}$, $e \mid d'$, $\mathrm{M}(\alpha) \geq \varphi$); EV-F the parity-split floor scan (even-min $= \varphi$, odd-min $= 2$, no even below φ); EV-A $p_1 = q_2 = x^4{+}x^2{-}1$ and the unique-hit mechanism ($L_{2n} = 3 \iff n = 1$)	$13 + 6$
`ax_softcheck.py`	C=4/9 not an emission eigenvalue (Rem 9.4): AX-1 $\sqrt{3}$ is the unique catalog 3-carrier (w_1=2; all others 0); AX-2 emission monomials keep $w_1 \geq 0$; AX-3 $\frac{4}{9}$ has $w_1{=}{-}8 < 0$, excluded; AX-4 the test discriminates ($3{=}\psi^2(\sqrt{3})$, w_1=4, not excluded) — the exact prime-3 mirror of Prop 3.8	4

Discipline: totally-real / all-real decisions exact over $\mathbb{Z}$ (Sturm); Mahler measures at 40 dps as certified displays backed by the exact all-real gates; the six *falsifiers* (F1–F6) must *fire* — the exponent-d Schinzel reading (false for totally real, F2), the TP-2-only route ($\sqrt{2} < \varphi$, F3), an odd object fed to the even claim (F4), and the empty sub-φ even sweep (F5). *The theorem does not rest on the scan*: Theorem 4.20 is proved by the descent, and `ev` corroborates it and certifies the one external import (Schinzel), as `nd_softcheck.py` does for Theorem 4.17, and `ax_softcheck.py` (v2.4, the C=4/9 emission-eigenvalue kill of Remark 9.4) does for Proposition 3.8 at the prime 3. With them the *twenty-two* session harnesses total **495** exact/computed checks + **70** certified guards + **6** fail-first falsifiers + **13** PSLQ corroborations + **7** labelled walls (five after BL-G7 joins BL-G4 off the board), all exit 0.

Every boxed identity of the source reports and every displayed identity of the review thread appears above with a passing check ID; the eight findings (12.1, 12.2, 12.3, 12.4, 12.5, 12.6, 12.7, 12.8) are the complete defect list, extended in v2.1 by two — one corrected framing and one superseded harness artifact, neither touching a decision: the five v1.8 verifiers flagged only citation-hygiene in the derivation reports (companion-paper theorem numbers), corrected on the way in, and found no new display defect. The whitepaper's own harnesses (87+1 shipped; 89 independent) certify the upstream structure this paper conditions on.

14 The active blockers: status of the open problems

This closing section is the definitive, machine-certified statement of exactly what stands between the corpus and each remaining open question. The corpus's *forced* content is complete through v1.9; what is recorded here is not new mathematics but a precise *location of the walls*. Every classification below, and the single load-bearing forced fact it rests on, is re-certified in one place by `bl_softcheck.py` (groups BL-A–BL-H), which does *not* re-derive the corpus but isolates, for each open problem, the one forced fact that fixes its blocker class and shows each wall-detector *fail-first* by rejecting a planted false wall. The harness also *labels* — without checking, because no harness can settle them — the seven walls that lie outside the substrate (BL-G1–BL-G7). Pass line: `BL: EXACT 16/16 checks passed | 7 walls LABELLED`.

v2.1 motion. One row has since moved off the board. P3/P4 (the odd relational floor) is CLOSED by Theorem 4.17: *eight* of the nine rows below remain active, the NAMED-LEMMA class is empty, and BL-G4 is no longer a wall. The row and the harness output are retained verbatim rather than rewritten — the ledger is a record of what blocked what, and a blocker that fell is part of that record (Finding 12.8). The eight survivors are untouched by v2.1: nothing in the descent bears on a declared atom, a restatement wall, or a transcendence tool-gap.

v2.2 motion. A second row moves off the board. P1′ cross-shell ν (§14.9) is RESOLVED by Theorem 4.12: the UNFINISHED-COMPUTATION class is now empty — its sole occupant executed and its no-tie proved forced — `BL-G7` is no longer a wall, and *seven* of the nine rows remain active. The row is retained and marked, per the same record-keeping discipline. The seven survivors are untouched by v2.2: the cross-shell detector bears on no declared atom, no restatement or impossibility wall, and no external tool-gap.

The six blocker classes. Each open problem is filed under exactly one (for OP-RATE, a dual) of the following.

IMPOSSIBILITY
: a forced no-go: the resolution cannot come from the stated axioms. A hard wall, always scoped *relative* to D1–D3 (equivalently, relative to the discrete substrate), never an absolute claim.

DECLARED-ATOM
: an irreducible declaration the substrate does not force; resolvable only by adding a new principle.

RESTATEMENT-WALL
: every proposed reduction is logically equivalent to the axiom; the residual content is irreducible.

NAMED-LEMMA
: reduced to a single precisely-stated open lemma requiring new (internal) mathematics. *As of v2.1 this class is empty*: its sole occupant, ODD-2, is proved (Theorem 4.17). The class is retained because it is the one class whose occupancy the corpus could always have discharged by its own means — and, in the event, did.

EXTERNAL-TOOL-GAP
: blocked on a theorem that does not exist (no tool reaches the regime) or a named conjecture.

UNFINISHED-COMPUTATION
: tractable in principle, not yet executed. *As of v2.2 this class is empty*: its sole occupant, the P1′ cross-shell ν-criterion, is executed and its no-tie proved forced (Theorem 4.12).

Throughout, an IMPOSSIBILITY verdict asserts that the resolution cannot be a consequence of D1–D3 (or of the discrete substrate); it is a statement *about the axiom system*, never an absolute impossibility in mathematics at large.

open problem	active blocker class	load-bearing forced fact (id)	what would resolve it
OP-RATE (derive κ)	IMPOSSIBILITY (discrete) + DECLARED-ATOM (D3)	discrete rung state is k-invariant, so every functional of it is branch-blind `BL-A1`	a principle forcing the continuum rate D3 (none in the corpus)
OP-RADIUS ab initio ([OPEN] 1)	IMPOSSIBILITY (apex parity)	$v_z(r^2)$=1 is odd while the chart value group is $2\mathbb{Z}$ `BL-B1/BL-B2`	adjoin the odd generator (signed $\sqrt{z}$); it is outside D1–D3
OP-RADIUS profile (given the bridge)	DECLARED-ATOM (D4)	orientation is a free $\mathbb{Z}/2$ sign `BL-C1`; cap onset is a free datum `BL-C2`	a formation-dynamics principle fixing onset + orientation
W[OPEN]2 / D2 (χ)	RESTATEMENT-WALL	naming the K-seed clock is logically equivalent to the quarter-turn selection `BL-D2`	a principle more primitive than the quarter-turn (none)
P3/P4 (odd floor = 2)	CLOSED in v2.1 — was NAMED-LEMMA (ODD-2)	TP-2: $\mathrm{M}(N) \geq 2^r$ for N tot. positive, $N \neq 1$, elementary `ND-E1--ND-E3`; descent $\mathrm{M}(\alpha) \geq \mathrm{M}(N)^{d'/e}$, $e \mid d'$ `ND-N3, ND-N4`	*nothing* — *resolved* by Thm. 4.17; row retained per Finding 12.8
L_{res} transcendence	EXTERNAL-TOOL-GAP	m transcendental iff κ transcendental `BL-F1`; labelled `BL-G1`	a transcendence theorem at a transcendental modulus
$(\pi, \ln\varphi)$ independence	EXTERNAL-TOOL-GAP (conjecture)	$\{\ln\varphi, i\pi\}$ $\mathbb{Q}$-linearly independent is forced `BL-F3`; labelled `BL-G2`	Schanuel's conjecture
$(\kappa, L_{\mathrm{res}})$ independence	EXTERNAL-TOOL-GAP (periods)	L_{res} an elliptic value outside the exp/log framework; labelled `BL-G3`	Kontsevich–Zagier period-relation theory
P1′ cross-shell ν	RESOLVED in v2.2 — was UNFINISHED-COMPUTATION	the no-tie is *proved* (Thm 4.12): multi-shell cross-shell coherent $\iff$ admissible; corroborated 855/855, 0 exceptions `XS-C2`; `BL-G7` discharged	*nothing* — *resolved*

14.1 OP-RATE: deriving the winding rate

Status (register): *closed as a classification* (v1.8), open only as a continuum derivation. **Blocker:** dual — IMPOSSIBILITY (discrete) + DECLARED-ATOM (D3). **Forced fact:** the discrete rung state is k-invariant — every branch $\kappa_k = (4k+1)\kappa$ yields the identical rung values i^n (`BL-A1`; origin `cl/dx`, Lemma 2.16, Theorem 2.17) — so *every* functional of the discrete state is branch-blind and cannot single out $k = 0$. This is an IMPOSSIBILITY *relative to the discrete substrate*: no discrete-level principle derives the rate (Corollary 2.18). The residual is then the continuum rate D3, a DECLARED-ATOM. **Missing object:** a principle forcing D3 from something more primitive than a declaration; none exists in the corpus. A continuum-referencing derivation is not excluded (it would live outside the discrete substrate), so the classification is closed while the derivation stays open.

14.2 OP-RADIUS ab initio

Status (register): *open as an ab-initio derivation*; whitepaper [OPEN] 1 is *not* closed. **Blocker:** IMPOSSIBILITY (apex parity). **Forced fact:** at the apex the inherited radius-squared has odd valuation, $v_z(r^2) = 1$, whereas every chart quantity has valuation in the value group $2\mathbb{Z}$ (`BL-B1/BL-B2`; origin `rd/rx`, Theorem 10.9, Theorem 10.6); a planted "r^2 is even" is rejected fail-first (`BL-B3`).

Missing object: the odd generator — the signed square root $\sqrt{z}$ — is provably outside the chart's differential closure and must be *adjoined*; deriving r from D1–D3 alone is therefore impossible *relative to those axioms.* **Honest scope:** this impossibility is scoped to D1–D3; r *itself* is an elementary function ($r = (K/\sqrt{z_c})\,(z^2)^{1/4}$, Proposition 10.14). The wall is not that r is exotic, but that the substrate's even-parity chart cannot reach it. **Scope caveat (v2.3).** The no-go is proved at the *chart-rational* level ($r^2 \notin \mathbb{Q}(\sqrt{5})(z^2)$; odd apex valuation). W-§23.1 asks the *gate-family / elementary-differential* question one level up, and there r *is* elementary (Proposition 10.14), so parity — which $\frac{d}{dz}$ shifts and so does not preserve — is not the deciding invariant at that level. The statement that answers the question actually asked lives at the *unramified* elementary-differential closure, where r's apex monodromy order 2 (a branch point) excludes it — but that is *conditional on the unit-log hypothesis* (Proposition 10.14), and closing W[OPEN] 1 by IMPOSSIBILITY at that strength needs a localized structure theorem (a Rosenlicht/Risch localization carrying a monodromy-order invariant), an *internal* named missing object. Parity settles the chart level; the gate-family level stays conditional.

14.3 OP-RADIUS profile, given the bridge

Status (register): *reduced* to the minimal declared atom D4 (v1.9); given D4 and the inherited chart-ties the profile $r(z) = K\sqrt{z/z_c}$ (capped) is uniquely forced (Theorem 10.12). **Blocker:** DECLARED-ATOM (D4). **Forced fact:** D4 has two irreducible halves, each a free datum the substrate does not fix — the *orientation* of the apex double cover is a free $\mathbb{Z}/2$ sign ($+r$ and $-r$ both square to r^2 yet differ, BL-C1), and the *cap onset* is a free choice (z_c versus, e.g., 4/5 give distinct capped profiles, BL-C2); origin `ra`. Each half is *necessary* (Theorem 10.12). **Missing object:** a formation-dynamics principle that forces the cap onset at z_c, plus an orientation of the apex double cover (BL-G5). Until such a principle is added, D4 is declared, not derived; and "D4 is *the* minimal axiomatization" is itself only a candidate (the kink-slope-plus-sign atom is equivalent, Remark 10.13).

14.4 W[open]2 / D2: the chi-selection theorem

Status (register): *advanced and substrate-grounded, still PARTIAL as a derivation* (v1.9); D2's content is relocated to a substrate-grounded, g1-disjoint atom, an advance, not a closure (Remark 3.13). **Blocker:** RESTATEMENT-WALL. **Forced fact:** the K-seed rotation pair $\pm i\beta$ sits *off* the unit circle ($|i\beta|^2 = \varphi^4 - 1 > 1$, BL-D1), so it is genuinely disjoint from the on-circle-image obstruction g1; but naming that K-seed clock is logically *equivalent* to the quarter-turn selection over the competing quanta {terrain $\pi : \mathbb{Z}/2$, pentagon $2\pi/5 : \mathbb{Z}/5$, quarter $\pi/2 : \mathbb{Z}/4$} (BL-D2; origin `qt` / QT-C3/C5, Proposition 3.12). Under the relocated atom $\chi = \pm\pi/2$ becomes the *forced* argument of a forced object and the $\mathbb{Z}/4\mathbb{Z}$ completion a forced output — but the atom is not thinner than D2. **Missing object:** a principle forcing the clock-choice from something more primitive than the quarter-turn selection; none exists in the corpus (BL-G6). Every candidate reduction is logically equivalent to D2, so its residual content is irreducible. **Audit not run (v2.3).** The drop-one-witness test is applied to the keystone's four constraints (W-§3) and to the $\lambda = 2c$ atoms, but not to D1–D4 themselves. Theorem 3.3's unconditional $J = \pm UA$ already forces the quarter-turn's *existence* and the angle lattice $(\pi/2)\mathbb{Z}$; whether it also forces D2's residual *selection* given D3's rate — i.e. whether a model of {D1, D3, keystone algebra} can violate D2 — is a distinct question the restatement-wall verdict does not answer. The wall says every proposed *reduction* restates D2; an *independence* audit (a drop-one model) has not been run, and if D2 proved redundant the row would *retire* rather than close. Flagged as an open audit, not a resolution.

14.5 P3/P4: the odd relational floor equals two — closed

Status (register): *proved* (v2.1). $\mu_{\text{rel}}(\text{odd } m) = 2$ for every odd m, unconditionally (Theorem 4.17); P4 inherits it exactly (Remark 4.19). **Blocker:** NONE. This row is retained, and the v2.0 text below preserved in substance, because the way it was mis-classified is the useful part of the record (Finding 12.7).

What was true, and stays true. The $= 2$ *ceiling* is forced — $\mathrm{M}(x^m - 2) = 2$ for odd m ($x^m - 2$ is Eisenstein at 2, all roots at modulus $2^{1/m} > 1$, `BL-E1`); Smyth's theorem alone gives only $\mu_S < \varphi < 2$ (the plastic number $\mu_S = 1.3247$, `BL-E2`); the real-reciprocal sub-case is forced $\geq \varphi^2 > 2$ (`BL-E3`); and the naive "non-reciprocal + Smyth forces ≥ 2" idea is false. **What was wrong.** v2.0 recorded the missing object as "a Mahler-measure lower bound of 2 in that charge-constrained regime — new number theory strengthening Smyth", and labelled the residual lift $\mu_S \to 2$ an unsettleable wall (`BL-G4`). Both inferences ran through the realification route, which discards the ray constraint; they are properties of that route, not of the lemma (Finding 12.7). **What supplied it.** Not a strengthening of Smyth — a change of route. The relative norm $N = N_{K/F}(\alpha)$, $F = \mathbb{Q}(\alpha^m)$, carries the ray constraint forward: N is totally positive because $\pi \notin (2\pi/m)\mathbb{Z}$ for odd m, and $N \neq 1$ because α is not a root of unity. A totally positive algebraic integer $N \neq 1$ has $\mathrm{M}(N) \geq 2$ by an elementary argument (Lemma 4.15: $|g(1)| \geq 1$ plus convexity of $\log(1+e^t)$), and $e \mid d'$ makes the descent exponent d'/e a positive integer, so $\mathrm{M}(\alpha) \geq \mathrm{M}(N)^{d'/e} \geq 2$. Sharp at $x^m - 2$. Certified over a forced population of 565 objects, 0 failures, including the 20 objects with $1 < k < m$ that no degree split reaches `ND-N1--ND-N5`. **Cost to the corpus:** none — no declared atom, no external theorem, no new axiom.

14.6 Transcendence of L_{res}

Status (register): *external-open*, with a named tool-gap; not resolved here. **Blocker:** EXTERNAL-TOOL-GAP. **Forced fact:** via the Möbius correspondence $m = \kappa^2/(1+\kappa^2)$, the elliptic modulus m is transcendental if and only if κ is transcendental (`BL-F1`; origin `pf`, Remark 2.21); and κ is transcendental (Theorem 2.8), so the modulus is transcendental. **Missing object:** a transcendence theorem for elliptic periods at a *transcendental* modulus. No such theorem exists — Schneider's and Chudnovsky's results require an *algebraic* modulus, and ours is provably transcendental. This is labelled `BL-G1`, not checked: the paper names the tool-gap and does not claim the transcendence.

14.7 Algebraic independence of π and $\ln \varphi$

Status (register): *conditional on Schanuel*; not resolved here. **Blocker:** EXTERNAL-TOOL-GAP (a named conjecture). **Forced fact:** the premise $\{\ln \varphi, i\pi\}$ is $\mathbb{Q}$-linearly independent is forced, and Baker's theorem delivers the $\overline{\mathbb{Q}}$-linear independence of $\{1, \ln 2, \ln 3, \ln \varphi\}$ (`BL-F3`; origin `pf`, Remark 2.23) — but linear independence is all Baker reaches. **Missing object:** Schanuel's conjecture, which would upgrade the forced linear independence to algebraic independence of π and $\ln \varphi$. Labelled `BL-G2`; the conclusion is Schanuel-conditional and is *not* asserted.

14.8 Algebraic independence of κ and L_{res}

Status (register): *external-open*, beyond Schanuel; not resolved here. **Blocker:** EXTERNAL-TOOL-GAP (period-relation theory). **Forced fact:** L_{res} is an elliptic value (a period) lying *outside* the exponential/logarithm framework in which Schanuel operates (origin `pf`, Remark 2.21); the certified PSLQ search finds no integer relation among the corpus bases at 230 dps, height $\leq 10^{12}$ ([COMPUTED] corroboration only). **Missing object:** Kontsevich–Zagier period-relation theory (conjectural) — the framework that would decide relations among periods. Labelled BL-G3; the independence is neither proved nor claimed, and the PSLQ non-relation corroborates but never proves it.

14.9 P1′ cross-shell nu-criterion — resolved (v2.2)

Status (register): *resolved* (v2.2). Within-shell classification delivered (v1.8); the cross-shell ν-criterion half is now *proved* a no-tie (Theorem 4.12). **Former blocker:** UNFINISHED-COMPUTATION — a cross-shell coherence detector, tractable in principle and simply not yet executed. **Resolution:** the detector was built, validated fail-first against Theorem 4.6 (it reproduces the generator's $\nu \equiv 1$ and discriminates the Salem/Pisot non-coherent objects), and run; and the equivalence it computes was *proved forced* — for a multi-shell object, cross-shell coherence $\iff$ charge-admissibility, by conjugation-closure ($\theta_\alpha \pm \theta_\beta \in \pi\mathbb{Q} \Rightarrow \theta_\alpha \in \pi\mathbb{Q}$). The within-shell OTHER $= 0$ verdict is thereby the complete cross-shell statement: no object carries a cross-shell mechanism absent from its within-shell class. Corroborated over the same 855-object forced window with 0 exceptions (`xs_softcheck.py`, XS-C2). BL-G7 is discharged; this was the sole UNFINISHED-COMPUTATION *blocker row*, and that class is now *empty*. This is distinct from the *general* within-shell taxonomy over unbounded degree, which is a [COMPUTED] corroboration (Remark 4.11), *never* a §14 row, and stays [COMPUTED]: a wider scan corroborates but does not *settle* an infinite family. Extending Remark 4.11's window past its degree-6 cap (e.g. to Lehmer's degree 10) is thus a runnable corroboration, not a blocker whose non-execution the ledger tracks.

Closing. The corpus's forced content is complete through v1.9; v2.1 adds the odd relational floor, proved $= 2$ (Theorem 4.17), and v2.2 adds the cross-shell ν-criterion no-tie (Theorem 4.12). What remains falls into *four* of the six blocker classes above — both the NAMED-LEMMA class (discharged from inside in v2.1, by an elementary argument requiring no external theorem and no new axiom) and the UNFINISHED-COMPUTATION class (executed, and its output proved forced, in v2.2) are now *empty*. *Three are hard walls internal to the axiom system* — the two impossibilities (OP-RATE at discrete resolution and OP-RADIUS ab initio) and the restatement wall (D2); *two are single declared atoms* (D4's cap-onset and orientation); and *three are external tool-gaps* (the transcendence and algebraic-independence questions, blocked respectively on a nonexistent transcendental-modulus theorem, on Schanuel's conjecture, and on Kontsevich–Zagier period theory). *Seven rows, not eight.* No remaining §14 row is a mere oversight or an unattempted derivation. The honest scope of that claim is *by the routes the corpus had reached for*, and v2.3 shows the audit is never finished: outside the seven-row ledger, the rung-1 seed's floor hit was a settleable identity ($p_1 = q_2$, Remark 8.6) carried three versions as a coincidence-candidate, and the *even* relational floor (Theorem 4.20) was reachable by the odd descent with Schinzel in place of TP-2 — both settleable, both unsettled until named. That is Finding 12.7's lesson one level up: a completeness pass over the whitepaper's tags against these rows is itself an open audit. Each of the seven survivors is a precisely located wall, and for each this paper states the load-bearing forced fact and the exact missing object rather than claiming a resolution — and, per the honesty gate, every IMPOSSIBILITY is scoped relative to D1–D3 and no external-open item is asserted to be resolved. One caution the ODD-2 and cross-shell closures earn their own ledger: a blocker classification is a statement about the routes in hand — §14.5 sat at NAMED-LEMMA for three versions on the strength of a framing (Finding 12.7) rather than a theorem, and §14.9 sat at UNFINISHED-COMPUTATION until the detector was actually built, validated, and its output proved forced. The remaining seven are argued from forced facts, not from framings — but the distinction is exactly what this section exists to keep visible.